Exercise J-M

1. Let $\vec{u}=\hat{i}-\hat{j}-2\hat{k}$, $\vec{v}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{v}.\vec{w}=2$ and $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$. Then $\vec{u}.\vec{w}$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{u}=\hat{i}-\hat{j}-2\hat{k}$, $\vec{v}=2\hat{i}+\hat{j}-\hat{k}$, $\vec{v}.\vec{w}=2$ तथा $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$. तो $\vec{u}.\vec{w}$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $-\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Ans. D

Sol. $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$

$(\vec{v}\times\vec{w})\vec{v}=\vec{u}\cdot\vec{v}+\lambda\vec{v}\cdot\vec{v}$

$0=\vec{u}\cdot\vec{v}+\lambda|v|^2$

$\lambda=-\left(\frac{\vec{u}\cdot\vec{v}}{|v|^2}\right)=-\left(\frac{2-1+2}{6}\right)=-\frac{1}{2}$

Now, $(\vec{v}\times\vec{w})\vec{w}=\vec{u}\cdot\vec{w}+\lambda\vec{v}\cdot\vec{w}$

$\Rightarrow 0=\vec{u}\cdot\vec{w}+2\lambda\Rightarrow\vec{u}\cdot\vec{w}=-2\lambda$

$\Rightarrow\vec{u}\cdot\vec{w}=-2\left(-\frac{1}{2}\right)=1$

2. Let $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ and $\beta=\hat{i}+2\hat{j}-4\hat{k}$. Let $\vec{\beta}_1$ be parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ be perpendicular to $\vec{\alpha}$. If $\vec{\beta}=\vec{\beta}_1+\vec{\beta}_2$, then the value of $5\vec{\beta}_2\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)$ is : **[JEE(Main)Jan.2023]**

माना $\vec{\alpha}=4\hat{i}+3\hat{j}+5\hat{k}$ एवं $\beta=\hat{i}+2\hat{j}-4\hat{k}$ है। माना $\vec{\beta}_1$, $\vec{\alpha}$ के समान्तर कोण तथा $\vec{\beta}_2$, $\vec{\alpha}$ के लम्बवत है यदि $\vec{\beta}=\vec{\beta}_1+\vec{\beta}_2$ है, r c $5\vec{\beta}_2\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)$ का मान है : **[JEE(Main)Jan.2023]**

(A) 7 (B) 6 (C) 9 (D) 11

Ans. A

Sol. Let $\vec{\beta}_1=\lambda\vec{\alpha}$

Now $\vec{\beta}_2=\vec{\beta}-\vec{\beta}_1$

$=(\hat{i}+2\hat{j}-4\hat{k})-\lambda(4\hat{i}+3\hat{j}+5\hat{k})$

$=(1-4\lambda)\hat{i}+(2-3\lambda)\hat{j}-(5\lambda+4)\hat{k}$

$\vec{\beta}_2\cdot\vec{\alpha}=0$

$\Rightarrow 4(1-4\lambda)+3(2-3\lambda)-5(5\lambda+4)=0$

$\Rightarrow 4-16\alpha+6-9\lambda-25\lambda-20=0$

$\Rightarrow 50\lambda=-10$

$\Rightarrow \boxed{\lambda = \frac{-1}{5}}$

$\vec{\beta}_2 = \left(1+\frac{4}{5}\right)\hat{i} + \left(2+\frac{3}{5}\right)\hat{j} - (-1+4)\hat{k}$

$\vec{\beta}_2 = \frac{9}{5}\hat{i} + \frac{13}{5}\hat{j} - 3\hat{k}$

$5\vec{\beta}_2 = 9\hat{i} + 13\hat{j} - 15\hat{k}$

$5\vec{\beta}_2 \cdot (\hat{i}+\hat{j}+\hat{k}) = 9+13-15 = 7$

3. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non zero vectors such that $\vec{b}.\vec{c} = 0$ and $\vec{a}\times\left(\vec{b}\times\vec{c}\right) = \frac{\vec{b}-\vec{c}}{2}$. If $\vec{d}$ be a vector such that $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$, then $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{a}, \vec{b}$ और $\vec{c}$ तीन शून्येत्तर सदिश हैं जबकि $\vec{b}.\vec{c} = 0$ और $\vec{a}\times\left(\vec{b}\times\vec{c}\right) = \frac{\vec{b}-\vec{c}}{2}$ । यदि $\vec{d}$ एक सदिश ऐसा है कि $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$ तब $\left(\vec{a}\times\vec{b}\right)\cdot\left(\vec{c}\times\vec{d}\right)$ बराबर होगा– **[JEE(Main)Jan.2023]**

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $-\frac{1}{4}$ (D) $\frac{1}{2}$

Ans. A

Sol. $\vec{a}\times(\vec{b}\times\vec{c}) = \frac{\vec{b}-\vec{c}}{2} \Rightarrow (\vec{a},\vec{c})\vec{b} - (\vec{a},\vec{b})\vec{c} = \frac{\vec{b}}{2} - \frac{\vec{c}}{2}$

$\Rightarrow \vec{a}.\vec{c} = \frac{1}{2}, \vec{a}\cdot\vec{b} = \frac{1}{2} (\because \vec{b}\cdot\vec{c} = 0)$

Given $\vec{b}\cdot\vec{d} = \vec{a}\cdot\vec{b}$

Now $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c}) = \frac{1}{4}$

4. The vector $\vec{a} = -\hat{i}+2\hat{j}+\hat{k}$ is rotated through a right angle, passing through the y-axis in its way and the resulting vector is $\vec{b}$. Then the projection of $3\vec{a}+\sqrt{2}\vec{b}$ on $\vec{c} = 5\hat{i}+4\hat{j}+3\hat{k}$ is : **[JEE(Main)Jan.2023]**

सदिश $\vec{a} = -\hat{i}+2\hat{j}+\hat{k}$ को y-अक्ष से होकर ले जाते हुए एक समकोण तक घुमाया जाता है और इसके परिणामस्वरूप सदिश $\vec{b}$ प्राप्त gks kgsSA rks $3\vec{a}+\sqrt{2}\vec{b}$ का सदिश $\vec{c} = 5\hat{i}+4\hat{j}+3\hat{k}$ पर प्रक्षेप है : **[JEE(Main)Jan.2023]**

(A) $\sqrt{6}$ (B) $2\sqrt{3}$ (C) 1 (D) $3\sqrt{2}$

Ans. D

Sol. $\vec{b}=\lambda\vec{a}\times(\vec{a}\times\hat{j})$

$\Rightarrow\vec{b}=\lambda(-2\hat{i}-2\hat{j}+2\hat{k})$

$|\vec{b}|=|\vec{a}| \quad \therefore\sqrt{6}=\sqrt{12}\,|\lambda|\Rightarrow\lambda=\pm\frac{1}{\sqrt{2}}$

$\left(\lambda=\frac{1}{\sqrt{2}}\text{ rejected }\because\vec{b}\text{ makes acute angle with y axis}\right)$

$\vec{b}=-\sqrt{2}(-\hat{i}-\hat{j}+\hat{k})$

$\frac{(3\vec{a}+\sqrt{2}\vec{b})\cdot\vec{c}}{|\vec{c}|}=3\sqrt{2}$

5. If the four points, whose position vectors are $3\hat{i}-4\hat{j}+2\hat{k},\hat{i}+2\hat{j}-\hat{k},-2\hat{i}-\hat{j}+3\hat{k}$ and $5\hat{i}-2\alpha\hat{j}+4\hat{k}$ are coplanar, then α is equal to : **[JEE(Main)Jan.2023]**

यदि चार बिंदु, जिनके स्थिति सदिश $3\hat{i}-4\hat{j}+2\hat{k},\hat{i}+2\hat{j}-\hat{k},-2\hat{i}-\hat{j}+3\hat{k}$ तथा $5\hat{i}-2\alpha\hat{j}+4\hat{k}$ है, सहतलीय है, तो α बराबर है **[JEE(Main)Jan.2023]**

(A) $-\frac{73}{17}$ (B) $\frac{107}{17}$ (C) $\frac{73}{17}$ (D) $-\frac{107}{17}$

Ans. C

Sol. $\begin{vmatrix}3-5 & -4+2\alpha & 2-4\\ 3-1 & -4-2 & 2+1\\ 3+2 & -4+1 & 2-3\end{vmatrix}=\begin{vmatrix}-2 & 2\alpha-4 & -2\\ 2 & -6 & 3\\ 5 & -3 & -1\end{vmatrix}=0$

$\begin{vmatrix}0 & 2\alpha-4 & -2\\ -1 & -6 & 3\\ 6 & -3 & -1\end{vmatrix}=0\Rightarrow-(2\alpha-4)(1-18)-2(3+36)=0$

$(\alpha-2)(-17)+39=0$

$\alpha-2=\frac{39}{17}\Rightarrow\alpha=2+\frac{39}{17}=\frac{34+39}{17}$

$\alpha=\frac{73}{17}$

6. Let $\vec{a}=-\hat{i}-\hat{j}+\hat{k},\vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{i}-\hat{j}$ then $\vec{a}-6\vec{b}$ is equal to : **[JEE(Main)Jan.2023]**

माना $\vec{a}=-\hat{i}-\hat{j}+\hat{k},\vec{a}\cdot\vec{b}=1$ तथा $\vec{a}\times\vec{b}=\hat{i}-\hat{j}$ तो $\vec{a}-6\vec{b}$ बराबर है **[JEE(Main)Jan.2023]**

(A) $3\left(\hat{i}+\hat{j}+\hat{k}\right)$ (B) $3\left(\hat{i}-\hat{j}-\hat{k}\right)$ (C) $3\left(\hat{i}+\hat{j}-\hat{k}\right)$ (D) $3\left(\hat{i}-\hat{j}+\hat{k}\right)$

Ans. A

Sol. $\vec{a}\times\vec{b}=(\hat{i}-\hat{j})$

Taking cross product with $\vec{a}$

$\Rightarrow \vec{a}\times(\vec{a}\times\vec{b})=\vec{a}\times(\hat{i}-\hat{j})$

$\Rightarrow (\vec{a}\cdot\vec{b})\vec{a}-(\vec{a}\cdot\vec{a})\vec{b}=\hat{i}+\hat{j}+2\hat{k}$

$\Rightarrow \vec{a}-3\vec{b}=\hat{i}+\hat{j}+2\hat{k}$

$\Rightarrow 2\vec{a}-6\vec{b}=2\hat{i}+2\hat{j}+4\hat{k}$

$\Rightarrow \vec{a}-6\vec{b}=3\hat{i}+3\hat{j}+3\hat{k}$

7. If the vectors $\vec{a}=\lambda\hat{i}+\mu\hat{j}+4\hat{k},\vec{b}=-2\hat{i}+4\hat{j}-2\hat{k}$ and $\vec{c}=2\hat{i}+3\hat{j}+\hat{k}$ are coplanar and the projection of $\vec{a}$ on the vector $\vec{b}$ is $\sqrt{54}$ units, then the sum of all possible values of $\lambda+\mu$ is equal to

[JEE(Main)Jan.2023]

यदि सदिश $\vec{a}=\lambda\hat{i}+\mu\hat{j}+4\hat{k},\vec{b}=-2\hat{i}+4\hat{j}-2\hat{k}$ तथा $\vec{c}=2\hat{i}+3\hat{j}+\hat{k}$ समतलीय हैं एवं $\vec{a}$ का सदिश $\vec{b}$ पर प्रक्षेप $\sqrt{54}$ इकाई है, तब $\lambda+\mu$ के सभी संभव मानों का योग है : **[JEE(Main)Jan.2023]**

(A) 6 (B) 18 (C) 24 (D) 0

Ans. C

Sol. $\begin{vmatrix}\lambda & \mu & 4\\ -2 & 4 & -2\\ 2 & 3 & 1\end{vmatrix}=0$

$\lambda(10)-\mu(2)+4(-14)=0$

$10\lambda-2\mu=56$

$5\lambda-\mu=28$ ... (1)

$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=\sqrt{54}$

$\frac{-2\lambda+4\mu-8}{\sqrt{24}}=\sqrt{54}$

$-2\lambda+4\mu-8=\sqrt{54\times24}$ ... (2)

By solving equation (1) and (2)

$\Rightarrow \lambda+\mu=24$

8. If $\vec{a}=\hat{i}+2\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$, $\vec{c}=7\hat{i}-3\hat{j}+4\hat{k}$, $\vec{r}\times\vec{b}+\vec{b}\times\vec{c}=\vec{0}$ and $\vec{r}\cdot\vec{a}=0$. Then $\vec{r}\cdot\vec{c}$ is equal to :

[JEE(Main)Jan.2023]

यदि $\vec{a}=\hat{i}+2\hat{k}$, $\vec{b}=\hat{i}+\hat{j}+\hat{k}$, $\vec{c}=7\hat{i}-3\hat{j}+4\hat{k}$, $\vec{r}\times\vec{b}+\vec{b}\times\vec{c}=\vec{0}$ तथा $\vec{r}\cdot\vec{a}=0$ है। तो $\vec{r}\cdot\vec{c}$ बराबर है :

[JEE(Main)Jan.2023]

(A) 30 (B) 34 (C) 36 (D) 32

Ans. B

Sol. $\vec{r}\times\vec{b}-\vec{c}\times\vec{b}=0$

$\Rightarrow(\vec{r}-\vec{c})\times\vec{b}=0$

$\Rightarrow\vec{r}-\vec{c}=\lambda\vec{b}$

$\Rightarrow\vec{r}=\vec{c}+\lambda\vec{b}$

And given that $\vec{r}\cdot\vec{a}=0$

$\Rightarrow(\vec{c}+\lambda\vec{b})\cdot\vec{a}=0$

$\Rightarrow\vec{c}\cdot\vec{a}+\lambda\vec{b}\cdot\vec{a}=0$

$\Rightarrow\lambda=\dfrac{-\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}$

Now $\vec{r}\cdot\vec{c}=(\vec{c}++\lambda\vec{b})\cdot\vec{c}$

$=\left(\vec{c}-\dfrac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}\vec{b}\right)\cdot\vec{c}$

$=|\vec{c}|-\left(\dfrac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}\right)(\vec{b}\cdot\vec{c})$

$=74-\left[\dfrac{15}{3}\right]8$

$= 74 - 40 = 34$

9. Let $\vec{a}=4\hat{i}+3\hat{j}$ and $\vec{b}=3\hat{i}-4\hat{j}+5\hat{k}$. If $\vec{c}$ is a vector such that $\vec{c}\cdot(\vec{a}\times\vec{b})+25=0$, $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})=4$, and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals : **[JEE(Main)Jan.2023]**

माना $\vec{a}=4\hat{i}+3\hat{j}$ तथा $\vec{b}=3\hat{i}-4\hat{j}+5\hat{k}$ है। यदि एक सदिश $\vec{c}$ के लिए $\vec{c}\cdot(\vec{a}\times\vec{b})+25=0$, $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})=4$ हैं तथा $\vec{c}$ का $\vec{a}$ पर प्रक्षेप 1 है, तो $\vec{c}$ का $\vec{b}$ पर प्रक्षेप बराबर है : **[JEE(Main)Jan.2023]**

(A) $\dfrac{1}{\sqrt{2}}$ (B) $\dfrac{1}{5}$ (C) $\dfrac{3}{\sqrt{2}}$ (D) $\dfrac{5}{\sqrt{2}}$

Ans. D

Sol. $\vec{a}\times\vec{b}=15\hat{i}-20\hat{j}-25\hat{k}$

Let $\vec{c}=x\hat{i}+y\hat{j}+z\hat{k}$

$\Rightarrow 15x - 20y - 25z + 25 = 0$

$\Rightarrow 3x - 4y - 5z = -5$

Also $x + y + z = 4$

and $\frac{\vec{c}\cdot\vec{a}}{|\vec{a}|}=1 \Rightarrow 4x+3y=5$

$\Rightarrow \vec{c}=2\hat{i}-\hat{j}+3\hat{k}$

Projection of $\vec{c}$ or $\vec{b}=\frac{25}{5\sqrt{2}}=\frac{5}{\sqrt{2}}$

10. If $\vec{a},\vec{b},\vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a}=\alpha\vec{b}-\hat{n}$, $(\alpha\neq 0)$ and $\vec{b}\cdot\vec{c}=12$, then $|\vec{c}\times(\vec{a}\times\vec{b})|$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\vec{a},\vec{b},\vec{c}$ तीन शून्येतर सदिश है तथा $\vec{c}$ के लंबवत एक इकाई सदिश $\hat{n}$ है, जिनके लिए $\vec{a}=\alpha\vec{b}-\hat{n}$, $(\alpha\neq 0)$ तथा $\vec{b}\cdot\vec{c}=12$ है, तो $|\vec{c}\times(\vec{a}\times\vec{b})|$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 12 (B) 9 (C) 6 (D) 15

Ans. A

Sol. $\vec{n}\cdot\vec{c}=0,|\vec{n}|=1$

$\vec{a}=\alpha\vec{b}-\hat{n}$ & $\vec{b}\cdot\vec{c}=12$

$\vec{a}\cdot\vec{c}=\alpha\vec{b}\cdot\vec{c}-0$

$\vec{a}\cdot\vec{c}=12\alpha$

$\left|\vec{c}\times\left(\vec{a}\times\vec{b}\right)\right|$

$\left|\left(\vec{c}\cdot\vec{b}\right)\cdot\vec{a}-\left(\vec{c}\cdot\vec{a}\right)\cdot\vec{b}\right|$

$\left|12\vec{a}-12\alpha\vec{b}\right|$

$12\left|\vec{a}-\alpha\vec{b}\right|$

$12|\hat{n}|=12$

11. Let $\lambda\in R$, $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$. If $\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)\times\left(\vec{a}-\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$, then $\left|\lambda\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\right|^2$ is equal to : **[JEE(Main)Jan.2023]**

माना $\lambda\in R$, $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$ है। यदि $\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)\times\left(\vec{a}-\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$ है, तब $\left|\lambda\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)\right|^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 136 (B) 140 (C) 144 (D) 132

Ans. B

Sol. $\vec{a}=\lambda\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\lambda\hat{j}+2\hat{k}$

$\Rightarrow\left(\vec{b}-\vec{a}\right)\times\left(\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}\times\vec{b}\right)\right)=8\hat{i}-40\hat{j}-24\hat{k}$

$\Rightarrow\left(\left(\vec{a}-\vec{b}\right)\cdot\left(\vec{a}+\vec{b}\right)\right)\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

$\Rightarrow 8\left(\vec{a}\times\vec{b}\right)=8\hat{i}-40\hat{j}-24\hat{k}$

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ \lambda & 2 & -3\\ 1 & -\lambda & 2\end{vmatrix}$

$=\left(4-3\lambda\right)\hat{i}-\left(2\lambda+3\right)\hat{j}+\left(-\lambda^2-2\right)\hat{k}$

$\Rightarrow\lambda=1$

$\therefore\vec{a}=\hat{i}+2\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}-\hat{j}+2\hat{k}$

$\Rightarrow\vec{a}+\vec{b}=2\hat{i}+\hat{j}-\hat{k},\vec{a}-\vec{b}=3\hat{j}-5\hat{k}$

$\Rightarrow\left(\vec{a}+\vec{b}\right)\times\left(\vec{a}-\vec{b}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 0 & 3 & -5\end{vmatrix}=2\hat{i}+10\hat{j}+6\hat{k}$

$\therefore$ required answer = 4 + 100 + 36 = 140

12. A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60°, to the y-axis at 45° and to the z-axis at an acute angle. If a plane passing through the points $\left(\sqrt{2},-1,1\right)$ and (a,b,c), is normal to $\vec{v}$, then :

[JEE(Main)Jan.2023]

प्रथम अष्टांशक में एक सदिश $\vec{v}$ का झुकाव x अक्ष से 60°, y-अक्ष से 45° तथा z-अक्ष से एक न्यून कोण है। यदि बिंदु $\left(\sqrt{2},-1,1\right)$ व (a,b,c) से होकर जाने वाला समतल $\vec{v}$ पर लंबवत् है, तो **[JEE(Main)Jan.2023]**

(A) $a+\sqrt{2}b+c=1$ (B) $\sqrt{2}a+b+c=1$ (C) $a+b+\sqrt{2}c=1$ (D) $\sqrt{2}a-b+c=1$

Ans. A

Sol. $\therefore l=\cos 60^o=\frac{1}{2}$ Let $\vec{v}=l\hat{i}+m\hat{j}+n\hat{k}$

$m=\cos 45^o=\frac{1}{\sqrt{2}}$

$\therefore l^2 + m^2 + n^2 = 1$

$\Rightarrow n^2 = \frac{1}{4}$

$\Rightarrow n = \frac{1}{2}, -\frac{1}{2}$ reject as $\vec{v}$ is in first octant

So, dr's of normal to the plane can be taken as 1, $\sqrt{2}$, 1

Hence

$1\left(a-\sqrt{2}\right)+\sqrt{2}\left(b+1\right)+1\left(c-1\right)=0 \qquad \Rightarrow a+\sqrt{2}b+c=1$

13. Let $\vec{a}$ and $\vec{b}$ be two vectors, Let $|\vec{a}|=1$, $|\vec{b}|=4$ and $\vec{a}\cdot\vec{b}=2$. If $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$, then the value of $\vec{b}\cdot\vec{c}$ is:

[JEE(Main)Jan.2023]

माना $\vec{a}$ तथा $\vec{b}$ दो सदिश है। माना $|\vec{a}|=1$, $|\vec{b}|=4$ तथा $\vec{a}\cdot\vec{b}=2$ है। यदि $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ है, तब $\vec{b}\cdot\vec{c}$ का मान है :

[JEE(Main)Jan.2023]

(A) -60 (B) -24 (C) -84 (D) -48

Ans. D

Sol. $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ .....(1)

$\vec{b}$..(1)

$\Rightarrow \vec{b}.\vec{c}=0-3\vec{b}.\vec{b}=-48$

14. Let $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$, and $\vec{b}$ and $\vec{c}$ be two non zero vectors such that $|\vec{a}+\vec{b}+\vec{c}|=|\vec{a}+\vec{b}-\vec{c}|$ and $\vec{b}\cdot\vec{c}=0$. Consider the following two statements : **[JEE(Main)Jan.2023]**

(A) $|\vec{a}+\lambda\vec{c}|\geq|\vec{a}|$ for all $\lambda\in R$. (B) $\vec{a}$ and $\vec{c}$ are always parallel.

Then,

(A) bot (A) and (B) are correct (B) only (A) is correct

(C) only (B) is correct (D) neither (A) nor (B) is correct.

माना $\vec{a}=2\hat{i}+\hat{j}+\hat{k}$ तथा $\vec{b}$ और $\vec{c}$ दो शून्येतर सदिश हैं जिनके लिए $|\vec{a}+\vec{b}+\vec{c}|=|\vec{a}+\vec{b}-\vec{c}|$ तथा $\vec{b}\cdot\vec{c}=0$ है। निम्न दो d Hkuksdkfopkj dhft , : **[JEE(Main)Jan.2023]**

(A) $\lambda\in R$ के लिए $|\vec{a}+\lambda\vec{c}|\geq|\vec{a}|$ (B) $\vec{a}$ तथा $\vec{c}$ सदा समांतर है

rks,

; । द्ध; । द्धतथा (B) सही है (B) केवल (A) सही है

(C) केवल (B) सही है (D) न तो (A) न ही (B) सही है

Ans. B

Sol. $\left|\vec{a}+\vec{b}+\vec{c}\right|^2=\left|\vec{a}+\vec{b}-\vec{c}\right|^2$

$2\vec{a}.\vec{b}+2\vec{b}.\vec{c}+2\vec{c}.\vec{a}=2\vec{a}.\vec{b}-2\vec{b}.\vec{c}-2\vec{c}.\vec{a}$

$4\vec{a}.\vec{c}=0$

B is incorrect.

$\left|\vec{a}+\lambda\vec{c}\right|^2\geq\left|\vec{a}\right|^2$

$\lambda^2c^2\geq 0$

True $\forall\ \lambda\in R$ (A) is correct.

15. Let : $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$ be there vectors. If $\vec{r}$ is a vector such that, $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{r}\cdot\vec{a}=0$, then $25\left|\vec{r}\right|^2$ is equal to : **[JEE(Main)Jan.2023]**

माना तीन सदिश $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}-\hat{j}+2\hat{k}$ तथा $\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$ है। यदि एक सदिश $\vec{r}$ के लिए $\vec{r}\times\vec{b}=\vec{c}\times\vec{b}$ तथा $\vec{r}\cdot\vec{a}=0$ है, तो $25\left|\vec{r}\right|^2$ बराबर है **[JEE(Main)Jan.2023]**

(A) 336 (B) 560 (C) 339 (D) 449

Ans. C

Sol. $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{b}=\hat{i}-\hat{j}+2\hat{k}$

$\vec{c}=5\hat{i}-3\hat{j}+3\hat{k}$

$(\vec{r}-\vec{c})\times\vec{b}=0,\vec{r}\cdot\vec{a}=0$

$\Rightarrow\vec{r}-\vec{c}=\lambda\vec{b}$

Also, $(\vec{c}+\lambda\vec{b})\cdot\vec{a}=0$

$\Rightarrow\vec{a}\cdot\vec{c}+\lambda(\vec{a}\cdot\vec{b})=0$

$\therefore\lambda=\dfrac{\vec{a}\cdot\vec{c}}{\vec{a}\cdot\vec{b}}=\dfrac{-8}{5}$

$\vec{r}=\dfrac{5(5\hat{i}-3\hat{i}+3\hat{k})-8(\hat{i}-\hat{j}+2\hat{k})}{5}$

$\vec{r}=\dfrac{17\hat{i}-7\hat{j}+\hat{k}}{5}$

$|\vec{r}|^2=\dfrac{1}{25}(289+50)$

$25|\vec{r}|^2=339$

16. Let $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}+5\hat{k}$. Then which one of the following statements is TRUE?

[JEE(Main)Jan.2023]

(A) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-17}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

(B) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(C) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{-17}{\sqrt{35}}$ and the direction of the projection vector is opposite to the direction of $\vec{b}$.

(D) Projection of $\vec{a}$ on $\vec{b}$ is $\frac{17}{\sqrt{35}}$ and the direction of the projection vector is same as of $\vec{b}$.

माना दो सदिश $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ तथा $\vec{b}=\hat{i}+3\hat{j}+5\hat{k}$ है। तो निम्न में से कौन सा कथन सही है ?

[JEE(Main)Jan.2023]

(A) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{-17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा में समान है

(B) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा के विपरीत है

(C) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{-17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा में विपरीत है

(D) $\vec{a}$ का $\vec{b}$ पर प्रक्षेप $\frac{17}{\sqrt{35}}$ है तथा प्रक्षेप सदिश की दिशा $\vec{b}$ की दिशा के समान है

Ans. Drop

Sol. $\vec{a}=5i-j-3k$, $\vec{b}=i+3j+5k$

(A) Projection of $\vec{a}$ on $\vec{b}$ = $\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{(5-3-15)}{\sqrt{1+9+25}}=\frac{-13}{\sqrt{35}}$

17. Let $\vec{a}=2\hat{i}-7\hat{j}+5\hat{k}$, $\vec{b}=\hat{i}+\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ and $\vec{r}\cdot\vec{b}=0$, then $|\vec{r}|$ is equal to : **[JEE(Main)Jan.2023]**

माना तीन सदिश $\vec{a}=2\hat{i}-7\hat{j}+5\hat{k}$, $\vec{b}=\hat{i}+\hat{k}$ तथा $\vec{c}=\hat{i}+2\hat{j}-3\hat{k}$ है। यदि एक सदिश $\vec{r}$ के लिए $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ तथा $\vec{r}\cdot\vec{b}=0$ है, तो $|\vec{r}|$ बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{11}{7}\sqrt{2}$ (B) $\frac{\sqrt{914}}{7}$ (C) $\frac{11}{5}\sqrt{2}$ (D) $\frac{11}{7}$

Ans. A

Sol. $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$

$\Rightarrow(\vec{r}-\vec{c})\times\vec{a}=0$

$\Rightarrow(\vec{r}-\vec{c})=\lambda\vec{a}$

$\Rightarrow \vec{r} = \vec{c} + \lambda\vec{a}$

$\Rightarrow \vec{r}.\vec{b} = \vec{c}.\vec{b} + \lambda\vec{a}.\vec{b}$

$\Rightarrow 0 = (1-3) + \lambda(2+5)$

$2 = 7\lambda \Rightarrow \lambda = \frac{2}{7}$

$\vec{r} = \vec{c} + \frac{2}{7}\vec{a}$

$= \hat{i} + 2\hat{j} - 3\hat{k} + \frac{2}{7}\left(2\hat{i} - 7\hat{j} + 5\hat{k}\right)$

$\vec{r} = \frac{11\hat{i} - 11\hat{k}}{7}$

$|\vec{r}| = \frac{\sqrt{121+121}}{7}$

$|\vec{r}| = \frac{11\sqrt{2}}{7}$

18. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{31}$, $4|\vec{b}| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$. If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना तीन सदिशों $\vec{a}, \vec{b}, \vec{c}$ के लिए $|\vec{a}| = \sqrt{31}$, $4|\vec{b}| = |\vec{c}| = 2$ तथा $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$ है। यदि $\vec{b}$ तथा $\vec{c}$ के बीच कोण $\frac{2\pi}{3}$ है, तो $\left(\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right)^2$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 3

Sol. Given $|\vec{a}| = \sqrt{31}, 4|\vec{b}| = |\vec{c}| = 2$

$2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$ $\quad \vec{b} \wedge \vec{c} = \frac{2\pi}{3}$

$3(\vec{c} \times \vec{a}) + 2(\vec{b} \times \vec{a}) = 0$

$(3\vec{c} + 2\vec{b}) \times \vec{a} = 0$

$3\vec{c} + 2\vec{b} = \lambda\vec{a}$

Squaring both sides

$9|\vec{c}|^2 + 4|\vec{b}|^2 + 12(\vec{b}.\vec{c}) = \lambda^2|\vec{a}|^2$

$$36+1+12\times\frac{1}{2}\times2\left(\cos\left(\frac{2\pi}{3}\right)\right)=\lambda^2(31)$$

$\lambda^2=1$

$\lambda^2=\pm1$

$3\vec{c}+2\vec{b}=\pm\vec{a}$ _____(1)

Dot with $\vec{b}=3\left(\vec{b}.\vec{c}\right)+2\left(\vec{b}.\vec{b}\right)=\pm\vec{a}.\vec{b}$

$$\vec{a}.\vec{b}=\pm\left(-\frac{3}{2}+\frac{1}{2}\right)=\pm(-1)$$

$$\left(\vec{a}.\vec{b}\right)^2=1$$

$$3\left(\vec{c}\times\vec{a}\right)=2\left(\vec{a}\times\vec{b}\right)$$

$$\left(\vec{c}\times\vec{a}\right)^2=\frac{4}{9}\left(\vec{a}\times\vec{b}\right)^2$$

$$=\frac{4}{9}\left[|\vec{a}|^2|\vec{b}|^2-\left(\vec{a}.\vec{b}\right)^2\right]$$

$$=\frac{4}{9}\left[\frac{31}{4}-(1)\right]$$

$$=\frac{4}{9}\times\frac{27}{4}=3$$

19. If $\lambda_1<\lambda_2$ are two values of λ such that the angle between the planes $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ and $P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1,10\lambda_2,2)$ to the plane P_1 is _____. **[JEE(Main)Jan.2023]**

यदि λ के दो मान, जिनके लिए समतलों $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ तथा $P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$ के बीच का कोण $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$ है, λ_1, λ_2,$(\lambda_1<\lambda_2)$ है, तो बिंदु $(38\lambda_1,10\lambda_2,2)$ से समतल P_1 पर लंब की लंबाई का वर्ग बराबर _____ है।

[JEE(Main)Jan.2023]

Ans. 315

Sol. $P_1:\vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ and

$P_2:\vec{r}\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$

$$\cos\theta = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}$$

$$\cos\theta = \frac{(3\lambda - 5 - 3)}{\sqrt{35}\cdot\sqrt{\lambda^2 + 10}}$$

$$\cos\theta = \frac{(3\lambda - 8)}{\sqrt{35}\cdot\sqrt{\lambda^2 + 10}}$$

$$\therefore \theta = \sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$$

$$\sin\theta = \frac{2\sqrt{6}}{5}$$

$$\sin^2\theta = \frac{24}{25}$$

$$1 - \cos^2\theta = \frac{24}{25}$$

$$1 - \frac{(3\lambda - 8)^2}{35\cdot(\lambda^2 + 10)} = \frac{24}{25}$$

$25\,[35\,(\lambda^2 + 10) - (3\lambda - 8)^2] = 35\cdot 24\,(\lambda^2 + 10)$
$5\,[35\lambda^2 + 350 - 9\lambda^2 - 64 + 48\lambda] = 168\lambda^2 + 1680$
$175\lambda^2 + 1750 - 45\lambda^2 - 320 + 240\lambda - 168\lambda^2 - 1680 = 0$
$-38\lambda^2 + 240\lambda - 250 = 0$
$19\lambda^2 - 120\lambda + 125 = 0$

$$\lambda = 5, \frac{25}{19}$$

$$\because \lambda_1 < \lambda_2$$

$$\therefore \lambda_2 = 5, \lambda_1 = \frac{25}{19}$$

$$\text{then point } \left(38\cdot\frac{25}{19}, 10\cdot 52\right) = (50, 50, 2)$$

$P_1 : 3x - 5y + z = 7$

$$d = \left|\frac{3\times 50 - 5\times 50 + 1\cdot 2 - 7}{\sqrt{9 + 25 + 1}}\right|$$

$$d = \left|\frac{150 - 250 + 2 - 7}{\sqrt{35}}\right|$$

$$d = \left|\frac{-100-5}{\sqrt{35}}\right| = \frac{105}{\sqrt{35}}$$

$$d^2 = \frac{105\times 105}{35}$$

$= 315$

20. Let $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$, $\vec{a}\cdot\vec{c} = 7, 2\vec{b}\cdot\vec{c} + 43 = 0$, $\vec{a}\times\vec{c} = \vec{b}\times\vec{c}$. Then $|\vec{a}\cdot\vec{b}|$ is equal to _____.

[JEE(Main)Jan.2023]

माना $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}$, $\vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}$, $\vec{a}\cdot\vec{c} = 7, 2\vec{b}\cdot\vec{c} + 43 = 0$, $\vec{a}\times\vec{c} = \vec{b}\times\vec{c}$ है। तब $|\vec{a}\cdot\vec{b}|$ बराबर है।

[JEE(Main)Jan.2023]

Ans. 8

Sol. $\vec{a} = \hat{i} + 2\hat{j} + \lambda\hat{k}, \quad \vec{b} = 3\hat{i} - 5\hat{j} - \lambda\hat{k}, \vec{a}\cdot\vec{c} = 7$

$\vec{a}\times\vec{c} - \vec{b}\times\vec{c} = \vec{0}$

$(\vec{a} - \vec{b})\times\vec{c} = 0 \Rightarrow (\vec{a} - \vec{b})$ is paralleled to $\vec{c}$

$\vec{a} - \vec{b} = \mu\vec{c}$, where μ is a scalar

$-2\hat{i} + 7\hat{j} + 2\lambda\hat{k} = \mu\cdot\vec{c}$

Now $\vec{a}\cdot\vec{c} = 7$ gives $2\lambda^2 + 12 = 7\mu$

And $\vec{b}\cdot\vec{c} = -\frac{43}{2}$ gives $4\lambda^2 + 82 = 43\mu$

$\mu = 2$ and $\lambda^2 = 1$

$|\vec{a}\cdot\vec{b}| = 8$

21. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero non-coplanar vectors. Let the position vectors of four points A, B, C and D be $\vec{a} - \vec{b} + \vec{c}$, $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$, $-\vec{a} + 2\vec{b} - 3\vec{c}$ and $2\vec{a} - 4\vec{b} + 6\vec{c}$ respectively. If $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ are coplanar, then λ is equal to _____. **[JEE(Main)Jan.2023]**

माना $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ तीन शून्येत्तर असहतलीय सदिश है। माना चार बिंदुओं A, B, C व D के स्थिति सदिश क्रमशः $\vec{a} - \vec{b} + \vec{c}$, $\lambda\vec{a} - 3\vec{b} + 4\vec{c}$, $-\vec{a} + 2\vec{b} - 3\vec{c}$ व $2\vec{a} - 4\vec{b} + 6\vec{c}$ है, यदि $\overrightarrow{AB}$, $\overrightarrow{AC}$ तथा $\overrightarrow{AD}$ समतलीय है। तब λ बराबर _____ है।

[JEE(Main)Jan.2023]

Ans. 2

Sol. $\overline{AB} = (\lambda - 1)\bar{a} - 2\bar{b} + 3\bar{c}$

$\overline{AC} = 2\bar{a} + 3\bar{b} - 4\bar{c}$

$\overline{AD} = \bar{a} - 3\bar{b} + 5\bar{c}$

$$\begin{vmatrix} \lambda-1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$\Rightarrow (\lambda -1)(15-12) + 2(-10+4) + 3(6-3) = 0$

$\Rightarrow (\lambda - 1) = 1 \Rightarrow \lambda = 2$

22. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ and $|\vec{a} \times \vec{b}| = \sqrt{48}$. Then $\left(\vec{a} \cdot \vec{b}\right)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार है कि $|\vec{a}| = \sqrt{14}$, $|\vec{b}| = \sqrt{6}$ तथा $|\vec{a} \times \vec{b}| = \sqrt{48}$, तो $\left(\vec{a} \cdot \vec{b}\right)^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 36

Sol. $\left(\vec{a}.\vec{b}\right)^2 + \left(\vec{a} \times \vec{b}\right) = |a|^2 |b|^2$

$\left(\vec{a}.\vec{b}\right)^2 + 48 = 14 \times 6$

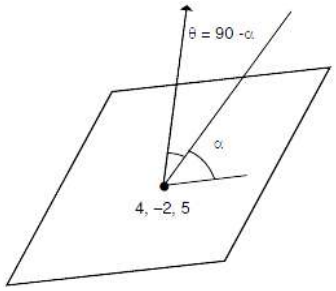
$\left(\vec{a}.\vec{b}\right)^2 = 84 - 48 = 36$

23. Let θ be the angle between the planes $P_1 : \vec{r} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 9$ and $P_2 : \vec{r} \cdot \left(2\hat{i} - \hat{j} + \hat{k}\right) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2. If α is the angle between L and P_2, then $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ is equal to _____. **[JEE(Main)Jan.2023]**

माना समतलों $P_1 : \vec{r} \cdot \left(\hat{i} + \hat{j} + 2\hat{k}\right) = 9$ तथा $P_2 : \vec{r} \cdot \left(2\hat{i} - \hat{j} + \hat{k}\right) = 15$ के बीच का कोण θ है। माना रेखा L समतल P_2 को बिंदु $(4, -2, 5)$ पर मिलती है तथा P_2 के अभिलंब से कोण θ बनाती है। यदि L तथा P_2 के बीच कोण α है तो $\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol.



$$\cos\theta = \frac{\left(\hat{i}+\hat{j}+2\hat{k}\right)\cdot\left(2\hat{i}-\hat{j}+\hat{k}\right)}{6} = \frac{2-1+2}{6} = \frac{1}{2}$$

$\theta = \frac{\pi}{3}$ $\alpha = \frac{\pi}{6}$

$\therefore (\tan^2\theta)(\cot^2\alpha) = (3)(3) = 9$

24. Let $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{u}$ be a vector such that $|\vec{u}| = \alpha > 0$. If the minimum value of the scalar triple product $[\vec{u}\ \vec{v}\ \vec{w}]$ is $-\alpha\sqrt{3401}$, and $\left|\vec{u}\cdot\hat{i}\right|^2 = \frac{m}{n}$ where m and n are coprime natural numbers, then m + n is equal to _____. **[JEE(Main)Jan.2023]**

माना $\vec{v} = \alpha\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{w} = 2\alpha\hat{i} + \hat{j} - \hat{k}$ है तथा एक सदिश $\vec{u}$ के लिए $|\vec{u}| = \alpha > 0$ है। यदि अदिश त्रिक गुणनफल $[\vec{u}\ \vec{v}\ \vec{w}]$ का न्यूनतम मान $-\alpha\sqrt{3401}$ है, और $\left|\vec{u}\cdot\hat{i}\right|^2 = \frac{m}{n}$ है जहाँ m एवं n सहअभाज्य धन पूर्णांक हैं, तब m + n का मान है।

[JEE(Main)Jan.2023]

Ans. 3501

Sol. $[\vec{u}\vec{v}\vec{w}] = \vec{u}.(\vec{v}\times\vec{w})$

$\left(|u||\vec{v}\times\vec{w}|\cos\theta\right) = -\alpha\sqrt{3401}$

$\Rightarrow \cos\theta = -1$

$|u| = \alpha$ (Given)

$|\vec{v}\times\vec{w}| = \sqrt{3401}$

$$\vec{v}\times\vec{w} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 2 & -3 \\ 2\alpha & 1 & -1 \end{vmatrix}$$

$\vec{v}\times\vec{w} = \hat{i} - 5\alpha j - 3\alpha k$

$|\vec{v}\times\vec{w}| = \sqrt{1+25\alpha^2+9\alpha^2} = \sqrt{3401}$

$34\alpha^2 = 3400$

$\alpha^2 = 100$

$\alpha = 10$ (as $\alpha > 0$)

So $\vec{u} = \lambda\left(\hat{i} - 5\alpha j - 3\alpha k\right)$

$\vec{u} = \sqrt{\lambda^2 + 25\alpha^2\lambda^2 + 9\alpha^2\lambda}$

$\alpha^2 = \lambda^2\left(1+25\alpha^2+9\alpha^2\right)$

$100 = \lambda^2(1 + 34 \times 100)$

$$\lambda^2 = \frac{100}{3401} = \frac{m}{n}$$

25. The distance of the point(7, –3, –4) from the plane passing through the points (2, –3, 1), (–1, 1, –2) and (3, –4, 2) is : **[JEE(Main)Jan.2023]**

बिंदु (7, –3, –4) की बिंदुओं (2, –3, 1), (–1, 1, –2) और (3, –4, 2) से होकर जाने वाले समतल से दूरी है : **[JEE(Main)Jan.2023]**

(A) 5 (B) $5\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$

Ans. B

Sol. Equation of Plane is

$$= \begin{vmatrix} x-2 & y+3 & z-1 \\ -3 & 4 & -3 \\ 4 & -5 & 4 \end{vmatrix}$$

$x - z - 1 = 0$

Distance of P(7, –3, –4) from Plane is

$$d = \left|\frac{7+4-1}{\sqrt{2}}\right| = 5\sqrt{2}$$

26. The distance of the point (–1, 9, –16) from the plane $2x + 3y - z = 5$ measured parallel to the line $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is : **[JEE(Main)Jan.2023]**

बिंदु (–1, 9, –16) की समतल $2x + 3y - z = 5$ से रेखा $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ के समांतर मापी दूरी है : **[JEE(Main)Jan.2023]**

(A) $20\sqrt{2}$ (B) $13\sqrt{2}$ (C) 31 (D) 26

Ans. D

Sol. Equation of line

$$\frac{x+1}{3} = \frac{y-9}{-4} = \frac{z+16}{12}$$

G.P on line $(3\lambda - 1, -4\lambda + 9, 12\lambda - 16)$

point of intersection of line and plane

$6\lambda - 2 - 12\lambda + 27 - 12\lambda + 16 = 5$

$\lambda = 2$

Point (5, 1, 8)

Distance = $\sqrt{36 + 64 + 576} = 26$

27. If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel

to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to :

[JEE(Main)Jan.2023]

यदि बिंदु (3, 2, 1) से होकर जाने वाली तथा समतलों $x + 2y + z = 0$ और $3y - z = 3$ के समान्तर रेखा पर बिंदु (1, 9, 7) से डाले x, y Ec d ki ln (α, β, γ) है, तब $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 1 (B) 3 (C) –1 (D) 5

Ans. D

Sol. D.R's of line be

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$i(-2-3) - j(-1) + \hat{k}(3)$$

$$= -5i + j + 3\hat{k}$$

Equation of line

$$\frac{x-3}{-5} = \frac{y-2}{1} = \frac{z-1}{3} = \lambda$$

$N(-5\lambda + 3, \lambda + 2, 3\lambda + 1)$

D.R's of AN, : $-5\lambda + 2, \lambda - 7, 3\lambda - 6$

Now, $(-5\lambda + 2)(-5) + (\lambda - 7)(1) + (3\lambda - 6)3 = 0$

$25\lambda - 10 + \lambda - 7 + 9\lambda - 18 = 0$

$35\lambda = 35 \Rightarrow \lambda = 1$

N (–2, 3, 4)

` $\alpha + \beta + \gamma = -2 + 3 + 4 = 5$

28. Let the plane containing the line of intersection of the planes $P_1 : x + (\lambda + 4)y + z = 1$ and $P_2 : 2x + y + z = 2$ pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P2 is : **[JEE(Main)Jan.2023]**

माना समतल, जिसमें समतलों $P_1 : x + (\lambda + 4)y + z = 1$ तथा $P_2 : 2x + y + z = 2$ की प्रतिच्छेदन रेखा स्थित है, बिंदुओं (0, 1, 0) तथा (1, 0, 1) से होकर जाता है। तो बिंदु $(2\lambda, \lambda, -\lambda)$ की समतल P2 से दूरी है : **[JEE(Main)Jan.2023]**

(A) $5\sqrt{6}$ (B) $2\sqrt{6}$ (C) $4\sqrt{6}$ (D) $3\sqrt{6}$

Ans. D

Sol. Eq. of plane containing the line of intersection of planes is given by

$p_1 + k\, p_2 = 0$

$X + (\lambda + 4)\, y + z - 1 + k\,(2x + y + z - 2) = 0$

It passes through (0, 1, 0) & (1, 0, 1)

$\lambda + 3 + k\,(1 - 2) = 0 \Rightarrow \lambda - k = -3$ .....(1)

And $(1 + 1 - 1) + k\,(2 + 1 - 2) = 0 \Rightarrow 1 + k = 0$

$\Rightarrow k = -1, \lambda = -4$

Now given point (–8, –4, 4)

Its distance from plane $p_2 = \left|\frac{-16-4+4-2}{\sqrt{6}}\right| = 3\sqrt{6}$

29. The distance of the point P(4, 6, –2) from the line passing through the point (–3, 2, 3) and parallel to a line with direction ratios 3, 3, –1 is equal to : **[JEE(Main)Jan.2023]**

बिंदु (–3, 2, 3) से होकर जाने वाली तथा दिक् अनुपात 3, 3, –1 की एक रेखा के समांतर रेखा से बिंदु P(4, 6, –2) की दूरी है : **[JEE(Main)Jan.2023]**

(A) $2\sqrt{3}$ (B) $\sqrt{14}$ (C) $\sqrt{6}$ (D) 3

Ans. B

Sol. Equation of line passing through (–3, 2, 3) with direction ratios 3, 3, –1 is

$\frac{x+3}{3} = \frac{y-2}{3} = \frac{z-3}{-1}$ _____ (1)

Consider Q (3r – 3, 3r + 2, –r + 3) on the line (1) such that PQ is $\perp$ to line (1)

D.R. of PQ = 3r – 7, 3r – 4, – r + 5

PQ $\perp$ line (1)

So (3r – 7) (3) + (3r – 4) (3) + (– r + 5) (–1) = 0

$\Rightarrow r = 2$

So Q (3, 8, 1)

$\Rightarrow \quad PQ = \sqrt{1+4+4} = \sqrt{14}$

30. Consider the lines L_1 and L_2 given by

$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$

$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

A line L_3 having direction ratios 1, –1, –2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is : **[JEE(Main)Jan.2023]**

निम्न रेखाओं L_1 तथा L_2 का विचार कीजिए।

$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$

$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$

दिक् अनुपात 1, –1, –2 की एक रेखा L_3 रेखाओं L_1 तथा L_2 को क्रमशः बिंदुओं P तथा Q पर काटती है। तो रेखाखंड PQ की लंबाई gS **[JEE(Main)Jan.2023]**

(A) $2\sqrt{6}$ (B) $4\sqrt{3}$ (C) $3\sqrt{2}$ (D) 4

Ans. A

Sol. Let P on L_1 be P $(2r, +1, r_1 + 3, 2r_1 + 2)$

and Q on L_2 be Q $(r_2 + 2, 2r_2 + 2, 3r_2 + 3)$

$\Rightarrow$ D.R. of PQ = $2r_1 - r_2 - 1, r_1 - 2r_2 + 1, 2r_1 - 3r_2 - 1$

which are equivalent to $-1, 1, 2$

i.e. $\dfrac{2r_1 - r_2 - 1}{-1} = \dfrac{r_1 - 2r_2 + 1}{1} = \dfrac{2r_1 - 3r_2 - 1}{2}$

$\Rightarrow r_1 = 3 = r_2$

$\Rightarrow$ P(7, 6, 8) and Q (5, 8, 12) $\Rightarrow$ length PQ = $2\sqrt{6}$

31. The foot of perpendicular of the point (2,0,5) on the line $\dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1}$ is (α, β, γ). then :

[JEE(Main)Jan.2023]

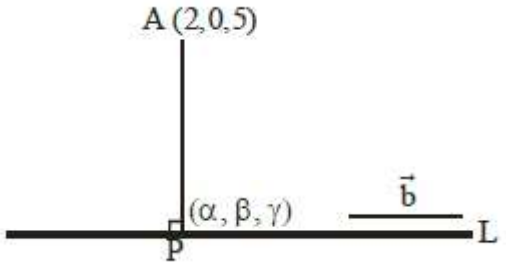
बिंदु (2,0,5) से रेखा $\dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1}$ पर लंब का पाद (α, β, γ) है। तो निम्न में से कौन सा सही नहीं है :

[JEE(Main)Jan.2023]

(A) $\dfrac{\beta}{\gamma} = -5$ (B) $\dfrac{\gamma}{\alpha} = \dfrac{5}{8}$ (C) $\dfrac{\alpha}{\beta} = -8$ (D) $\dfrac{\alpha\beta}{\gamma} = \dfrac{4}{15}$

Ans. A

Sol. $L : \dfrac{x+1}{2} = \dfrac{y-1}{5} = \dfrac{z+1}{-1} = \lambda$ (let)



Let food of perpendicular is

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1)$

$\overrightarrow{PA} = (3 - 2\lambda)\hat{i} - (5\lambda + 1)\hat{j} + (6 + \lambda)\hat{k}$

Direction ratio of line $\Rightarrow \vec{b} = 2\hat{i} + 5\hat{j} - \hat{k}$

Now, $\Rightarrow \overrightarrow{PA} \cdot \vec{b} = 0$

$\Rightarrow 2(3 - 2\lambda) - 5(5\lambda + 1) - (6 + \lambda) = 0$

$\Rightarrow \lambda = \dfrac{-1}{6}$

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1) \equiv P(\alpha, \beta, \gamma)$

$\Rightarrow \alpha = 2\left(-\frac{1}{6}\right)-1 = -\frac{4}{3} \Rightarrow \alpha = -\frac{4}{3}$

$\Rightarrow \beta = 5\left(-\frac{1}{6}\right)+1 = \frac{1}{6} \Rightarrow \beta = \frac{1}{6}$

$\Rightarrow \gamma = -\lambda - 1 = \frac{1}{6} - 1 \Rightarrow \gamma = -\frac{5}{6}$

$\therefore$ Check options

32. The shortest distance between the lines $x + 1 = 2y = -12z$ and $x = y + 2 = 6z - 6$ is :

[JEE(Main)Jan.2023]

रेखाओं $x + 1 = 2y = -12z$ तथा $x = y + 2 = 6z - 6$ के बीच न्यूनतम दूरी है **[JEE(Main)Jan.2023]**

(A) $\frac{3}{2}$ (B) 3 (C) 2 (D) $\frac{5}{2}$

Ans. C

Sol. $\frac{x+1}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{\frac{-1}{12}}$ and $\frac{x}{1} = \frac{y+2}{1} = \frac{z-1}{\frac{1}{6}}$

$\Rightarrow$ Shortest distance $= \frac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$

$$\text{S.D.} = (-\hat{i}+2\hat{j}-\hat{k})\cdot\frac{(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\left\{\vec{p}\times\vec{q} \equiv \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \frac{1}{2} & \frac{-1}{12} \\ 1 & 1 & \frac{1}{6} \end{vmatrix} = \frac{1}{6}\hat{i} - \frac{1}{4}\hat{j} + \frac{1}{2}\hat{k} \text{ or } 2\hat{i} - 3\hat{j} + 6\hat{k}\right\}$$

$$\text{S.D.} = \frac{(-\hat{i}+2\hat{j}-\hat{k})\cdot(2\hat{i}-3\hat{j}+6\hat{k})}{\sqrt{2^2+3^2+6^2}} = \left|\frac{-14}{7}\right| = 2$$

33. If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z = a$ is : **[JEE(Main)Jan.2023]**

यदि रेखाऐं $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ तथा $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ बिंदु P पर मिलती हैं, तो बिंदु P की समतल $z = a$ से दूरी है : **[JEE(Main)Jan.2023]**

(A) 28 (B) 16 (C) 10 (D) 22

Ans. A

Sol. Point on $L_1 \equiv (\lambda + 1, 2\lambda + 2, \lambda - 3)$

Point on $L_2 \equiv (2\mu + a, 3\mu - 2, \mu + 3)$

$\lambda - 3 = \mu + 3 \quad \Rightarrow \lambda = \mu + 6 \quad ...(1)$

$2\lambda + 2 = 3\mu - 2 \quad \Rightarrow 2\lambda = 3\mu - 4 \quad ...(2)$

Solving, (1) and (2)

$\Rightarrow \lambda = 22$ & $\mu = 16$

$\Rightarrow P \equiv (23, 46, 19)$

$\Rightarrow a = -9$

Distance of P from $z = -9$ is 28

34. The plane $2x - y + z = 4$ intersects the line segment joining the points A $(a, -2, 4)$ and B $(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to : **[JEE(Main)Jan.2023]**

समतल $2x - y + z = 4$ बिंदुओं A $(a, -2, 4)$ तथा B $(2, b, -3)$ को मिलाने वाले रेखाखंड को 2 : 1 के अनुपात में बिंदु C पर d kVr kgS Hkkx ewy fcanq ls fcanq C की दूरी $\sqrt{5}$ है। यदि $ab < 0$ है तथा P, बिंदु $(a - b, b, 2b - a)$ है, तो CP^2 बराबर है :

[JEE(Main)Jan.2023]

(A) $\frac{16}{3}$ (B) $\frac{97}{3}$ (C) $\frac{17}{3}$ (D) $\frac{73}{3}$

Ans. C

Sol. A(a, –2, 4), B(2, b, –3)

AC : CB = 2 : 1

$$\Rightarrow C \equiv \left(\frac{a+4}{3}, \frac{2b-2}{3}, \frac{-2}{3}\right)$$

C lies on $2x - y + 2 = 4$

$$\Rightarrow \frac{2a+8}{3} - \frac{2b-2}{3} - \frac{2}{3} = 4$$

$\Rightarrow a - b = 2 \quad ....(1)$

Also $OC = \sqrt{5}$

$$\Rightarrow \left(\frac{a+4}{3}\right)^2 + \left(\frac{2b-2}{3}\right)^2 + \frac{4}{9} = 5 \quad ....(2)$$

Solving, (A) and (2)

$(b + 6)^2 + (2b - 2)^2 = 41$

$\Rightarrow 5b^2 + 4b - 1 = 0$

$\Rightarrow b = -1$ or $\frac{1}{5}$

$\Rightarrow$ a = 1 or $\frac{11}{5}$

But ab < 0 $\Rightarrow$ (a, b) = (1, –1)

$C \equiv \left(\frac{5}{3}, \frac{-4}{3}, \frac{-2}{3}\right)$, $P \equiv (2,-1,-3)$

$CP^2 = \frac{1}{9} + \frac{1}{9} + \frac{49}{9} = \frac{51}{9} = \frac{17}{3}$

35. The shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ is :

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ तथा $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2023]**

(A) $4\sqrt{3}$ (B) $3\sqrt{3}$ (C) $5\sqrt{3}$ (D) $2\sqrt{3}$

Ans. A

Sol. $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$ $\qquad \vec{a} = \hat{i} - 8\hat{j} + 4\hat{k}$

$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$ $\qquad \vec{b} = \hat{i} + 2\hat{j} + 6\hat{k}$

$\vec{p} = 2\hat{i} - 7\hat{i} + 5\vec{\hat{k}}, \vec{q} = 2\hat{i} + \hat{j} - 3\hat{k}$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix}$$

$= \hat{i}(16) - \hat{j}(-16) + \hat{k}(16)$

$= 16(\hat{i} + \hat{j} + \hat{k})$

$$d = \left|\frac{(a-b)\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right| = \left|\frac{\left(-10\hat{j} - 2\hat{k}\right)\cdot 16\left(\hat{i} + \hat{j} + \hat{k}\right)}{16\sqrt{3}}\right|$$

$$= \left|\frac{-12}{\sqrt{3}}\right| = 4\sqrt{3}$$

36. Let a unit vector $\widehat{OP}$ make angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{OP}$ is perpendicular to the plane through points (1,2,3), (2,3,4) and (1,5,7), then which one of the following is true? **[JEE(Main)Jan.2023]**

माना एक इकाई सदिश $\widehat{OP}$ निर्देशांक अक्षों OX, OY, OZ की धनात्मक दिशाओं से क्रमशः कोण α,β,γ बनाते हैं, जहाँ $\beta \in \left(0,\frac{\pi}{2}\right)$ है। यदि $\widehat{OP}$ बिंदुओं (1,2,3), (2,3,4) तथा (1,5,7) से होकर जाने वाले समतल के लंबवत है, तो निम्न में से कौनसा एक सत्य है ?

[JEE(Main)Jan.2023]

(A) $\alpha \in \left(\frac{\pi}{2},\pi\right)$ and $\gamma \in \left(0,\frac{\pi}{2}\right)$ (B) $\alpha \in \left(0,\frac{\pi}{2}\right)$ and $\gamma \in \left(0,\frac{\pi}{2}\right)$

(C) $\alpha \in \left(\frac{\pi}{2},\pi\right)$ and $\gamma \in \left(\frac{\pi}{2},\pi\right)$ (D) $\alpha \in \left(0,\frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2},\pi\right)$

Ans. C

Sol. Equation of plane :-

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 1 & 1 \\ 0 & 3 & 4 \end{vmatrix} = 0$$

$\Rightarrow [x-1] - 4[y-2] + 3[z-3] = 0$

$\Rightarrow x - 4y + 3z = 2$

D.R's of normal of plane <1, –4, 3>

D.C's of $\left\langle \pm\frac{1}{\sqrt{26}}, \mp\frac{4}{\sqrt{26}}, \pm\frac{3}{\sqrt{26}} \right\rangle$

$\cos\beta = \frac{4}{\sqrt{26}}$

$\cos\alpha = \frac{-1}{\sqrt{26}}$ $\quad \frac{\pi}{2} < \alpha < \pi$

$\cos\gamma = \frac{-3}{\sqrt{26}}$ $\quad \frac{\pi}{2} < \gamma < \pi$

37. The line l_1 passes through the point (2,6,2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is : **[JEE(Main)Jan.2023]**

रेखा l_1 बिंदु (2,6,2) से होकर जाती है ताथा समतल $2x + y - 2z = 10$ पर लंबवत है। तो l_1 तथा रेखा $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ के बीच

U;wre nyjhgS **[JEE(Main)Jan.2023]**

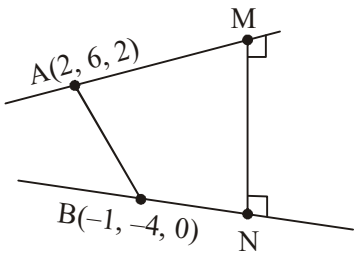
(A) 7 (B) $\frac{13}{3}$ (C) 9 (D) $\frac{19}{3}$

Ans. C

Sol. Line *l*, is given by $L_1 : \frac{x-2}{2} = \frac{y-6}{1} = \frac{z-2}{-2}$

Given,

$L_3 : \frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{-2}$



Shortest distance = $\left|\frac{\overrightarrow{AB}.\overrightarrow{MN}}{MN}\right|$

$\overrightarrow{AB} = 3\hat{i} + 10\hat{j} + 2\hat{k}$

$\overrightarrow{MN} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 2 & -3 & 2 \end{vmatrix} = -4\hat{i} - 8\hat{j} - 8\hat{k}$

$\therefore\ MN = \sqrt{16+64+64} = 12$

$\therefore$ Shortest distnace $= \left|\frac{-12-80-16}{12}\right| = 9$

$\therefore$ Option (3) is correct

38. If a plane passes through the points $(-1, k, 0)$, $(2, k, -1)$, $(1,1,2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ **[JEE(Main)Jan.2023]**

यदि समतल बिंदुओं $(-1, k, 0)$, $(2, k, -1)$, $(1,1,2)$ से होकर जाता है एवं रेखा $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$ के समान्तर है, तब $\frac{k^2+1}{(k-1)(k-2)}$ का मान है : **[JEE(Main)Jan.2023]**

(A) $\frac{13}{6}$ (B) $\frac{5}{17}$ (C) $\frac{17}{5}$ (D) $\frac{6}{13}$

Ans. A

Sol. Line is parallel to the

vector $\hat{i} + \hat{j} - \hat{k}$

Let A (–1, k, 0), B (2, k, –1), C (1, 1, 2)

So $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 0 & -1 \\ 2 & 1-k & 2 \end{vmatrix}$

$\Rightarrow \vec{n} = (1-k)\hat{i} - 8\hat{j} + (3-3)\hat{k}$

$\therefore \hat{n}.\left(\hat{i}+\hat{j}-\hat{k}\right)=0$

$\Rightarrow 1-k-8-3+3k=0$

$\Rightarrow k=5$

hence $\dfrac{k^2+1}{(k-1)(k-2)}=\dfrac{26}{(4)(3)}=\dfrac{13}{6}$

39. Let the shortest distance between the lines $L:\dfrac{x-5}{-2}=\dfrac{y-\lambda}{0}=\dfrac{z+\lambda}{1},\lambda\geq 0$ and $L_1: x+1=y-1=4-z$ be $2\sqrt{6}$. If (α,β,γ) lies on L, then which of the following is **NOT** possible? **[JEE(Main)Jan.2023]**

माना रेखाओं $L:\dfrac{x-5}{-2}=\dfrac{y-\lambda}{0}=\dfrac{z+\lambda}{1},\lambda\geq 0$ तथा $L_1: x+1=y-1=4-z$ के बीच न्यूनतम दूरी $2\sqrt{6}$ है। यदि (α,β,γ) रेखा L पर है, तो निम्न में से कौन सा संभव मान नहीं है ? **[JEE(Main)Jan.2023]**

(A) $2\alpha-\gamma=9$ (B) $\alpha+2\gamma=24$ (C) $2\alpha+\gamma=7$ (D) $\alpha-2\gamma=19$

Ans. B

Sol. $\overrightarrow{b_1}\times\overrightarrow{b_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&0&1\\1&1&-1\end{vmatrix}=-\hat{i}-\hat{j}-2\hat{k}$

$\overrightarrow{a_2}-\overrightarrow{a1}=6\hat{i}+(\lambda-1)\hat{j}+(-\lambda-4)\hat{k}$

$2\sqrt{6}=\left|\dfrac{-6-\lambda+1+2\lambda+8}{\sqrt{1+1+4}}\right|$

$|\lambda+3|=12\Rightarrow\lambda=9,-15$

$\alpha=-2k+5,\ \gamma=k-\lambda$ where $k\in R$

$\Rightarrow \alpha+2\gamma=5-2\lambda=-13,35$

40. If a point $P(\alpha,\beta,\gamma)$ satisfying $\begin{pmatrix}\alpha&\beta&\gamma\end{pmatrix}\begin{pmatrix}2&10&8\\9&3&8\\8&4&8\end{pmatrix}=\begin{pmatrix}0&0&0\end{pmatrix}$ lies on the plane $2x+4y+3z=5$, then $6\alpha+9\beta+7\gamma$ is equal to : **[JEE(Main)Jan.2023]**

यदि $\begin{pmatrix}\alpha&\beta&\gamma\end{pmatrix}\begin{pmatrix}2&10&8\\9&3&8\\8&4&8\end{pmatrix}=\begin{pmatrix}0&0&0\end{pmatrix}$ को संतुष्ट करने वाला एक बिंदु $P(\alpha,\beta,\gamma)$ समतल $2x+4y+3z=5$ पर है, तो $6\alpha+9\beta+7\gamma$ बराबर है : **[JEE(Main)Jan.2023]**

(A) -1 (B) 11 (C) $\frac{11}{5}$ (D) $\frac{5}{4}$

Ans. B

Sol. $(\alpha\ \beta\ \gamma)\begin{pmatrix}2 & 10 & 8\\ 9 & 3 & 8\\ 8 & 4 & 8\end{pmatrix}=(0\ 0\ 0)$

$2\alpha + 9\beta + 8\gamma = 0$ _____(1)
$10\alpha + 3\beta + 4\gamma = 0$ _____(2)
$8\alpha + 8\beta + 8\gamma = 0$ _____(3)
from (1) and (3) by cross multiplication

$\frac{\alpha}{1}=\frac{\beta}{6}=\frac{\gamma}{-7}=k \Rightarrow\ \alpha = k,\ \beta = 6k,\ \gamma = -7k$

Point P (α, β, γ) lie on the plane $2x + 4y + 3z = 5 \Rightarrow 2\alpha + 4\beta + 3\gamma = 5$
$\Rightarrow 2k + 24k - 21k = 5$
$\Rightarrow k = 1$

Hence $6\alpha + 9\beta + 7\gamma = 6k + 54k - 49k = 11k = 11$

41. The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points A, B, C is (2, a, 4), $a \in N$. If the volume of the tetrahedron OABC is 144 , then which of the following points is **NOT** on P? **[JEE(Main)Jan.2023]**

मूल बिंदु से एक समतल P, जो निर्देशांक अक्षों को बिंदुओं A,B,C पर मिलता है, पर डाले गये लंब का पाद (2, a, 4), $a \in N$ है। यदि चतुष्फलक OABC का आयतन 144 है, तो निम्न बिंदुओं में से कौन सा P पर नहीं है ? **[JEE(Main)Jan.2023]**

(A) (0,4,4) (B) (2,2,4) (C) (0,6,3) (D) (3,0,4)

Ans. D

Sol. Equation of Plane :

$(2\hat{i} + a\hat{j} + 4\hat{k}) \cdot [(x-2)\hat{i} + (y-a)\hat{j} + (z-4)\hat{k}] = 0$

$\Rightarrow 2x + ay + 4z = 20 + a^2$

$\Rightarrow A \equiv \left(\frac{20+a^2}{2}, 0, 0\right)$

$B \equiv \left(0, \frac{20+a^2}{a}, 0\right)$

$C \equiv \left(0, 0, \frac{20+a^2}{4}\right)$

$\Rightarrow$ Volume of tetrahedron

$$= \frac{1}{6}[\vec{a} \cdot \vec{b}\vec{c}]$$

$$= \frac{1}{6}\vec{a}(\vec{b} \times \vec{c})$$

$$\Rightarrow \frac{1}{6}\left(\frac{20+a^2}{2}\right) \cdot \left(\frac{20+a^2}{a}\right) \cdot \left(\frac{20+a^2}{4}\right) = 144$$

$$\Rightarrow \left(20+a^2\right)^3 = 144 \times 48 \times a$$

$\Rightarrow a = 2$

$\Rightarrow$ Equation of plane is $2x + 2y + 4z = 24$

Or $x + y + 2z = 12$

$\Rightarrow (3, 0, 4)$ Not lies on the Plane $x + y + 2z = 12$

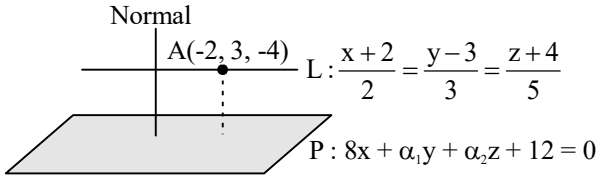
42. Let the plane $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to the line $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P on the y-axis is 1, then the distance between P and L is : **[JEE(Main)Jan.2023]**

माना समतल $P : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ रेखा $L : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$ के समांतर है। यदि P का y-अक्ष पर अंतःखंड 1 है, तो P तथा L के बीच दूरी है : **[JEE(Main)Jan.2023]**

(A) $\frac{6}{\sqrt{14}}$ (B) $\sqrt{\frac{2}{7}}$ (C) $\sqrt{\frac{7}{2}}$ (D) $\sqrt{14}$

Ans. D

Sol.



Normal $\perp$ L $\Rightarrow 8 \times 2 + 3\alpha_1 + 5\alpha_2 = 0$

For Y intercept of plane put $x = z = 0$ in equation of plane

$$\Rightarrow y = \frac{-12}{\alpha_1} = 1 \Rightarrow \alpha_1 = -12$$

Put α_1 in equation (1) we get $\alpha_2 = 4$

Hence equation of plane $P : 2x - 3y + z + 3 = 0$

Now distance of point A (–2, 3, –4) from the plane P is = $\left|\frac{2(-2)-3(3)-4+3}{\sqrt{(2)^2+(-3)^2+(1)^2}}\right| = \sqrt{14}$

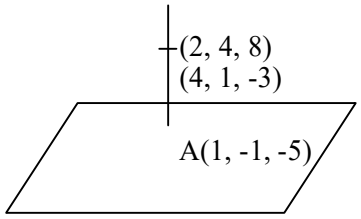
43. Let P be the plane, passing through the point $(1, -1, -5)$ and perpedicular to the line joining the points (4, $1, -3$) and (2, 4, 3). Then the distance of P from the point $(3, -2, 2)$ is : **[JEE(Main)Jan.2023]**

माना बिंदु $(1,-1,-5)$ से होकर जाने वाले तथा बिंदुओं $(4,1,-3)$ तथा $(2, 4, 3)$ को मिलाने वाली रेखा के लंबवत समतल P है। तो P की बिंदु $(3,-2, 2)$ से दूरी है। **[JEE(Main)Jan.2023]**

(A) 7 (B) 6 (C) 5 (D) 4

Ans. C

Sol.



D'r of Plane Normal : 2, –3, –6
Equation of Plane : $2(x-1)-3(j+1)-6(z+5)=0$
$2x-3y-6z-35=0$
Hence $\perp$ distance of plane from (3, –2, 2)

$$\perp \text{ distance} = \left|\frac{6+6-12-35}{\sqrt{2^2+(-3)^2+(-6)^2}}\right| = 5$$

44. Let the image of the point P(2, –1, 3) in the plane $x + 2y - z = 0$ be Q. Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is : **[JEE(Main)Jan.2023]**

माना समतल $x + 2y - z = 0$ में बिंदु P(2, –1, 3) का प्रतिबिंब Q है। तब बिंदु Q से समतल $3x + 2y + z + 29 = 0$ की दूरी है : **[JEE(Main)Jan.2023]**

(A) $2\sqrt{14}$ (B) $3\sqrt{14}$ (C) $\frac{24\sqrt{2}}{7}$ (D) $\frac{22\sqrt{2}}{7}$

Ans. B

Sol. Image of point P(2, –1, 3) in palne $x + 2y - z = 0$ be

$$\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-1}=\frac{-2(2-2-3)}{(1)^2+(2)^2+(-1)^2}$$

$$\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-1}=\frac{6}{6}=1$$

so, $x - 2 = 1$, $y + 1 = 2$, $z - 3 = -1$
$\therefore x = 3, y = 1, z = 2$
So : Q (3, 1, 2)

$$\therefore d=\left|\frac{3(3)+2(1)+2+29}{\sqrt{3^2+2^2+1^2}}\right|$$

$$d=\left|\frac{9+4+29}{\sqrt{14}}\right|$$

$$d = \frac{42}{\sqrt{14}} = 3\sqrt{14}$$

45. The shortest distance between the lines $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ and $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ is :

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ एवं $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2023]**

(A) $5\sqrt{3}$ (B) $7\sqrt{3}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

Ans. D

Sol. Shortest distance between two lines

$\frac{x-x_1}{a_1} = \frac{y-y_1}{a_2} = \frac{z-z_1}{a_3}$ &

$\frac{x-x_2}{b_1} = \frac{y-y_2}{b_2} = \frac{z-z_2}{b_3}$ is given as

$$\frac{\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}}{\sqrt{(a_1b_3-a_3b_2)^2+(a_1b_3-a_3b_1)^2+(a_1b_2-a_2b_1)^2}}$$

$$\frac{\begin{vmatrix} 5-(3) & 2-(-5) & 4-1 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(-10+12)^2+(-5+3)^2+(4-2)^2}}$$

$$\frac{\begin{vmatrix} 8 & 7 & 3 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(2)^2+(2)^2+(2)^2}}$$

$$= \frac{|8(-10+12)-7(-5+3)+3(4-2)|}{\sqrt{4+4+4}}$$

$$= \frac{16+14+6}{\sqrt{12}} = \frac{36}{\sqrt{12}} = \frac{36}{2\sqrt{3}}$$

$$= \frac{18}{\sqrt{3}} = 6\sqrt{3}$$

46. Let the plane P pass through the intersection of the planes $2x + 3y - z$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point (–7,1,1), then d^2 is equal to :

[JEE(Main)Jan.2023]

माना समतल P समतलों $2x + 3y - z = 2$ तथा $x + 2y + 3z = 6$ के प्रतिच्छेदन से होकर जाता है तथा समतल $2x + y - z + 1 = 0$ के y a or g S; fn P की बिंदु (–7,1,1) से दूरी d है, तो d^2 बराबर है : **[JEE(Main)Jan.2023]**

(A) $\frac{25}{83}$ (B) $\frac{250}{82}$ (C) $\frac{250}{83}$ (D) $\frac{15}{53}$

Ans. C

Sol. A plane through intersection of $2x + 3y - z = 2$ & $x + 2y + 3z = 6$ is

$2x + 3y - z - 2 + \lambda (x + 2y + 3z - 6) = 0$

$\Rightarrow (2 + \lambda)x + (3 + 2\lambda)y + (3\lambda - 1) z - (2 + 6\lambda) = 0$

Plane is $\perp$ to $2x + y - z + 1 = 0$

$\therefore 2(2 + \lambda) + 1 (3 + 2\lambda) - (3\lambda - 1) = 0$

$4 + 2\lambda + 3 + 2\lambda - 3\lambda + 1 = 0$

$\lambda = -8$

$\therefore$ plane is $6x + 13y + 25z - 46 = 0$

$$\therefore d = \frac{|-42 + 13 + 25 - 46|}{\sqrt{36 + 169 + 625}} = \frac{50}{\sqrt{830}}$$

$$\therefore d^2 = \frac{50 \times 50}{830} = \frac{250}{83}$$

47. The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____.

[JEE(Main)Jan.2023]

रेखाओं $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ तथा $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ के बीच न्यूनतम दूरी है _____

[JEE(Main)Jan.2023]

Ans. 14

Sol. Shortest distance between the lines

$$= \frac{\begin{vmatrix} 4 & 2 & -14 \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & k \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}$$

$$= \frac{16+12+168}{|-4\hat{i}+6\hat{j}-12k|} = \frac{196}{14} = 14$$

48. Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point(a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____. **[JEE(Main)Jan.2023]**

माना रेखा $x - 2y - z - 5 = 0 = x + y + 3z - 5$ से होकर जाने वाले तथा रेखा $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ के समांतर समतल का समीकरण $ax + by + cz = 65$ है। तो बिंदु (a, b, c) की समतल $2x + 2y - z + 16 = 0$ से दूरी है। **[JEE(Main)Jan.2023]**

Ans. 9

Sol. since plane passing through the line

$x - 2y - z - 5 = 0 = x + y + 3z - 5$

So equation of plane is

$(x - 2y - z - 5) + \lambda (x + y + 3z - 5) = 0$

$(1 + \lambda) x + (\lambda - 2) y + (3\lambda - 1) z - 5 - 5\lambda = 0$ ___ (1)

And this plane parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ with direction ratio (–5, 3, 1)

Now

$-5 (1 + \lambda) + (\lambda - 2). 3 + (3\lambda - 1). 1 = 0$ [since plane & line are parallel]

$\Rightarrow \lambda = 12$

Hence equation of plane is $13 x + 10y + 35z = 65$

Given equation of place is $ax + by + cz = 65$

so, (a, b, c) = (13, 10, 35)

Now we have to find the distance of (13, 10, 35) from $2x + 2y - z + 16 = 0$

$$d = \frac{|26+20-35+16|}{\sqrt{4+4+1}} = 9$$

49. If the shortest distance between the lines $\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is 6, then the square of sum of all possible values of λ is _____. **[JEE(Main)Jan.2023]**

यदि $\frac{x+\sqrt{6}}{2}=\frac{y-\sqrt{6}}{3}=\frac{z-\sqrt{6}}{4}$ और $\frac{x-\lambda}{3}=\frac{y-2\sqrt{6}}{4}=\frac{z+2\sqrt{6}}{5}$ के बीच न्यूनतम दूरी 6 है, तब λ के सभी संभव मानों

d k; kx d koxZgS_____ **[JEE(Main)Jan.2023]**

Ans. 384

Sol. In vector form

$\vec{r}=\left(-\sqrt{6}\hat{i}+\sqrt{6}\hat{j}+\sqrt{6}\hat{k}\right)+A\left(2\hat{i}+3\hat{j}+4\hat{k}\right)$

$\vec{r}=\left(\lambda\hat{i}+2\sqrt{6}\hat{j}-2\sqrt{6}\hat{k}\right)+B\left(3\hat{i}+4\hat{j}+5\hat{k}\right)$

$\overrightarrow{a_2}-\overrightarrow{a_1}=\left(\lambda+\sqrt{6}\right)\hat{i}+\sqrt{6}\hat{j}-3\sqrt{6}\hat{k}$

$\vec{b}_1\times\vec{b}_2=-\hat{i}+2\hat{j}-\hat{k}$

$\left|\vec{b}_1\times\vec{b}_2\right|=\sqrt{6}$

$\left(\vec{a}_2-\vec{a}_1\right)\left(\vec{b}_1\times\vec{b}_2\right)=\left(4\sqrt{6}-\lambda\right)$

$\text{S.D.}=\left|\frac{\left(\vec{a}_2-\vec{a}_1\right)\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}\right|=6$

$\frac{4\sqrt{6}-\lambda}{\sqrt{6}}=\pm6$

$\lambda_1=-2\sqrt{6},\lambda_2=10\sqrt{6}$

$\lambda_1+\lambda_2=8\sqrt{6}$

$\left(\lambda_1+\lambda_2\right)^2=384$

$=384$

50. If the shortest distance between the line joining the points (1,2,3) and (2,3,4), and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि बिंदुओं (1,2,3) तथा (2,3,4) को मिलाने वाली रेखा तथा रेखा $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ के बीच न्यूनतम दूरी α है, तो $28\alpha^2$ बराबर _____ है। **[JEE(Main)Jan.2023]**

Ans. 18

Sol. $\frac{x+1}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{\frac{-1}{12}}$ and $\frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{\frac{1}{6}}$

$\Rightarrow$ Shortest distance $= \dfrac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$

$$\text{S.D.} = (-\hat{i}+2\hat{j}-\hat{k})\cdot\frac{(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\left\{\vec{p}\times\vec{q}\equiv\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & \frac{1}{2} & \frac{-1}{12}\\ 1 & 1 & \frac{1}{6}\end{vmatrix}=\frac{1}{6}\hat{i}-\frac{1}{4}\hat{j}+\frac{1}{2}\hat{k} \text{ or } 2\hat{i}-3\hat{j}+6\hat{k}\right\}$$

$$\text{S.D.} = \frac{(-\hat{i}+2\hat{j}-\hat{k})\cdot(2\hat{i}-3\hat{j}+6\hat{k})}{\sqrt{2^2+3^2+6^2}} = \left|\frac{-14}{7}\right| = 2$$

51. Let the equation of the plane P containing the line $x+10=\dfrac{8-y}{2}=z$ be ax + by + 3z = 2 (a + b) and the distance of the plane P from the point (1,27,7) be . Then $a^2 + b^2 + c^2$ is equal to _____.

[JEE(Main)Jan.2023]

माना समतल P, जिस पर रेखा $x+10=\dfrac{8-y}{2}=z$ स्थित है, का समीकरण ax + by + 3z = 2 (a + b) है एवं बिंदु (1,27,7) से समतल P की दूरी c है। तब $a^2 + b^2 + c^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 355

Sol. $\dfrac{x+10}{1}=\dfrac{y-8}{-2}=\dfrac{z}{1}$ .....(1)

and ax + by + 3z = 2 (a + b) .....(2)

$\because$ (2) contains the line (1)

$\therefore$ (–10, 8, 0) will lie on (2)

$\Rightarrow$ –10a + 8b = 2a + 2b

b = 2a .....(3)

and a – 2b + 3 = 0 .....(4)

from (3) and (4), we get

a = 1 & b = 2

$\therefore$ eq^n of plane (2) is x + 2y + 3z – 6 = 0

$$\therefore \left|\frac{1+54+21-6}{\sqrt{1+4+5}}\right| = 5\sqrt{14}$$

$\therefore$ $a_2 + b_2 + c_2 = 1 + 4 + 25 \times 14 = 5 + 350 = 355$

52. Let the co-ordinates of one vertex of ΔABC be $A(0,2,\alpha)$ and the other two vertices lie on the line

$\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha\in\mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____. **[JEE(Main)Jan.2023]**

माना ΔABC के एक शीर्ष के निर्देशांक $A(0,2,\alpha)$ है तथा अन्य दो शीर्ष रेखा $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ पर स्थित है। $\alpha\in\mathbb{Z}$ के लिए, यदि ΔABC का क्षेत्रफल 21 वर्ग इकाई है एवं रेखा खण्ड BC की लम्बाई $2\sqrt{21}$ इकाई है, तब α^2 बराबर है।

[JEE(Main)Jan.2023]

Ans. 9

Sol. A. $(O_1\ 2,\ \alpha)$



$$\left|\frac{1}{2}.2\sqrt{21}.\begin{vmatrix} i & j & k \\ \alpha & 1 & \alpha+4 \\ 5 & 2 & 3 \end{vmatrix}\frac{1}{\sqrt{25+4+9}}\right|=21\sqrt{21}$$

$$\sqrt{(2\alpha+5)^2+(2\alpha+20)^2+(2\alpha-5)^2}=\sqrt{21}\sqrt{38}$$

$\Rightarrow 12\alpha^2+80\alpha+450=798$

$\Rightarrow 12\alpha^2+80\alpha-348=0$

$\Rightarrow \alpha=3 \Rightarrow \alpha^2=9$

53. If the equation of the plane passing through the point (1,1,2) and are perpendicular to the line $x-3y+2z-1=0=4x-y+z$ is $Ax+By+Cz=1$, then 140 (C − B + A) is equal to _____.

[JEE(Main)Jan.2023]

यदि बिंदु (1,1,2) से होकर जाने वाला तथा रेखा $x-3y+2z-1=0=4x-y+z$ के लंबवत् समतल का समीकरण $Ax+By+Cz=1$ है, तो 140 (C − B + A) बराबर है। **[JEE(Main)Jan.2023]**

Ans. 15

Sol. $x-3y+2z-1=0$

$4x-y+z=0$

$$\vec{n}_1\times\vec{n}_2=\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 2 \\ 4 & -1 & 1 \end{vmatrix}$$

$=-\hat{i}+7\hat{j}+11\hat{k}$

D.R. of normal to the plane is –1, 7, 11

Equation of plane

$-1(x-1)+7(y-1)+11(z-2)=0$

$-x+7y+11z=28$

$$-\frac{1}{28}x+\frac{7}{28}y+\frac{11}{28}z=1$$

$Ax + By + Cz = 1$

$$140(C-B+A)=140\left(\frac{11}{28}-\frac{7}{28}-\frac{1}{28}\right)$$

$$=140\times\frac{3}{28}=15$$

54. Let a line L pass through the point P (2,3,1) and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L of the point (5,3,8) is α, then $3\alpha^2$ is equal to _____.

[JEE(Main)Jan.2023]

माना रेखा L बिंदु P (2,3,1) से होकर जाती है तथा रेखा $x + 3y - 2z - 2 = 0 = x - y + 2z$ के समान्तर है। यदि बिंदु (5,3,8) की रेखा L से दूरी α है, तो $3\alpha^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 158

Sol. $$\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 3 & -2\\ 1 & -1 & 2\end{vmatrix}=4\hat{i}-4\hat{j}-4\hat{k}$$

$\therefore$ Equation of line is $\frac{x-2}{1}=\frac{y-3}{-1}=\frac{z-1}{-1}$

Let Q be (5, 3, 8) and foot of $\perp$ from Q on this line be R.

Now, $R \equiv (k + 2, -k + 3, -k + 1)$

DR of QR are $(k - 3, -k, -k - 7)$

$\therefore$ $(1)(k - 3) + (-1)(-k) + (-1)(-k - 7) = 0$

$$\Rightarrow k=-\frac{4}{3}$$

$$\therefore \alpha^2=\left(\frac{13}{3}\right)^2+\left(\frac{4}{3}\right)^2+\left(\frac{17}{3}\right)^2=\frac{474}{9}$$

$\therefore$ $3\alpha^2 = 158$

55. Let the line $L:\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point R(1,-1,-3) on the line L. If α is the area of triangle PQR, then α^2 is equal to _____.

[JEE(Main)Jan.2023]

माना रेखा $L:\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{1}$ तथा समतल $2x + y + 3z = 16$ का प्रतिच्छेदन बिंदु P है। माना बिंदु R(1,-1,-3) से रेखा L पर लंब का पाद Q है। यदि त्रिभुज PQR का क्षेत्रफल α है तो α^2 बराबर है। **[JEE(Main)Jan.2023]**

Ans. 180

Sol. Any point on L $((2\lambda + 1), (-\lambda - 1), (\lambda + 3))$

$2(2\lambda + 1) + (-\lambda - 1) + 3(\lambda + 3) = 16$

$6\lambda + 10 = 16 \Rightarrow \lambda = 1$

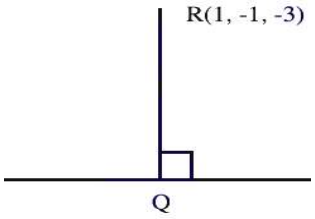
$\therefore P = (3, -2, 4)$

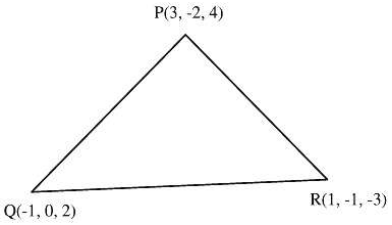
DR of QR $= \langle 2\lambda, -\lambda, \lambda + 6 \rangle$

DR of L $= \langle 2, -1, 1 \rangle$

$4\lambda + \lambda + \lambda + 6 = 0$ $\qquad$ $6\lambda + 6 = 0 \Rightarrow \lambda = -1$

$Q = (-1, 0, 2)$





$\overrightarrow{QR} = 2\hat{i} - \hat{j} - 5\hat{k}$ $\qquad$ $\overrightarrow{QP} = 4\hat{i} - 2\hat{j} + 2\hat{k}$

$$\overrightarrow{QR} \times \overrightarrow{QP} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & -5 \\ 4 & -2 & 2 \end{vmatrix} = -12i66 - 24\hat{j}$$

$$\alpha = \frac{1}{2} \times \sqrt{144 + 576} \Rightarrow \alpha^2 = \frac{720}{4} = 180$$

56. A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1) and D(4, 5, 0), $|\lambda| \le 5$ are the vertices of a quadrilateral ABCD. If its area is 18 square units, then $5 - 6\lambda$ is equal to _____. **[JEE(Main)Jan.2023]**

A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1) तथा D(4, 5, 0), $|\lambda| \le 5$ एक चतुर्भुज ABCD के शीर्ष है। यदि इसका क्षेत्रफल 18 वर्ग इकाई है, तो $5 - 6\lambda$ है। **[JEE(Main)Jan.2023]**

Ans. 11

Sol. A(2, 6, 2), B(–4, 0, λ), C(2, 3, –1), D(4, 5, 0)

$$\text{Area} = \frac{1}{2}\left|\overrightarrow{BD} \times \overrightarrow{AC}\right| = 18$$

$$\overrightarrow{AC}\times\overrightarrow{BD}=\begin{vmatrix}\hat{i} & j & k\\ 0 & -3 & -3\\ 8 & 5 & -\lambda\end{vmatrix}$$

$$=(3\lambda+15)\hat{i}-j(-24)+k(-24)$$

$$\overrightarrow{AC}\times\overrightarrow{BD}=(3\lambda+15)\hat{i}+24j-24k$$

$$=\sqrt{(3\lambda+15)^2+(24)^2+(24)^2}=36$$

$= \lambda^2 + 10\lambda + 9 = 0$

$= \lambda = -1,-9$

$|\lambda| \le 5 \Rightarrow \lambda = -1$

$5 - 6\lambda = 5 - 6(-1) = 11$

57. The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points A(–3,–6,1) and B(2,4,–3) divides the line segment AB internally in the ratio k : 1. If a, b, c (|a|, |b|, |c| are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1}=\frac{y+4}{2}=\frac{z+2}{3}$, then |a + b + c| is equal to _____. **[JEE(Main)Jan.2023]**

समतल $8x + y + 2z = 0$ तथा बिंदुओं A(–3,–6,1) और B(2,4,–3) को मिलाने वाली रेखा का प्रतिच्छेदन बिंदु C रेखाखंड AB को k : 1 के अनुपात में अंतः विभाजित करता है। यदि बिंदु C से रेखा $\frac{1-x}{1}=\frac{y+4}{2}=\frac{z+2}{3}$ पर अभिलंब के दिक् अनुपात a, b, c ;ंद्यद्यद्य

CD $\perp$ to line : $-1(2-3\lambda)+2(6\lambda-14)+3(9\lambda-1)=0$

$\Rightarrow -2+3\lambda+12\lambda-28+27\lambda-3=0$

$42\lambda=33 \Rightarrow \lambda=\frac{33}{42}=\frac{11}{14}$

$\therefore$ drs of CD $= \frac{2-\frac{33}{14}}{3}, \frac{-14+\frac{66}{14}}{3}, \frac{-1+\frac{99}{14}}{3}$

$= -5, -130, 85$

$= -1, -26, 17$

$\therefore (-1 - 26 + 17) = 10$

58. Let $\alpha x+\beta y+\gamma z=1$ be the equation of a plane passing through the point (3,-2,5) and perpendicular to the line joining the points (1,2,3) and (-2,3,5). Then the value of $\alpha\beta\gamma$ is equal to _____.

[JEE(Main)Jan.2023]

माना बिंदु (3, –2, 5) से होकर जाने वाले तथा बिंदुओं (1,2,3) और (3,-2,5) को मिलाने वाली रेखा के लंबवत समतल का समीकरण $\alpha x+\beta y+\gamma z=1$ है। तो $\alpha\beta\gamma$ का मान बराबर है। **[JEE(Main)Jan.2023]**

Ans. 6

Sol. Given Equation is not equation of plane as γz is present. If we consider y is γ then answer would be 6.

Normal vector of plane = $3\hat{i}-\hat{j}-2\hat{k}$

Plane : $3x - y - 2z + \lambda = 0$

Point (3 –2, 5) satisfies the plane

$\lambda = -1$

$3x - y - 2z = 1$

$\alpha\beta\gamma = 6$

59. Let the position vectors of the points A, B, C and D be $5\hat{i}+5\hat{j}+2\lambda\hat{k}, \hat{i}+2\hat{j}+3\hat{k}, -2\hat{i}+\lambda\hat{j}+4\hat{k}$ and $-\hat{i}+5\hat{j}+6\hat{k}$. Let the set $S=\{\lambda\in\mathbb{R}$: the points A, B, C and D are coplanar$\}$. Then $\sum_{\lambda\in S}(\lambda+2)^2$ is equal to :

[JEE(Main)April,2023]

माना बिंदुओं A, B, C व D के स्थिति सदिश $5\hat{i}+5\hat{j}+2\lambda\hat{k}, \hat{i}+2\hat{j}+3\hat{k}, -2\hat{i}+\lambda\hat{j}+4\hat{k}$ व $-\hat{i}+5\hat{j}+6\hat{k}$ है। माना समुच्चय $S=\{\lambda\in\mathbb{R}$: बिंदु A, B, C व D सहतलीय हैं$\}$ है, तब $\sum_{\lambda\in S}(\lambda+2)^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 25 (B) 41 (C) $\frac{37}{2}$ (D) 13

Ans. B

Sol. $\vec{a}=5\hat{i}+5\hat{j}+2\lambda\hat{k}$

$\hat{b}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{c}=-2\hat{i}+\lambda\hat{j}+4\hat{k}$

$\hat{d} = -\hat{i} + 5\hat{j} + 6\hat{k}$

$\overrightarrow{BA} \cdot (\overrightarrow{BC} \times \overrightarrow{BD}) = 0$

$$\begin{vmatrix} 4 & 3 & 2\lambda - 3 \\ -3 & \lambda - 2 & 1 \\ -2 & 3 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 2, 3$$

$\Sigma(\lambda + 2)^2 = 16 + 25 = 41$

60. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a} \cdot \vec{d} = 18$, then $|\vec{a} \times \vec{d}|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ व $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$ है। यदि एक सदिश $\vec{d}$ सदिशों $\vec{b}$ व $\vec{c}$ दोनों के लम्बवत् है और $\vec{a} \cdot \vec{d} = 18$ है, तब $|\vec{a} \times \vec{d}|^2$ का मान है : **[JEE(Main)April,2023]**

(A) 640 (B) 760 (C) 720 (D) 680

Ans. C

Sol. $\vec{b} \times \vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$

$\therefore \quad \vec{d} = \lambda(2\hat{i} - \hat{j} + 2\hat{k})$

$\vec{a} \cdot \vec{d} = 18$

$\Rightarrow \lambda = 2$

$\therefore \quad |\vec{a} \times \vec{d}|^2 = |\vec{a}|^2 |\vec{d}|^2 - (\vec{a} \cdot \vec{d})^2$

$= 720$

61. One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are 3, 4 and 5 units respectively. Let P be the vertex (3,4,5). Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is : **[JEE(Main)April,2023]**

एक आयताकार समांतर षट्फलक का एक शीर्ष मूल बिंदु O पर है और x, y तथा z अक्षों के अनुदिश इसके किनारों (edges) की लम्बाईयाँ क्रमशः 3, 4 तथा 5 इकाई है। माना इसका शीर्ष P बिंदु (3,4,5) पर है। तब विकर्ण OP तथा z अक्ष के समांतर इसके एक fdukjs] tks O या P से होकर नहीं जाता है, के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) $\frac{12}{5\sqrt{5}}$ (B) $\frac{12}{\sqrt{5}}$ (C) $12\sqrt{5}$ (D) $\frac{12}{5}$

Ans. D

Sol. Equation of OP is $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$

$a_1 = (0,0,0)$ $a_2 = (3,0,5)$

$b_1 = (3,4,5)$ $b_2 = (0,0,1)$

Equation of edge parallel to z axis

$$\frac{x-3}{0}=\frac{y-0}{0}=\frac{z-5}{1}$$

$$S.D=\frac{\left(\vec{a}_2.\vec{a}_1\right).\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}$$

$$\frac{\begin{vmatrix}3&0&5\\3&4&5\\0&0&1\end{vmatrix}}{\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&4&5\\0&0&1\end{vmatrix}}=\frac{3(4)}{\left|4\hat{i}-3\hat{j}\right|}=\frac{12}{5}$$

62. A plane P contains the line of intersection of the plane $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=6$ and $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k})=-5$. If P passes through the point (0, 2, –2), then the square of distance of the point (12, 12, 18) from the plane P is: **[JEE(Main)April,2023]**

एक समतल P में समतलों $\vec{r}\cdot(\hat{i}+\hat{j}+\hat{k})=6$ तथा $\vec{r}\cdot(2\hat{i}+3\hat{j}+4\hat{k})=-5$ की प्रतिच्छेदन रेखा स्थित है। यदि P, बिंदु (0, 2, –2) से gkdj t kr kgS rks fcanq (12, 12, 18) की समतल P से दूरी का वर्ग है : **[JEE(Main)April,2023]**

(A) 310 (B) 1240 (C) 620 (D) 155

Ans. C

Sol. Given planes

$P_1: x+y+z-6=0$

$P_2: 2x+3y+4z+5=0$

Equation of plane passing through line of intersection of

$P_1=0$ & $P_2=0$ is $P_1+\lambda P_2=0$

$(1+2\lambda)x+(1+3\lambda)y+(1+4\lambda)z+(5\lambda-6)=0$

It passes through (0, 2,–2)

$2(1+3\lambda)+6(1+4\lambda)(-2)+5\lambda-6=0$

$2+6\lambda-2-8\lambda+5\lambda-6=0$

$3\lambda-6=0, \lambda=2$

So plane : 5x +7y + 9z + 4 = 0

distance of (12,12,18) = $\frac{5\times12+7\times12+9\times18+4}{\sqrt{25+49+81}}$

$d=\frac{310}{\sqrt{155}}=2\sqrt{155}$

$d^2=4\times155=620$

63. Let the vectors $\vec{a},\vec{b},\vec{c}$ represent three coterminous edges of a parallelopiped of volume V. Then the volume of the parallelopiped, whose coterminous edges are represented by $\vec{a},\vec{b}+\vec{c}$ and $\vec{a}+2\vec{b}+3\vec{c}$ is equal to : **[JEE(Main)April,2023]**

माना सदिश $\vec{a},\vec{b},\vec{c}$ आपतन V के एक समांतर षट्फलक के सहावसानी किनारों को निरूपित करते है। तो उस समांतर षटफक, ft l d sl gkol kuhfd ukjsa fn' kkvk$\vec{a},\vec{b}+\vec{c}$ तथा $\vec{a}+2\vec{b}+3\vec{c}$ से निरूपित होते हैं, का आयतन बराबर है : **[JEE(Main)April,2023]**

(A) V (B) 3V (C) 6V (D) 2V

Ans. A

Sol. $=\left[\vec{a}\quad\vec{b}\quad\vec{c}\right]v_1$

$=\left[\vec{a}\quad\vec{b}+\vec{c}\quad\vec{a}+2\vec{b}+3\vec{c}\right]v_1$

$=\begin{bmatrix}1&0&0\\0&1&1\\1&2&3\end{bmatrix}\left[\vec{a}\,\vec{b}\,\vec{c}\right]v_1$

$=(3-2)v$

$=v$

64. The sum of all values of α, for which the points whose position vectors are $\hat{i}-2\hat{j}+3\hat{k},2\hat{i}-3\hat{j}+4\hat{k}$, $(\alpha+1)\hat{i}+2\hat{k}$ and $9\hat{i}+(\alpha-8)\hat{j}+6\hat{k}$ are coplanar, is equal to : **[JEE(Main)April,2023]**

a के सभी मानों, जिनके लिए स्थिति सदिशों $\hat{i}-2\hat{j}+3\hat{k},2\hat{i}-3\hat{j}+4\hat{k}$, $(\alpha+1)\hat{i}+2\hat{k}$ तथा $9\hat{i}+(\alpha-8)\hat{j}+6\hat{k}$ के बिंदु सहतलीय g$Sd k; kx cj kcj gS **[JEE(Main)April,2023]**

(A) 4 (B) –2 (C) 6 (D) 2

Ans. D

Sol. Let the points be A, B, C, D

$\overrightarrow{AB}=\hat{i}-\hat{j}+\hat{k}$

$\overrightarrow{AC}=\alpha\hat{i}+2\hat{j}-\hat{k}$

$\overrightarrow{AD}=8\hat{i}+(\alpha-6)\hat{j}+3\hat{k}$

So, vectors $\overrightarrow{AB},\overrightarrow{AC},\overrightarrow{AD}$ are coplanar.

$$\begin{vmatrix}1&-1&1\\\alpha&2&-1\\8&(\alpha-6)&3\end{vmatrix}=0$$

$(6+\alpha-6)+(3\alpha+8)+\left(\alpha^2-6\alpha-16\right)=0$

$\alpha^2-2\alpha-14=0$

Sum of values of $\alpha = 2$

65. If the points with position vectors $\alpha\hat{i}+10\hat{j}+13\hat{k}$, $6\hat{i}+11\hat{j}+11\hat{k}$, $\frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ are collinear, then $(19\alpha - 6\beta)^2$ is equal to : **[JEE(Main)April,2023]**

यदि स्थिति सदिश $\alpha\hat{i}+10\hat{j}+13\hat{k}$, $6\hat{i}+11\hat{j}+11\hat{k}$, $\frac{9}{2}\hat{i}+\beta\hat{j}-8\hat{k}$ के बिंदु एक रेखा में हैं, तो $(19\alpha - 6\beta)^2$ बराबर है :

[JEE(Main)April,2023]

(A) 16 (B) 36 (C) 49 (D) 25

Ans. B

Sol. If $A(\vec{a}), B(\vec{b}), C(\vec{c})$ are collinear then

$\overrightarrow{AB} \| \overrightarrow{BC} \| \overrightarrow{AC}$

Now $\overrightarrow{AB} = (6-\alpha)\hat{i}+\hat{j}-2\hat{k}$

$\overrightarrow{BC} = -\frac{3}{2}\hat{i}+(\beta-11)\hat{j}-19\hat{k}$

$\overrightarrow{AB} \| \overrightarrow{BC} \Rightarrow \frac{6-\alpha}{-\frac{3}{2}} = \frac{1}{\beta-11} = +\frac{2}{19}$

$\Rightarrow 6-\alpha = \frac{-3}{19} \Rightarrow \alpha = 6+\frac{3}{19} = \frac{117}{19} \Rightarrow 19\alpha = 117$

$\beta-11 = +\frac{19}{2} \Rightarrow 11+\frac{19}{2} = \frac{41}{2} \Rightarrow 6\beta = 123$

$(19\alpha-6\beta)^2 = (-6)^2 = 36$

66. Let the vectors $\vec{u}_1 = \hat{i}+\hat{j}+a\hat{k}$, $\vec{u}_2 = \hat{i}+b\hat{j}+\hat{k}$ and $\vec{u}_3 = c\hat{i}+\hat{j}+\hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\vec{v}_2 = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ and $\vec{v}_3 = b\hat{i}+b\hat{j}+(c+a)\hat{k}$ are also coplanar, then $6(a + b + c)$ is equal to : **[JEE(Main)April,2023]**

माना सदिश $\vec{u}_1 = \hat{i}+\hat{j}+a\hat{k}$, $\vec{u}_2 = \hat{i}+b\hat{j}+\hat{k}$ तथा $\vec{u}_3 = c\hat{i}+\hat{j}+\hat{k}$ सह–तलीय है। यदि सदिश $\vec{v}_1 = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\vec{v}_2 = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ तथा $\vec{v}_3 = b\hat{i}+b\hat{j}+(c+a)\hat{k}$ भी सह–तलीय हैं, तो $6(a + b + c)$ बराबर है :

[JEE(Main)April,2023]

(A) 6 (B) 4 (C) 12 (D) 0

Ans. C

Sol. Given : $\overrightarrow{u_1} = \hat{i}+\hat{j}+a\hat{k}, \overrightarrow{u_2} = \hat{i}+b\hat{j}+\hat{k}$ and $\overrightarrow{u_3} = c\hat{i}+\hat{j}+\hat{k}$

and $\overrightarrow{v_1} = (a+b)\hat{i}+c\hat{j}+c\hat{k}$, $\overrightarrow{v_2} = a\hat{i}+(b+c)\hat{j}+a\hat{k}$ and $\overrightarrow{v_3} = b\hat{i}+b\hat{j}+(c+a)\hat{k}$

$$\text{Now}\begin{vmatrix}1 & 1 & a\\ 1 & b & 1\\ c & 1 & 1\end{vmatrix}=0\Rightarrow(b-1)-(1-c)+a(1-bc)=0$$

$a+b+c=2+abc$ …(i)

$$\text{and}\begin{vmatrix}a+b & c & c\\ a & b+c & a\\ b & b & c+a\end{vmatrix}=0$$

$$\Rightarrow\begin{vmatrix}a+b & c & c\\ a & b+c & a\\ -2a & -2c & 0\end{vmatrix}=0\quad\left(R_3\to R_3-\left(R_1+R_2\right)\right)$$

$\Rightarrow abc=0$

$\therefore a+b+c=2$

$\therefore\quad 6(a+b+c)=12$

67. Let O be the origin and the position vector of the point P be $-\hat{i}-2\hat{j}+3\hat{k}$. If the position vectors of the points A, B and C are $-2\hat{i}+\hat{j}-3\hat{k}$, $2\hat{i}+4\hat{j}-2\hat{k}$ and $-4\hat{i}+2\hat{j}-\hat{k}$ respectively, then the projection of the vector $\overrightarrow{OP}$ on a vector perpendicular to the vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$ is : **[JEE(Main)April,2023]**

माना O मूल बिंदु है तथा बिंदु P का स्थिति सदिश $-\hat{i}-2\hat{j}+3\hat{k}$ है। यदि बिंदुओं A, B तथा C के स्थिति सदिश क्रमशः $-2\hat{i}+\hat{j}-3\hat{k}$, $2\hat{i}+4\hat{j}-2\hat{k}$ तथा $-4\hat{i}+2\hat{j}-\hat{k}$ हैं, तो सदिशों $\overrightarrow{AB}$ तथा $\overrightarrow{AC}$ के लंबवत् एक सदिश पर, सदिश $\overrightarrow{OP}$ का प्रक्षेप है : **[JEE(Main)April,2023]**

(A) $\frac{10}{3}$ (B) $\frac{8}{3}$ (C) 3 (D) $\frac{7}{3}$

Ans. C

Sol. $\overrightarrow{OP}=-\hat{i}-2\hat{j}+3k$

$\overrightarrow{AB}=4\hat{i}+3\hat{j}+k$

$\overrightarrow{AC}=-2\hat{i}+\hat{j}+2k$

$$\overrightarrow{AB}\times\overrightarrow{AC}=\begin{vmatrix}\hat{i} & \hat{j} & k\\ 4 & 3 & 1\\ -2 & 1 & 2\end{vmatrix}$$

$=5\hat{i}-10\hat{j}+10k$

Projection of $\overrightarrow{OP}$ on

$$\overrightarrow{AB}\times\overrightarrow{AC}=\left|\frac{\overrightarrow{OP}\cdot(\overrightarrow{AB}\times\overrightarrow{AC})}{|\overrightarrow{AB}\times\overrightarrow{AC}|}\right|$$

$$=\frac{5(-\hat{i}-2\hat{j}+3k)(\hat{i}-2\hat{j}+2k)}{5\sqrt{1+4+4}}$$

$= 3$

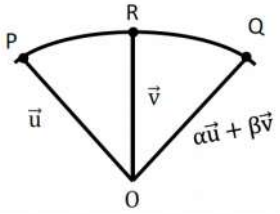
68. An arc PQ of a circle subtends a right angle at its centre O. The mid point of the arc PQ is R. If $\overrightarrow{OP}=\vec{u},\overrightarrow{OR}=\vec{v}$ and $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$, then α,β^2 are the roots of the equation : **[JEE(Main)April,2023]**

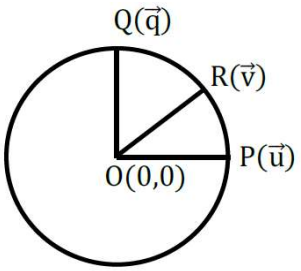
एक वृत्त का एक चाप PQ इसके केन्द्र पर समकोण बनाता है। चाप PQ का मध्य बिंदु R है। यदि $\overrightarrow{OP}=\vec{u},\overrightarrow{OR}=\vec{v}$ तथा $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$ है, तो α,β^2 किस समीकरण के मूल है ? **[JEE(Main)April,2023]**

(A) $x^2 + x - 2 = 0$ (B) $3x^2 - 2x - 1 = 0$ (C) $x^2 - x - 2 = 0$ (D) $3x^2 + 2x - 1 = 0$

Ans. C

Sol.





Let $\overrightarrow{OP}=\vec{u}=\hat{i}$

$\overrightarrow{OQ}=\vec{q}=\hat{j}$

$\because$ R is the mid-point of $\overrightarrow{PQ}$

$$\overrightarrow{OR}=\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$$

Now $\overrightarrow{OQ}=\alpha\vec{u}+\beta\vec{v}$

$$\hat{j}=\alpha\hat{i}+\beta\left(\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}\right)$$

$$\beta=\sqrt{2},\alpha+\frac{\beta}{\sqrt{2}}=0\Rightarrow\alpha=-1$$

Now equation $x^2-\left(\alpha+\beta^2\right)x+\alpha\beta^2=0$

$x^2 - (-1 + 2)x + (-1)(2) = 0$

$x^2 - x - 2 = 0$

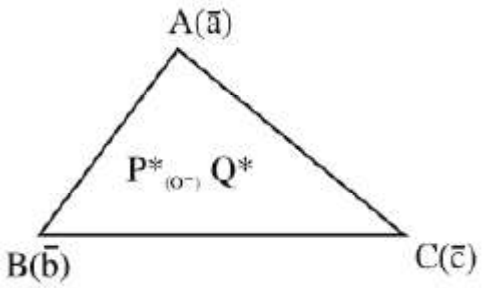
69. If the points P and Q are respectively the circumcenter and the orthocentre of a ΔABC, then $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ is : **[JEE(Main)April,2023]**

यदि एक त्रिभुज ABC के परिकेन्द्र तथा लंबकेन्द्र क्रमशः P तथा Q है, तो $\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ बराबर है : **[JEE(Main)April,2023]**

(A) $2\overrightarrow{PQ}$ (B) $\overrightarrow{QP}$ (C) $2\overrightarrow{QP}$ (D) $\overrightarrow{PQ}$

Ans. D

Sol.



$\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}=\vec{a}+\vec{b}+\vec{c}$

$\overrightarrow{PG}=\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

$\Rightarrow \vec{a}+\vec{b}+\vec{c}=3\overrightarrow{PG}=\overrightarrow{PQ}$

70. Let $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}+5\hat{k}$ and $\vec{c}=\hat{i}-\hat{j}+2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c}\cdot\vec{d}=12$. Then $(-\hat{i}+\hat{j}-\hat{k})\cdot(\vec{c}\times\vec{d})$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$, $\vec{b}=3\hat{i}+5\hat{k}$ तथा $\vec{c}=\hat{i}-\hat{j}+2\hat{k}$ है। माना सदिशों $\vec{a}$ तथा $\vec{b}$ के लंबवत् एक सदिश $\vec{d}$ है तथा $\vec{c}\cdot\vec{d}=12$ है। तो $(-\hat{i}+\hat{j}-\hat{k})\cdot(\vec{c}\times\vec{d})$ बराबर है : **[JEE(Main)April,2023]**

(A) 48 (B) 42 (C) 24 (D) 44

Ans. D

Sol. $\vec{a}=2\hat{i}+7\hat{j}-\hat{k}$

$\vec{b}=3\hat{i}+5\hat{k}$

$\vec{c}=\hat{i}-\hat{j}+2\hat{k}$

$$\vec{d}=\lambda(\vec{a}\times\vec{b})=\lambda\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&7&-1\\3&0&5\end{vmatrix}$$

$\vec{d}=\lambda(35\hat{i}-13\hat{j}-21\hat{k})$

$\lambda(35+13-42)=12$

$\lambda = 2$

$\vec{d} = 2(35\hat{i} - 13\hat{j} - 21\hat{k})$

$(\hat{i} + \hat{j} - \hat{k})(\vec{c} \times \vec{d})$

$$= \begin{vmatrix} -1 & 1 & -1 \\ 1 & -1 & 2 \\ 70 & -26 & -42 \end{vmatrix} = 44$$

71. For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10|a_i| < 1$, i = 1, 2, 3, consider the following statements : **[JEE(Main)April,2023]**

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \le |\vec{a}|$

(B) : $|\vec{a}| \le 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) Both (A) and (B) are true (B) Only (B) is true

(C) Only (A) is true (D) Neither (A) nor (B) is true

किसी सदिश $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $10|a_i| < 1$, i = 1, 2, 3 के लिए, निम्नलिखित कथनों पर विचार कीजिए : **[JEE(Main)April,2023]**

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \le |\vec{a}|$

(B) : $|\vec{a}| \le 3\max\{|a_1|, |a_2|, |a_3|\}$

(A) (A) तथा (B) दोनों सत्य है (B) केवल (B) सत्य है

(C) केवल (A) सत्य है (D) न (A) और न (B) सत्य है

Ans. A

Sol. $\vec{a} = a_1 i + a_2\hat{j} + a_3\hat{k}, |a_i| < \frac{1}{10}, i = 1, 2, 3$

$\because |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} \Rightarrow |\vec{a}| = \sqrt{|a_1|^2 + |a_2|^2 + |a_3|^2} \ge \max(|a_1|, |a_2|, |a_3|)$

Similarly $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2} < \sqrt{\frac{1}{100} + \frac{1}{100} + \frac{1}{100}}$

$|\vec{a}| < \frac{\sqrt{3}}{10} < \frac{3}{10}$

$\Rightarrow |\vec{a}| < 3\max(|a_1|, |a_2|, |a_3|) \Rightarrow$ A and B both are correct

72. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$ and $\hat{i} - \hat{j}, \hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\vec{i} - 2\vec{j} + \vec{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a}$; $\hat{i}+\hat{j},\hat{i}+\hat{k}$ तथा $\hat{i}-\hat{j},\hat{j}-\hat{k}$ द्वारा वर्णित दो समतलो के दो प्रतिच्छेदन रेखा के समान्तर एक अशून्य सदिश है। यदि सदिश $\vec{a}$ तथा सदिश $\vec{b}=2\vec{i}-2\vec{j}+\vec{k}$ के बीच का कोण θ है तो क्रमित युग्म $(\theta,|\vec{a}\times\vec{b}|)$ का मान ज्ञात कीजिए :

[JEE(Main)April,2023]

(A) $\left(\frac{\pi}{4},6\right)$ (B) $\left(\frac{\pi}{3},6\right)$ (C) $\left(\frac{\pi}{4},3\sqrt{6}\right)$ (D) $\left(\frac{\pi}{3},3\sqrt{6}\right)$

Ans. A

Sol. $\vec{n}_1$ and $\vec{n}_2$ are normal vector to the plane $\hat{i}+\hat{j},\hat{i}+\hat{k}$ and $\hat{i}-\hat{j},\hat{i}-\hat{k}$ respectively

$$\vec{n}_1=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&0\\1&0&1\end{vmatrix}=\hat{i}-\hat{j}-\hat{k}$$

$$\vec{n}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&0\\1&0&-1\end{vmatrix}=\hat{i}+\hat{j}+\hat{k}$$

$$\vec{a}=\lambda|\vec{n}_2\times\vec{n}_2|$$

$$=\lambda\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-1&-1\\1&1&1\end{vmatrix}=\lambda(-2\hat{j}+2\hat{k})$$

$$\vec{a}\cdot\vec{b}=\lambda|0+4+2|=6$$

$$\Rightarrow\lambda=1$$

$$\vec{\alpha}=-2\hat{j}+2\hat{k}$$

$$\cos\theta=\frac{\vec{a}\cdot\vec{b}}{|a||b|}$$

$$\cos\theta=\frac{6}{2\sqrt{2}\times3}=\frac{1}{\sqrt{2}}$$

$$\theta=\frac{\pi}{4}$$

Now $|\vec{a}.\vec{b}|^2+|\vec{a}\times\vec{b}|^2=|a|^2|\vec{b}|^2$

$$36+\left|\vec{a}\times\mathbf{b}^2\right|=8\times9=72$$

$$|\vec{\mathbf{a}}\times\mathbf{b}|^2=36$$

$$|\vec{\mathbf{a}}\times\vec{b}|=6$$

73. If four distinct points with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are coplanar, then $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ is equal to :

[JEE(Main)April,2023]

यदि चार भिन्न बिंदु, जिनके स्थिति सदिश $\vec{a}$, $\vec{b}$, $\vec{c}$ तथा $\vec{d}$ हैं, सहतलीय हैं, तो $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ बराबर है :

[JEE(Main)April,2023]

(A) $\left[\vec{a}\,\vec{d}\,\vec{b}\right]+\left[\vec{d}\,\vec{c}\,\vec{a}\right]+\left[\vec{d}\,\vec{b}\,\vec{c}\right]$ (B) $\left[\vec{d}\,\vec{b}\,\vec{a}\right]+\left[\vec{a}\,\vec{c}\,\vec{d}\right]+\left[\vec{d}\,\vec{b}\,\vec{c}\right]$

(C) $\left[\vec{d}\,\vec{c}\,\vec{a}\right]+\left[\vec{b}\,\vec{d}\,\vec{a}\right]+\left[\vec{c}\,\vec{d}\,\vec{b}\right]$ (D) $\left[\vec{b}\,\vec{c}\,\vec{d}\right]+\left[\vec{d}\,\vec{a}\,\vec{c}\right]+\left[\vec{d}\,\vec{b}\,\vec{a}\right]$

Ans. C

Sol. $\vec{a},\vec{b},\vec{c},\vec{d}$ are coplanar point then

vectors $\vec{b}-\vec{a},\vec{c}-\vec{a},\vec{d}-\vec{a}$ are also coplanar

$$\left[\vec{b}-\vec{a}\ \ \vec{c}-\vec{a}\ \ \vec{d}-\vec{a}\right]=0$$

$$\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]-\left[\vec{b}\ \ \vec{c}\ \ \vec{a}\right]-\left[\vec{b}\ \ \vec{a}\ \ \vec{d}\right]-\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]=0$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]+\left[\vec{a}\ \ \vec{b}\ \ \vec{d}\right]+\left[\vec{a}\ \ \vec{d}\ \ \vec{c}\right]$$

74. Let $\lambda\in\mathbb{Z},\vec{a}=\lambda\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$. Let $\vec{c}$ be a vector such that $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=\vec{0}$, $\vec{a}\cdot\vec{c}=-17$ and $\vec{b}\cdot\vec{c}=-20$. Then $|\vec{c}\times(\lambda\hat{i}+\hat{j}+\hat{k})|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\lambda\in\mathbb{Z},\vec{a}=\lambda\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=3\hat{i}-\hat{j}+2\hat{k}$ है। माना $\vec{c}$ के लिए $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=\vec{0}$, $\vec{a}\cdot\vec{c}=-17$ तथा $\vec{b}\cdot\vec{c}=-20$ हैं। तो $|\vec{c}\times(\lambda\hat{i}+\hat{j}+\hat{k})|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 62 (B) 46 (C) 49 (D) 53

Ans. B

Sol. $(\vec{a}+\vec{b}+\vec{c})\times\vec{c}=0$

$(\vec{a}+\vec{b})\times\vec{c}=0$

$\vec{c}=\alpha(\vec{a}+\vec{b})=\alpha(\lambda+3)\hat{i}+\alpha\hat{k}$

$\vec{b}\cdot\vec{c}=-20\Rightarrow 3\alpha(\lambda+3)+2\alpha=-20$

$\vec{a}\cdot\vec{c}=-17\Rightarrow\alpha\lambda(\lambda+3)-\alpha=-17$

$\Rightarrow\alpha(3\lambda+9+2)=-20$

$\alpha\left(\lambda^2+3\lambda-1\right)=-17$

$17(3\lambda+11)=20\left(\lambda^2+3\lambda-1\right)$

$20\lambda^2+9\lambda-207=0$

$\lambda=3\quad(\lambda\in Z)$

$\Rightarrow \alpha = -1 \Rightarrow \vec{c} = -(6\hat{i} + \hat{k})$

$\vec{v} = \vec{c} \times (3\hat{i} + \hat{j} + \hat{k})$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -6 & 0 & -1 \\ 3 & 1 & 1 \end{vmatrix} = \hat{i} + 3\hat{j} - 6\hat{k}$$

$|\vec{v}|^2 = (-1)^2 + 3^2 + 6^2 = 46$

75. Let a, b, c be three distinct real numbers, none equal to one. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to : **[JEE(Main)April,2023]**

माना a, b, c तीन भिन्न वास्तविक संख्याएँ है तथा इनमें से कोई भी एक के बराबर नहीं है। यदि सदिश $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ तथा $\hat{i} + \hat{j} + c\hat{k}$ समतलीय हैं, तो $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ बराबर है : **[JEE(Main)April,2023]**

(A) –2 (B) 1 (C) 2 (D) –1

Ans. B

Sol. $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

$$\begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$$

$R_1 \to R_1 - R_2$
$R_2 \to R_2 - R_3$
$\Rightarrow (a-1)[c(b-1) - (1-c)] + 1[(1-b)(1-c)] = 0$
$\Rightarrow c(a-1)(b-1) - (a-1)(1-c) + (1-b)(1-c) = 0$

$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$

76. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$, then $|\vec{d}|^2$ is equal to : **[JEE(Main)April,2023]**

माना $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ तथा $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$ हैं। यदि सदिश $\vec{d}$, $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ तथा $\vec{d} \cdot \vec{a} = 24$ को संतुष्ट djr kgSतो $|\vec{d}|^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 313 (B) 323 (C) 423 (D) 413

Ans. D

Sol. $\vec{d}\times\vec{b}=\vec{c}\times\vec{b}$

$\Rightarrow\left(\vec{d}-\vec{c}\right)\times\vec{b}=0$

$\Rightarrow\vec{d}=\vec{c}+\lambda\vec{b}$

Also $\vec{d}.\vec{a}=24$

$\Rightarrow\left(\vec{c}+\lambda\vec{b}\right)\cdot\vec{a}=24$

$\lambda=\dfrac{24-\vec{a}\cdot\vec{c}}{\vec{b}\cdot\vec{a}}=\dfrac{24-6}{9}=2$

$\Rightarrow\vec{d}=\vec{c}+2\left(\vec{b}\right)$

$=8\hat{i}-5\hat{j}+18\hat{k}$

$\Rightarrow\left|\vec{d}\right|^2=64+25+324=413$

77. The distance of the point (–1, 2, 3) from the plane $\vec{r}\cdot(\hat{i}-2\hat{j}+3\hat{k})=10$ parallel to the line of the shortest distance between the lines $\vec{r}=(\hat{i}-\hat{j})+\lambda(2\hat{i}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j})+\mu(\hat{i}-\hat{j}+\hat{k})$ is :

[JEE(Main)April,2023]

बिन्दु (–1, 2, 3) की रेखाओं $\vec{r}=(\hat{i}-\hat{j})+\lambda(2\hat{i}+\hat{k})$ तथा $\vec{r}=(2\hat{i}-\hat{j})+\mu(\hat{i}-\hat{j}+\hat{k})$ के बीच न्यूनतम दूरी की रेखा के समान्तर l er y $\vec{r}\cdot(\hat{i}-2\hat{j}+3\hat{k})=10$ से दूरी है : **[JEE(Main)April,2023]**

(A) $2\sqrt{5}$ (B) $3\sqrt{6}$ (C) $3\sqrt{5}$ (D) $2\sqrt{6}$

Ans. D

Sol. Let DR's of line of shortest distance between given lines L_1 and L_2 are a, b, c then

$$a\hat{i}+b\hat{j}+c\hat{k}=t\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&0&1\\1&-1&1\end{vmatrix}=t\left(\hat{i}-\hat{j}-2\hat{k}\right)$$

Now equation of line passes through A (–1,2,3) and DR's are (1, –1, –2)

$\vec{r}=\left(-\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}-2\hat{k}\right)$ ...(1)

point of intersection of line (1) and plane $\vec{r}\cdot\left(\hat{i}-2\hat{j}+3\hat{k}\right)=10$ ...(2)

$(\lambda-1)-2(2-\lambda)+3(3-2\lambda)=10$

$\Rightarrow 4-3\lambda=10$

$\Rightarrow\lambda=-2$

So, B (–3, 4, 7)

Distance $AB=\sqrt{4+4+16}=\sqrt{24}=2\sqrt{6}$

78. Let for a triangle ABC,

$\overrightarrow{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$, then $\overrightarrow{CB} \cdot \overrightarrow{CA}$ is equal to :

[JEE(Main)April,2023]

माना एक त्रिभुज ABC के लिए,

$\overrightarrow{AB} = -2\hat{i} + \hat{j} + 3\hat{k}$

$\overrightarrow{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$

$\overrightarrow{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$

है। यदि $\delta > 0$ है तथा त्रिभुज ABC का क्षेत्रफल $5\sqrt{6}$ है, तो $\overrightarrow{CB} \cdot \overrightarrow{CA}$ बराबर है : **[JEE(Main)April,2023]**

(A) 60 (B) 54 (C) 120 (D) 108

Ans. A

Sol. $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$

$\alpha = 2, \beta = 4, \gamma - \delta = 3$

$\frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = 5\sqrt{6}$

$(\delta - 9)^2 + (2\delta + 12)^2 + 100 = 600$

$\Rightarrow \delta = 5, \gamma = 8$

Hence $\overrightarrow{CB} \cdot \overrightarrow{CA} = 60$

79. Let $|\vec{a}| = 2, |\vec{b}| = 3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})|^2$ is equal to:

[JEE(Main)April,2023]

माना $|\vec{a}| = 2, |\vec{b}| = 3$ है तथा सदिशों $\vec{a}$ और $\vec{b}$ के बीच का कोण $\frac{\pi}{4}$ है। तो $|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})|^2$ बराबर है :

[JEE(Main)April,2023]

(A) 882 (B) 841 (C) 441 (D) 482

Ans. A

Sol. $|\vec{a}| = 2, |\vec{b}| = 3$, $\vec{a} \wedge \vec{b} = \frac{\pi}{4}$

Now $\left|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})\right|^2 = \left|2\vec{a} \times \vec{a} - 3\vec{a} \times \vec{b} + 4\vec{b} \times \vec{a} - 6\vec{b} \times \vec{b}\right|^2$

$\left|7(\vec{a} \times \vec{b})\right|^2 = 49|\vec{a}|^2 |\vec{b}|^2 \sin^2\frac{\pi}{4} = 49 \cdot 4 \cdot 9 \frac{1}{2} = 882$

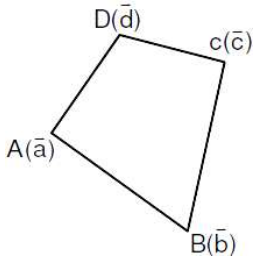
80. Let ABCD be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$, then k is equal to : **[JEE(Main)April,2023]**

माना ABCD एक चतुर्भुज है जिसके विकर्ण AC तथा BD के मध्य बिन्दु क्रमशः E तथा F है, तथा $(\overrightarrow{AB}-\overrightarrow{BC})+(\overrightarrow{AD}-\overrightarrow{DC})=k\overrightarrow{FE}$ हो तो k का मान होगा : **[JEE(Main)April,2023]**

(A) 4 (B) –2 (C) 2 (D) –4

Ans. D

Sol. Let P.V. & A, B, C, D are $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ resp. mid-point of AC is E.



$\Rightarrow$ P.V. of E are $\left(\frac{\vec{a}+\vec{c}}{2}\right)$

Mid point of BD is F

$\Rightarrow$ P.V. of F are $\left(\frac{\vec{b}+\vec{d}}{2}\right)$

$$\left(\overrightarrow{AB}-\overrightarrow{BC}\right)+\left(\overrightarrow{AD}-\overrightarrow{DC}\right)=\left(\vec{b}-\vec{a}\right)-\left(\vec{c}-\vec{b}\right)+\left(\vec{d}-\vec{a}\right)-\left(\vec{c}-\vec{d}\right)$$

$$=2\vec{b}-2\vec{a}-2\vec{c}+2\vec{d}$$

$$=2\left(\vec{b}+\vec{d}\right)-2\left(\vec{a}+\vec{c}\right)$$

$$=4\left[\left(\frac{\vec{b}+\vec{d}}{2}\right)-\left(\frac{\vec{a}+\vec{c}}{2}\right)\right]=4\overrightarrow{EF}=-4\overrightarrow{FE}$$

81. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+\mu\hat{k}$ and $3\hat{i}-4\hat{j}+5\hat{k}$, where λ – μ = 5, are coplanar, then $\sum_{(\lambda,\mu)\in S}80\left(\lambda^2+\mu^2\right)$ is equal to : **[JEE(Main)April,2023]**

यदि (λ, μ) का समुच्चय S इस प्रकार है जिसके लिए सदिश $\lambda\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+\mu\hat{k}$ तथा $3\hat{i}-4\hat{j}+5\hat{k}$ समतलीय है, जहाँ λ – μ = 5 हो तो $\sum_{(\lambda,\mu)\in S}80\left(\lambda^2+\mu^2\right)$ का मान होगा : **[JEE(Main)April,2023]**

(A) 2210 (B) 2370 (C) 2290 (D) 2130

Ans. C

Sol. $\begin{vmatrix}\lambda & -1 & 1\\ 1 & 2 & \mu\\ 3 & -4 & 5\end{vmatrix}=0$

$$\lambda(10+4\mu)+1(5-3\mu)+1(-4-6)=0$$

$$10\lambda+4\lambda\mu+5-3\mu-10=0$$

$10\lambda + 4\lambda\mu - 3\mu = 5$ ...(i)

And $\lambda = 5 + \mu$ ...(ii)

$10(5+\mu) + 4\mu(5+\mu) - 3\mu = 5$

$10\mu + 20\mu - 3\mu + 4\mu^2 + 50 = 5$

$4\mu^2 + 27\mu + 45 = 0$

$4\mu^2 + 12\mu + 15\mu + 45 = 0$

$4\mu(\mu+3) + 15(\mu+3) = 0$

$\mu = -3$ or $-\frac{15}{4}$

$\lambda = 5 - 3 = 2$ or $\lambda = 5 - \frac{15}{4} = \frac{5}{4}$

$$\sum_{\lambda,\mu} 80\left(\lambda^2 + \mu^2\right) = \left(9 + 4 + \frac{225}{16} + \frac{25}{16}\right)80 = \left(13 + \frac{250}{16}\right)80 = 1040 + 1250 = 2290$$

82. Let $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. If $\vec{a} \cdot \vec{c} = -12, \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$, then $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ is equal to _____. **[JEE(Main)April,2023]**

माना सदिश $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ तथा $\vec{c}$ इस प्रकार हैं कि $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$ है। यदि $\vec{a} \cdot \vec{c} = -12, \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$ हैं, तो $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 11

Sol. $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$ $\vec{a} \cdot \vec{b} = 6\alpha + 99 - 24 = 6\alpha + 75$

$\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ $|\vec{a}| = \sqrt{36 + 81 + 144} = \sqrt{261}$

$\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$, $\vec{a} \cdot \vec{c} = -12$, $\vec{c}.(\hat{i} - 2\hat{j} + \hat{k}) = 5$

$\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. $\Rightarrow \vec{a} \times \vec{c} - \vec{a} \times \vec{b}. = 0$

$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = 0$

$\Rightarrow \vec{c} - \vec{b} \parallel \vec{a}$

$\Rightarrow \vec{c} = \vec{b} + \lambda\vec{a}$

Now $\Rightarrow \vec{c} \cdot \vec{a} = \vec{b} \cdot \vec{a} + \lambda(\vec{a} \cdot \vec{a}) = -12$

$6\alpha + 75 + \lambda(261) = -12$

$6\alpha + 261\lambda = -87$ _____(i)

$\vec{c}.(\hat{i} - 2\hat{j} + \hat{k}) = 5, \Rightarrow (\alpha - 22 - 2) + \lambda(6 - 18 + 12) = 5$

$\Rightarrow \alpha = 29$

From equation (i) $\lambda = -1$

Hence $= \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = (\vec{b} - \vec{a}) \cdot (\hat{i} + \hat{j} + \hat{k}) = 11$

83. Let a line ℓ pass through the origin and be perpendicular to the lines

$\ell_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$ and

$\ell_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$.

If P is the point of intersection of ℓ and ℓ_1, and Q(α, β, γ) is the foot of perpendicular from P on ℓ_2, then $9(\alpha + \beta + \gamma)$ is equal to _____. **[JEE(Main)April,2023]**

माना एक रेखा ℓ मूल बिंदु से होकर जाती है तथा रेखाओं

$\ell_1 : \vec{r} = (\hat{i} - 11\hat{j} - 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k}), \lambda \in \mathbb{R}$,

$\ell_2 : \vec{r} = (-\hat{i} + \hat{k}) + \mu(2\hat{i} + 2\hat{j} + \hat{k}), \mu \in \mathbb{R}$

के लंबवत् है। यदि ℓ तथा ℓ_1 का प्रतिच्छेदन बिंदु P है तथा P से ℓ_2 पर लंब का पाद Q(α, β, γ) है, तो $9(\alpha + \beta + \gamma)$ बराबर है।

[JEE(Main)April,2023]

Ans. 5

Sol. Since $\ell \perp \ell_1$ and $\ell \perp \ell_2$

Let dr's of ℓ be a, b, c

$\Rightarrow a + 2b + 3c = 0$

$2a + 2b + c = 0$

$\Rightarrow \frac{a}{-4} = \frac{b}{5} = \frac{c}{-2}$

hence dr's of ℓ can be 4, –5, 2

So any point P on ℓ can be taken as $(4\alpha, -5\alpha, 2\alpha)$

$\Rightarrow 4\alpha = \lambda + 1$ and $2\lambda - 11 = -5\alpha, 3\lambda - 7 = 2\alpha$

$\Rightarrow 2(3\lambda - 7) = \lambda + 1$

$\Rightarrow 5\lambda = 15 \Rightarrow \lambda = 3$

so P(4, –5, 2)

dr's of PQ : 2μ-5, 2μ+5, μ-1

$\because$ PQ $\perp \ell_2$

$= 2(2\mu - 5) + 2(2\mu + 5) + \mu - 1 = 0$

$\Rightarrow 9\mu = 1 \Rightarrow \mu = \frac{1}{9}$

$\Rightarrow Q\left(\frac{-7}{9}, \frac{2}{9}, \frac{10}{9}\right)$

hence $9(\alpha + \beta + \gamma) = 5$

84. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11$, $\vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$,

then $|\vec{a}\times\vec{c}|^2$ is equal to _____. **[JEE(Main)April,2023]**

माना $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}-\hat{k}$ हैं। यदि एक सदिश $\vec{c}$ के लिए $\vec{a}\cdot\vec{c}=11$, $\vec{b}\cdot(\vec{a}\times\vec{c})=27$ तथा $\vec{b}\cdot\vec{c}=-\sqrt{3}|\vec{b}|$ है, r ks $|\vec{a}\times\vec{c}|^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 285

Sol. $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=\hat{i}+\hat{j}-\hat{k}$

$\vec{b}.(\vec{a}\times\vec{c})=27$, $\vec{a}.\vec{b}=0$

$\vec{b}\times(\vec{a}\times\vec{c})=-3\vec{a}$

Let θ be angle between $\vec{b}$, $\vec{a}\times\vec{c}$

Then $|\vec{b}|.|\vec{a}\times\vec{c}|\sin\theta=3\sqrt{14}$

$|\vec{b}|.|\vec{a}\times\vec{c}|\cos\theta=27$

$\Rightarrow \sin\theta=\dfrac{\sqrt{14}}{\sqrt{95}}$

$\therefore |\vec{b}|.|\vec{a}\times\vec{c}|=3\sqrt{95}$

$\Rightarrow |\vec{a}\times\vec{c}|=\sqrt{3}\times\sqrt{95}$

85. Let $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a}=\vec{b}\times\vec{c}$ and $|\vec{b}|^2=50$, then $\left|72-|\vec{b}+\vec{c}|^2\right|$ is equal to _____. **[JEE(Main)April,2023]**

माना $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$ हैं। यदि एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}=\vec{b}\times\vec{c}$ तथा $|\vec{b}|^2=50$ हैं, तो $\left|72-|\vec{b}+\vec{c}|^2\right|$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 66

Sol. $|\vec{a}|=|\vec{b}\times\vec{c}|$

$=\sqrt{11}=|\vec{b}|.\sqrt{22}.\sin\theta$

$\therefore \sin\theta=\dfrac{1}{10}$

$\text{or}\cos\theta=\dfrac{\sqrt{99}}{10}$

$\left|72-|\vec{b}+\vec{c}|^2\right|=\left|72-\left(50+22+2\times5\sqrt{2}.\sqrt{22}\dfrac{\sqrt{99}}{10}\right)\right|$

$$= \left| 72 - \left(72 + \frac{2\times5\times11\times3}{10} \right) \right|$$

$=|66| = 66$

86. If the equation of the plane passing through the line of intersection of the planes $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to : **[JEE(Main)April,2023]**

यदि समतलों $2x - y + z = 3$, $4x - 3y + 5z + 9 = 0$ की प्रतिच्छेदन रेखा से होकर जाने वाले तथा रेखा $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ के समांतर समतल का समीकरण $ax + by + cz + 6 = 0$ है, तब $a + b + c$ बराबर है : **[JEE(Main)April,2023]**

(A) 13 (B) 12 (C) 15 (D) 14

Ans. D

Sol. **Equation of family of plane**

$(2x - y + z - 3) + \lambda(4x - 3y + 5z + 9) = 0$

$x(2 + 4\lambda) - y(1 + 3\lambda) + z(1 + 5\lambda) - 3 + 9\lambda = 0$

Parallel to the line

$-2(2 + 4\lambda) - (1 + 3\lambda)4 + (1 + 5\lambda)5 = 0$

$5\lambda = 3$

$\lambda = \frac{3}{5}$

equation of plane

$11x - 7y + 10z + 6 = 0$

$a + b + c = 14$

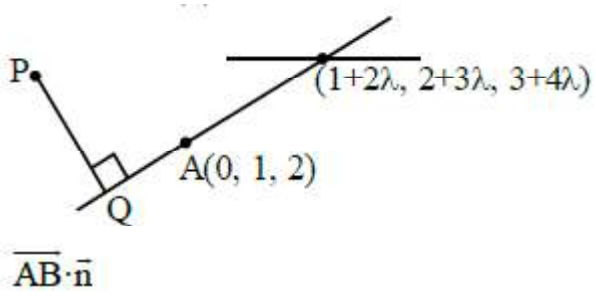
87. Let the line L pass through the point (0,1,2), intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point P(1, –9, 2) from the line L is : **[JEE(Main)April,2023]**

माना रेखा L, बिंदु (0,1,2) से होकर जाती है, रेखा $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ को काटती है तथा समतल $2x + y - 3z = 4$ समांतर gSa rsfca P(1, –9, 2) की रेखा L से दूरी है : **[JEE(Main)April,2023]**

(A) $\sqrt{74}$ (B) 9 (C) $\sqrt{69}$ (D) $\sqrt{54}$

Ans. A

Sol.



$\overline{AB}\cdot\vec{n}$

$\Rightarrow[(1+2\lambda)\hat{i}+(1+3\lambda)\hat{j}+(1+4\lambda)\hat{k}]\cdot(2\hat{i}+\hat{j}-3\hat{k})$

$2+4\lambda+1+3\lambda-3-12\lambda=0$

$5\lambda=0\Rightarrow\lambda=0$

Line $\overline{AB},\vec{r}=\hat{j}+2\hat{k}+\mu(\hat{i}+\hat{j}+\hat{k})$

General form : $Q(\mu,1+\mu,2+\mu)$

$\therefore\overrightarrow{PQ}\cdot\overrightarrow{AB}=0$

$(\mu-1)+(10+\mu)+\mu=0$

$3\mu=-9\Rightarrow\mu=-3$

$\therefore$ distance $=\sqrt{16+49+9}=\sqrt{74}$

88. The shortest distance between the lines $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ and $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ is :

[JEE(Main)April,2023]

रेखाओं $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ तथा $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) $3\sqrt{6}$ (B) $6\sqrt{3}$ (C) $6\sqrt{2}$ (D) $2\sqrt{6}$

Ans. A

Sol. $S_d=\left|\frac{(\vec{a}-\vec{b})\times(\vec{n}_1\times\vec{n}_2)}{|\vec{n}_1\times\vec{n}_2|}\right|$

$\bar{a}=(4,-2,-3)$

$\bar{b}=(1,3,4)$

$\bar{n}_1=(4,5,3)$

$\bar{n}_2=(3,4,2)$

$\bar{n}_1\times\bar{n}_2=\begin{vmatrix}i & j & k\\ 4 & 5 & 3\\ 3 & 4 & 2\end{vmatrix}=\hat{i}(-2)-\hat{j}(-1)+\hat{k}(1)=(-2,1,1)$

$$S_d = \frac{(3,-5,-7)\cdot(-2,1,1)}{\sqrt{6}} = \left|\frac{-6-5-7}{\sqrt{6}}\right| = 3\sqrt{6}$$

89. If the equation of the plane containing the line $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ is $ax + by + cz = 4$, then $(a - b + c)$ is equal to : **[JEE(Main)April,2023]**

यदि समतल, जिसमें रेखा $x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ स्थित है तथा जो समतल $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ के लंबवत् है, का समीकरण $ax + by + cz = 4$ है, तो $(a - b + c)$ बराबर है : **[JEE(Main)April,2023]**

(A) 18 (B) 22 (C) 20 (D) 24

Ans. B

Sol. Equation of plane containing line of intersection of plane $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ will be

$P_1 + \lambda P_2 = 0 \Rightarrow (1 + 2\lambda)x + (2 + \lambda)y + (3 - \lambda)z + (5\lambda - 4) = 0$ _____(1)

This plane is $\perp$ to plane

$\vec{r} = \left(\vec{i} - \vec{j}\right) + \lambda\left(\vec{i} + \vec{j} + \vec{k}\right) + \mu\left(\vec{i} - 2\vec{j} + 3\vec{k}\right)$ _____(2)

Normal of plane $(2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = 5\hat{i} - 2\hat{j} - 3\hat{k}$

Now both plane (1) and (2) $\perp$ Hence their normal also $\perp$

$5(1 + 2\lambda) - 2(2 + \lambda) - 3(3 - \lambda) = 0$

$-8 + 11\lambda = 0 \Rightarrow \lambda = \frac{8}{11}$

Now equation of plane $\frac{27}{11}x + \frac{30}{11}y + \frac{25}{11}z - \frac{4}{11} = 0$

$27x + 30y + 25z = 4$

Hence $a - b + c = 27 - 30 + 25 = 22$

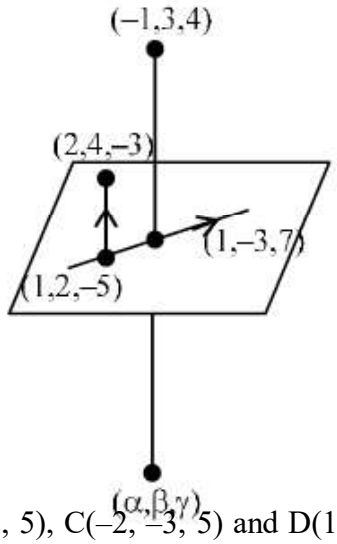
90. Let P be the plane passing through the line $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point(2, 4, –3). If the image of the point (–1, 3, 4) in the plane P is (α, β, γ) then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)April,2023]**

माना बिंदु (2, 4, –3) तथा रेखा $\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ से होकर जाने वाला समतल P है। यदि बिंदु (–1, 3, 4) का समतल P में प्रतिबिंब (α, β, γ) है, तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)April,2023]**

(A) 12 (B) 9 (C) 10 (D) 11

Ans. C

Sol. Equation of plane is given by



$$\begin{vmatrix} x-1 & y-2 & z+5 \\ 1 & 2 & 2 \\ 1 & -3 & 7 \end{vmatrix} = 0$$

$$4x - y - z = 7$$

$$\frac{\alpha+1}{4} = \frac{\beta-3}{-1} = \frac{\gamma-4}{-1} = \frac{-2(-4-3-4-7)}{16+1+1} = 2$$

$$\alpha = 7, \beta = 1, \gamma = 2$$

$$\alpha + \beta + \gamma = 10$$

91. The area of the quadrilateral ABCD with vertices A(2, 1, 1), B(1, 2, 5), C(–2, –3, 5) and D(1, –6, –7) is equal to : **[JEE(Main)April,2023]**

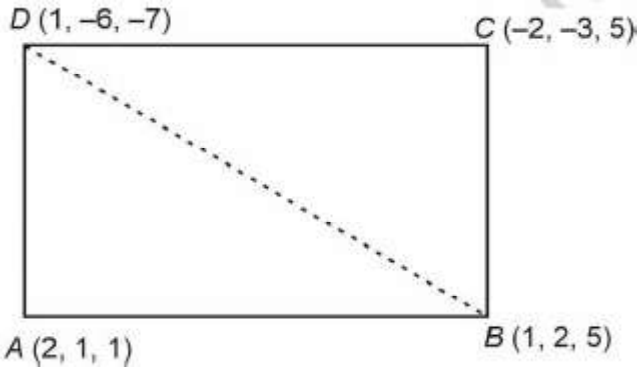
शीर्षों A(2, 1, 1), B(1, 2, 5), C(–2, –3, 5) तथा D(1, –6, –7) के चतुर्भुज ABCD का क्षेत्रफल है : **[JEE(Main)April,2023]**

(A) $8\sqrt{38}$ (B) 54 (C) $9\sqrt{38}$ (D) 48

Ans. A

Sol. $\overrightarrow{AB} = -\hat{i} + \hat{j} + 4\hat{k}$

$\overrightarrow{AD} = -\hat{i} - 7\hat{j} - 8\hat{k}$



∴ Area of ΔABD

$$= \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 4 \\ -1 & -7 & -8 \end{vmatrix}$$

$$= |10\hat{i} - 6\hat{j} + 4\hat{k}| = 2\sqrt{38}$$

and $\overrightarrow{CB} = 3\hat{i} + 5\hat{j}$ and $\overrightarrow{CD} = 3\hat{i} - 3\hat{j} - 12\hat{k}$

$$\therefore \text{ Area of } \Delta CBD = \frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -3 & -12 \\ 3 & 5 & 0 \end{vmatrix} |$$

$$= | 6(5\hat{i} - 3\hat{j} - 2\hat{k}) |$$

$$= 6\sqrt{38}$$

$\therefore$ Area of $\square ABCD = 8\sqrt{38}$ square units

92. For a, $b \in \mathbb{Z}$ and $|a-b| \le 10$, let the angle between the plane $P: ax + y - z = b$ and the line $l: x - 1 = a - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point (6, –6, 4) from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to : **[JEE(Main)April,2023]**

a, $b \in \mathbb{Z}$ तथा $|a-b| \le 10$ के लिए, माना समतल $P: ax + y - z = b$ तथा रेखा $l: x - 1 = a - y = z + 1$ के बीच का कोण $\cos^{-1}\left(\frac{1}{3}\right)$ है। यदि बिंदु (6, –6, 4) की समतल P से दूरी $3\sqrt{6}$ है, तो $a^4 + b^2$ बराबर है : **[JEE(Main)April,2023]**

(A) 25 (B) 85 (C) 48 (D) 32

Ans. D

Sol. $ax + y - z = b$

$$\frac{x-1}{1} = \frac{y-a}{-1} = \frac{z+1}{1}$$

$$\vec{n} = a\hat{i} + \hat{j} - \hat{k}$$

$$\vec{p} = \hat{i} - \hat{j} + \hat{k}$$

$$\sin\theta = \left|\frac{\vec{n}\cdot\vec{p}}{|\vec{n}|\vec{p}}\right|$$

$$\left|\frac{a-1-1}{\sqrt{a^2+2}\sqrt{3}}\right| = \frac{2\sqrt{2}}{3}$$

$$3(a-2)^2 = 8\times 3\left(a^2+2\right)$$

$$5a^2 + 12a + 4 = 0 \Rightarrow (5a+2)(a+2) = 0$$

$a = -2$ & $a = -2/5$ (reject)

$a = -2, \quad a - b \le 10$

$b \ge -8n$

Now distance of given point from the plane

$-2x + y - z = b$

$$\frac{12+6+4+b}{\sqrt{6}} = \pm 3\sqrt{6}$$

$b + 22 = \pm 18$

$b = -4$ and $b = -40$ (Reject)

$a^4 + b^2 = 16 + 16 = 32$

93. The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is :

[JEE(Main)April,2023]

रेखाओं $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ तथा $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ के बीच न्यूनतम दूरी है : **[JEE(Main)April,2023]**

(A) 6 (B) 7 (C) 9 (D) 8

Ans. C

Sol. $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$

$$\text{Shortest distance } (d) = \frac{\begin{Vmatrix} a_2 - a_1 & b_2 - b_1 & c_2 - c_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{Vmatrix}}{\begin{Vmatrix} \hat{i} & \hat{j} & k \\ l_1 & m_1 & n_1 \\ l_1 & m_2 & n_2 \end{Vmatrix}}$$

$$= \frac{\begin{Vmatrix} 4+2 & 1-0 & -3-5 \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{Vmatrix}}{\begin{Vmatrix} \hat{i} & \hat{j} & k \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{Vmatrix}}$$

$$= \frac{\begin{vmatrix} 6 & 1 & -8 \\ 1 & -2 & 2 \\ 1 & 2 & 0 \end{vmatrix}}{|\hat{i}(-4) - \hat{j}(-2) + k(2+2)|}$$

$$= \frac{|-54|}{|-4\hat{i} + 2\hat{j} + 4k|}$$

$$= \frac{54}{\sqrt{16+4+16}}$$

$$= \frac{54}{6}$$

$$= 9$$

94. Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q, then q and 2q are the roots of the equation :

[JEE(Main)April,2023]

माना रेखा $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ तथा समतल $x + y + z = 2$ का प्रतिच्छेदन बिंदु P है। यदि बिंदु P की समतल $3x - 4y + 12z = 32$ से दूरी q है, तो q तथा 2q किस समीकरण के मूल है ? **[JEE(Main)April,2023]**

(A) $x^2 - 18x - 72 = 0$ (B) $x^2 + 18x - 72 = 0$ (C) $x^2 + 18x + 72 = 0$ (D) $x^2 - 18x + 72 = 0$

Ans. D

Sol. $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2} = \lambda$

$A(3\lambda - 3, \lambda - 2, 1 - 2\lambda)$

$3\lambda - 3 + \lambda - 2 + 1 - 2\lambda = 2$

$2\lambda = 6$

$\lambda = 3$

$P(6, 1, -5)$

$$q = \left|\frac{18 - 4 - 60 - 32}{\sqrt{9+16+144}}\right| = \frac{78}{13} = 6$$

Equation with roots q and 2q is

$x^2 - 3qx + 2q^2 = 0$

$x^2 - 18x + 72 = 0$

95. Let two vertices of a triangle ABC be (2, 4, 6) and (0, –2, –5), and its centroid be (2, 1, –1). If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ), then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to :

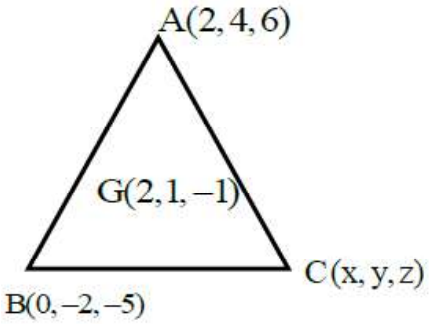
[JEE(Main)April,2023]

माना एक त्रिभुज ABC के दो शीर्ष (2, 4, 6) तथा (0, –2, –5) हैं तथा केन्द्रक (2, 1, –1) हैं यदि तीसरे शीर्ष का समतल $x + 2y + 4z = 11$ में प्रतिबिंब (α, β, γ) है, तो $\alpha\beta + \beta\gamma + \gamma\alpha$ बराबर है : **[JEE(Main)April,2023]**

(A) 70 (B) 76 (C) 74 (D) 72

Ans. C

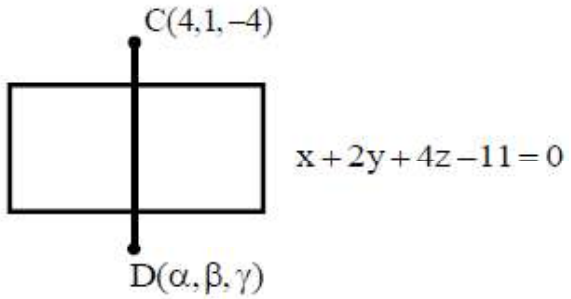
Sol.



Given Two vertices of Triangle A(2,4,6) and B(0, –2,–5) and if centroid G(2,1,–1)
Let Third vertices be (x, y,z)

Now $\frac{2+0+x}{3}=2, \frac{4-2+y}{3}=1, \frac{6-5+z}{3}=-1$

$x = 4, y = 1, z = -1$
Third vertices C(4,1,–4)
Now, Image of vertices C(4,1,–4) in the given plane is $D(\alpha,\beta,\gamma)$



Now

$$\frac{\alpha-4}{1}=\frac{\beta-1}{2}=\frac{\gamma+4}{4}=-2\frac{(4+2-16-11)}{1+4+16}$$

$$\frac{\alpha-4}{1}=\frac{\beta-1}{2}=\frac{\gamma+4}{4}=\frac{42}{21}\Rightarrow 2$$

$\alpha = 6, \beta = 5, \gamma = 4$

Then $\alpha\beta+\beta\gamma+\gamma\alpha$
$= (6 \times 5) + (5 \times 4) + (4 \times 6)$
$= 30 + 20 + 24$
$= 74$

96. Let the line $\frac{x}{1}=\frac{6-y}{2}=\frac{z+8}{5}$ intersect the lines $\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}$ and $\frac{x+3}{6}=\frac{3-y}{3}=\frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is : **[JEE(Main)April,2023]**

माना रेखा $\frac{x}{1}=\frac{6-y}{2}=\frac{z+8}{5}$ रेखाओं $\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}$ तथा $\frac{x+3}{6}=\frac{3-y}{3}=\frac{z-6}{1}$ को क्रमशः बिंदुओं A तथा B पर

काटती है। तो रेखाखंड AB के मध्य बिंदु की समतल $2x-2y+z=14$ से दूरी है : **[JEE(Main)April,2023]**

(A) $\frac{10}{3}$ (B) $\frac{11}{3}$ (C) 4 (D) 3

Ans. C

Sol. $\frac{x}{1}=\frac{y-6}{-2}=\frac{z+8}{5}=\lambda$ ....(1)

$\frac{x-5}{4}=\frac{y-7}{3}=\frac{z+2}{1}=\mu$ ....(2)

$\frac{x+3}{6}=\frac{y-3}{-3}=\frac{z-6}{1}=\gamma$ ....(3)

Intersection of (1) & (2) "A"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(4\mu+5,3\mu+7,\mu-2)$

$\lambda=1,\mu=-1$

$A(1,4,-3)$

Intersection (1) & (3) "B"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(4\mu+5,3\mu+7,\mu-2)$

$\lambda=1,\mu=-1$

$A(1,4,-3)$

Intersection (1) & (3) "B"

$(\lambda,-2\lambda+6,5\lambda-8)$ & $(6\gamma-3,-3\gamma+3,\gamma+6)$

$\lambda=3$

$\gamma=1$

$B(3,0,7)$

Mod point of A & B $\Rightarrow(2,2,2)$

Perpendicular distance from the plane

$2x-2y+z=14$

$\left|\frac{2(2)-2(2)+2-14}{\sqrt{4+4+1}}\right|=4$

97. Let the image of the point P(1, 2, 6) in the plane passing through the points A(1, 2, 0), B(1, 4, 1) and C(0, 5, 1) be Q(α, β, γ). Then $\alpha^2+\beta^2+\gamma^2$) is equal to : **[JEE(Main)April,2023]**

माना बिंदुओं A(1, 2, 0), B(1, 4, 1) तथा C(0, 5, 1) से होकर जाने वाले समतल में बिंदु P(1, 2, 6) का प्रतिबिंब Q(α, β, γ) है। r ks($\alpha^2+\beta^2+\gamma^2$) बराबर है : **[JEE(Main)April,2023]**

(A) 62 (B) 70 (C) 65 (D) 76

Ans. C

Sol. Equation of plane $A(x-1)+B(y-2)+C(z-0)=0$

Put $(1,4,1)\Rightarrow 2B+C=0$

Put $(0, 5, 1) \Rightarrow -A + 3B + C = 0$

Sub : $B - A = 0 \Rightarrow A = B, C = -2B$

$1(x - 1) + 1(y - 2) - 2(z - 0) = 0$

$x + y - 2z - 3 = 0$

Image is $(\alpha,\beta,\gamma)pt \equiv (1,2,6)$

$$\frac{\alpha-1}{1}=\frac{\beta-2}{1}=\frac{\gamma-6}{-2}=\frac{-2(1+2-12-3)}{6}$$

$$\frac{\alpha-1}{1}=\frac{\beta-2}{1}=\frac{\gamma-6}{-2}=4$$

$\alpha = 5, \beta = 6, \gamma = -2 \Rightarrow \alpha^2 + \beta^2 + \gamma^2$

$= 25 + 36 + 4 = 65$

98. Let (α, β, γ) be the image of the point P(2, 3, 5) in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to : **[JEE(Main)April,2023]**

माना बिंदु P(2, 3, 5) का समतल $2x + y - 3z = 6$ में प्रतिबिंब (α, β, γ) है। तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)April,2023]**

(A) 5 (B) 12 (C) 9 (D) 10

Ans. D

Sol. $$\frac{\alpha-2}{2}=\frac{\beta-3}{1}=\frac{\gamma-5}{-3}=-2\frac{(4+3-15-6)}{14}=2$$

$$\Rightarrow \frac{\alpha-2}{2}=2 \Rightarrow \alpha=6$$

$$\frac{\beta-3}{1}=2 \Rightarrow \beta=5$$

$$\frac{\gamma-5}{-3}=2 \Rightarrow \gamma=-1$$

$$\therefore \quad \alpha+\beta+\gamma=6+5-1=10$$

99. If equation of the plane that contains the point (–2, 3, 5) and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is $\alpha x + \beta y + \gamma z + 97 = 0$ then $\alpha + \beta + \gamma$: **[JEE(Main)April,2023]**

यदि समतल, जिसमें बिंदु (–2, 3, 5) स्थित है तथा जो दो समतलों $2x + 4y + 5z = 8$ तथा $3x - 2y + 3z = 5$ के लंबवत् है, का l ehdj.k $\alpha x + \beta y + \gamma z + 97 = 0$ है, तो $\alpha + \beta + \gamma =$ **[JEE(Main)April,2023]**

(A) 18 (B) 17 (C) 15 (D) 16

Ans. C

Sol. The equation of plane through (–2,3,5) is

$a(x + 2) + b(y - 3) + c(z - 5) = 0$

It is perpendicular to $2x + 4y + 5z = 8$ & $3x - 2y + 3z = 5$

$\therefore 2a+4b+5c=0$

$3a-2b+3c=0$

$$\therefore \frac{a}{\begin{vmatrix}4 & 5\\ -2 & 3\end{vmatrix}}=\frac{-b}{\begin{vmatrix}2 & 5\\ 3 & 3\end{vmatrix}}=\frac{c}{\begin{vmatrix}2 & 4\\ 3 & -2\end{vmatrix}}$$

$$\Rightarrow \frac{a}{22}=\frac{b}{9}=\frac{c}{-16}$$

$\therefore$ Equation of plane is

$22(x+2)+9(y-3)-16(z-5)=0$

$\Rightarrow 22x+9y-16z+97=0$

Comparing with $\alpha x+\beta y+\gamma x+97=0$

We get $\alpha+\beta+\gamma=22+9-16=15$

100. Let the line passing through the points P(2, –1, 2) and Q(5, 3, 4) meet the plane x – y + z = 4 at the point R. Then the distance of the point R from the plane x + 2y + 3z + 2 = 0 measured parallel to the line $\frac{x-7}{2}=\frac{y+3}{2}=\frac{z-2}{1}$ is equal to : **[JEE(Main)April,2023]**

माना बिंदुओं P(2, –1, 2) तथा Q(5, 3, 4) से होकर जाने वाली रेखा, समतल x – y + z = 4 को बिंदु R पर मिलती है। तो बिंदु R की समतल x + 2y + 3z + 2 = 0 से, रेखा $\frac{x-7}{2}=\frac{y+3}{2}=\frac{z-2}{1}$ के समांतर मापी गई, दूरी है :

[JEE(Main)April,2023]

(A) $\sqrt{89}$ (B) 3 (C) $\sqrt{31}$ (D) $\sqrt{61}$

Ans. B

Sol. Equation of line PQ.

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}=\lambda$$

Let R be $(3\lambda+2, 4\lambda-1, 2\lambda+2)$

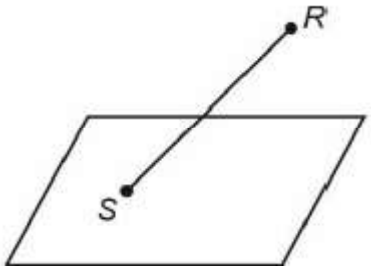
R lies on plane x – y + z = 4

$\therefore 3\lambda+2-4\lambda+1+2\lambda+2=4$

$\Rightarrow \lambda=-1$

$\therefore R(-1,-5,0)$

Let SR be : $\frac{x+1}{2}=\frac{y+5}{2}=\frac{z}{1}=k$



$S:(2k-1,2k-5,k)$

S lies on plane : $x + 2y + 3z + 2 = 0$

$\Rightarrow (2k-1)+(4k-10)+3k+2=0$

$\Rightarrow 9k-9=0 \Rightarrow k=1$

$S(1,-3,1) \quad \therefore SR=\sqrt{4+4+1}=3$

101. Let P be the plane passing through the points (5, 3, 0), (13, 3, –2) and (1, 6, 2). For $\alpha \in \mathbb{N}$, if the distances of the points A(3, 4, α) and B(2, α, a) from the plane P are 2 and 3 respectively, then the positive value of a is : **[JEE(Main)April,2023]**

माना समतल P, बिंदुओं (5, 3, 0), (13, 3, –2) तथा (1, 6, 2) से होकर जाता है। $\alpha \in \mathbb{N}$ के लिए यदि बिंदुओं A(3, 4, α) तथा B(2, α, a) की समतल P से दूरियाँ क्रमशः 2 तथा 3 है, तो a का धनात्मक मान है : **[JEE(Main)April,2023]**

(A) 5 (B) 3 (C) 4 (D) 6

Ans. C

Sol. $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 8 & 0 & -2 \\ 4 & -3 & -2 \end{vmatrix} = \hat{i}(-6)+8\hat{j}-24\hat{k}$

Normal of the plane $=3\hat{i}-4\hat{j}+12\hat{k}$

Plane : $3x - 4y + 12z = 3$

Distance from $A(3,4,\alpha)$

$$\left|\frac{9-16+12\alpha-3}{13}\right|=2$$

$\alpha = 3$

$\alpha = -8$ (rejected)

Distance from B(2,3,a)

$$\left|\frac{6-12+12a-3}{13}\right|=3$$

$a = 4$

102. Let the lines $l_1:\frac{x+5}{3}=\frac{y+4}{1}=\frac{z-\alpha}{-2}$ and $l_2:3x+2y+z-2=0=x-3y+2z-13$ be coplanar. If the point

P(a, b, c) on l_1 is nearest to the point Q(–4, –3, 2), then $|a|+|b|+|c|$ is equal to :

[JEE(Main)April,2023]

माना रेखाएँ $l_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ तथा $l_2 : 3x+2y+z-2=0=x-3y+2z-13$ सहतलीय हैं। यदि l_1 पर बिंदु P(a, b, c), बिंदु Q(–4, –3, 2) के निकटतम है, तो $|a|+|b|+|c|$ बराबर है : **[JEE(Main)April,2023]**

(A) 8 (B) 10 (C) 14 (D) 12

Ans. B

Sol. $(3x + 2y + z – 2) + \mu (x – 3y + 2z – 13) = 0$

$3(3 + \mu) + 1.(2 – 3\mu) – 2 (1+2\mu) = 0$

$9 – 4\mu = 0$

$\mu = \frac{9}{4}$

$4(–15 – 8 + \alpha – 2) + 9 (–5 + 12 + 2\alpha – 13) = 0$

$– 100 + 4\alpha – 54 + 18\alpha = 0$

$\Rightarrow \alpha = 7$

Let $P \equiv (3\lambda – 5, \lambda – 4, –2\lambda + 7)$

Directrion ratio of PQ $(3\lambda – 1, \lambda – 1, –2\lambda + 5)$

But $PQ \perp l_1$

$\Rightarrow 3(3\lambda-1)+1.(\lambda-1)-2(-2\lambda+5)=0$

$\Rightarrow \lambda = 1$

$P(-2,-3,5) \Rightarrow |a|+|b|+|c| = 10$

103. Let the plane P: $4x – y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y – z = 4$. If α is the distance of the point (2, 3, –4) from the new position of the plane P, then 35α is equal to : **[JEE(Main)April,2023]**

माना समतल P : $4x – y + z = 10$ को इसकी, समतल $x + y – z = 4$ से प्रतिच्छेदन रेखा के सापेक्ष $\frac{\pi}{2}$ के कोण तक घुमाया जाता gS; fn fcanq (2, 3, –4) की समतल P की नई स्थिति दूरी α है, तो 35α बराबर है : **[JEE(Main)April,2023]**

(A) 105 (B) 85 (C) 90 (D) 126

Ans. D

Sol. Let equation in new position is

$(4x – y + z – 10) + \lambda(x + y – z – 4) = 0$

$4(4 + \lambda) – 1.(–1 + \lambda) + 1.(1 – \lambda) = 0$

$\Rightarrow \lambda = – 9$

So equation in new position is

$–5x – 10y + 10z + 26 = 0$

$\Rightarrow \alpha = \frac{54}{15}$

104. Let the equation of plane passing through the line of intersection of the planes x + 2y + az = 2 and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point (a, –c, c) is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to : **[JEE(Main)April,2023]**

माना समतलों x + 2y + az = 2 तथा x – y + z = 3 की प्रतिच्छेदन रेखा से होकर जाने वाले समतल का समीकरण $5x - 11y + bz = 6a - 1$ है। $c \in \mathbb{Z}$ के लिए, यदि इस समतल की बिन्दु (a, –c, c) से दूरी $\frac{2}{\sqrt{a}}$ है, तो $\frac{a+b}{c}$ बराबर है : **[JEE(Main)April,2023]**

(A) –4 (B) –2 (C) 4 (D) 2

Ans. A

Sol. $(x + 2y + az - 2) + \lambda (x - y + z - 3) = 0$

$$\frac{1+\lambda}{5} = \frac{2-\lambda}{-11} = \frac{a+\lambda}{b} = \frac{2+3\lambda}{6a-1}$$

$$\lambda = -\frac{7}{2}, a = 3, b = 1$$

$$\frac{2}{\sqrt{a}} = \left|\frac{5a + 11c + bc - 6a + 1}{\sqrt{25+121+1}}\right|$$

$c = -1$

$$\therefore \frac{a+b}{c} = \frac{3+1}{-1} = -4$$

105. The plane, passing through the points (0, –1, 2) and (–1, 2, 1) and parallel to the line passing through (5, 1, –7) and (1, –1, –1), also passes through the point : **[JEE(Main)April,2023]**

बिन्दुओं (5, 1, –7) तथा (1, –1, –1) से होकर जाने वाली रेखा के समांतर तथा बिन्दुओं (0, –1, 2) तथा (–1, 2, 1) से होकर t kusokyk l er y] fuEu esa l sfdl fcUnq sgksdj t kr kgS8 **[JEE(Main)April,2023]**

(A) (0, 5, –1) (B) (–2, 5, 0) (C) (2, 0, 1) (D) (1, –2, 1)

Ans. B

Sol. Plane passing through (0, –1, 0) and (–1, 2, 1)

Then vector in plane $\langle -1, 3, -1 \rangle$ vector parallel to plane is $\langle 4, 2, -6 \rangle$

Normal vector to plane $(\vec{n}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 3 & -1 \\ 4 & 2 & -6 \end{vmatrix}$

$= \hat{i}(16) - \hat{j}(10) + \hat{k}(-14)$

$\vec{n} = \langle 8, 5, 7 \rangle$

Equation of plane

$8(x - 0) + 5(y + 1) + 7(z - 2) = 0$

$\Rightarrow 8x + 5y + 7z = 9$

From given options point (–2, 5, 0) lies on plane.

106. Let N be the foot of perpendicular from the point P(1, –2, 3) on the line passing through the points (4, 5, 8) and (1, –7, 5). Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is :

[JEE(Main)April,2023]

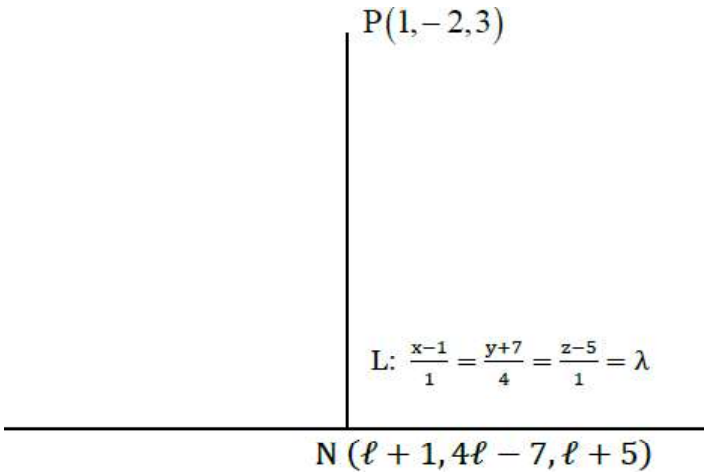
माना बिन्दुओं (4, 5, 8) तथा (1, –7, 5) से होकर जाने वाली रेखा पर बिन्दु P(1, –2, 3) से डाले गए लंब का पाद N है। तो समतल $2x - 2y + z + 5 = 0$ से N की दूरी है :

[JEE(Main)April,2023]

(A) 7 (B) 8 (C) 9 (D) 6

Ans. A

Sol.



$\overline{PN} = \langle \lambda, 4\lambda - 5, \lambda + 2 \rangle$

$\overline{PN} \cdot \langle 1, 4, 1 \rangle = 0$

$\Rightarrow \lambda + 16\lambda - 20 + \lambda + 2 = 0$

$\Rightarrow \lambda = 1$

N (2, –3, 6)

Distance of N from $2x - 2y + z + 5 = 0$

$$d = \left|\frac{2(2)-2(-3)+6+5}{\sqrt{2^2+(-2)^2+(1)^2}}\right|$$

$$= \left|\frac{21}{3}\right| = 7$$

107. The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is : **[JEE(Main)April,2023]**

रेखा $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, के सहतलीय रेखा का समीकरण है : **[JEE(Main)April,2023]**

(A) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$ (B) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$

(C) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ (D) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$

Ans. C

Sol. Check by option

$$\text{option 3} \Rightarrow \begin{vmatrix} -3+1 & 1-2 & 5-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = \begin{vmatrix} -2 & -1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & -1 & 0 \\ -2 & -1 & 0 \\ -1 & 2 & 5 \end{vmatrix} = 0$$

$\therefore$ option (3) is correct.

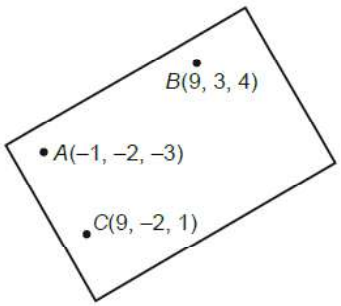
108. Let the foot of perpendicular of the point P(3, –2, –9) on the plane passing through the points (–1, –2, –3), (9, 3, 4), (9, –2, 1) be Q (α, β, γ). Then the distance of Q from the origin is : **[JEE(Main)April,2023]**

यदि बिन्दु P(3, –2, –9) का बिन्दुओं (–1, –2, –3), (9, 3, 4), (9, –2, 1) से गुजरने वाले समतल पर लम्ब पाद Q(α, β, γ) है, rlsQ की मूल बिन्दु से दूरी होगी : **[JEE(Main)April,2023]**

(A) $\sqrt{42}$ (B) $\sqrt{38}$ (C) $\sqrt{29}$ (D) $\sqrt{35}$

Ans. A

Sol.



$\overrightarrow{AC} = 10i + 4k$

$\overrightarrow{AB} = 10i + 5j + 7k$

$$\overrightarrow{AC} \times \overrightarrow{AB} = \begin{vmatrix} i & j & k \\ 10 & 0 & 4 \\ 10 & 5 & 7 \end{vmatrix}$$

$= -20i - 30j + 50k$

Equation of plane

$2x + 3y - 5z = d$

Put $(-1,-2,-3)$

$-2 - 6 + 15 = d$

$d = 7$

$\therefore\ 2x + 3y - 5z = 7$

Foot of $\perp r$

$$\frac{x-3}{2} = \frac{y+2}{3} = \frac{z+9}{-5} = -\left(\frac{38}{38}\right)$$

$x = 1, y = -5, z = -4$

$Q(1, -5, -4)$

Distance from origin $= \sqrt{1+25+16}$

$= \sqrt{42}$

109. Let S be the set of all values of λ, for which the shortest distance between the lines $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda \in S} \lambda\right|$ is equal to : **[JEE(Main)April,2023]**

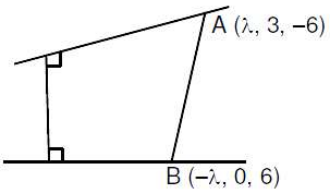
यदि λ के सभी मानों का समुच्चय S है, जिसके लिए रेखाओं $\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ तथा $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ के मध्य की Ur e nyh 13 है तो $8\left|\sum_{\lambda \in S} \lambda\right|$ का मान होगा : **[JEE(Main)April,2023]**

(A) 308 (B) 304 (C) 302 (D) 306

Ans. D

Sol. Given lines : $\frac{x-\lambda}{0}=\frac{y-3}{4}=\frac{z+6}{1}$ ...(i)

$\frac{x+\lambda}{3}=\frac{y}{-4}=\frac{z-6}{0}$ ...(ii)



DR of $\overrightarrow{AB}$: $2\lambda, 3, -12$

$\vec{p}\times\vec{q}$ is perpendicular both the lines

$$\vec{p}\times\vec{q}=\begin{vmatrix} i & j & k \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{vmatrix}$$

SD = 13

$$\left|\frac{\overrightarrow{AB}\cdot(\vec{p}\times\vec{q})}{\vec{p}\times\vec{q}}\right|=13$$

$$\Rightarrow\left|\frac{8\lambda+153}{13}\right|=13$$

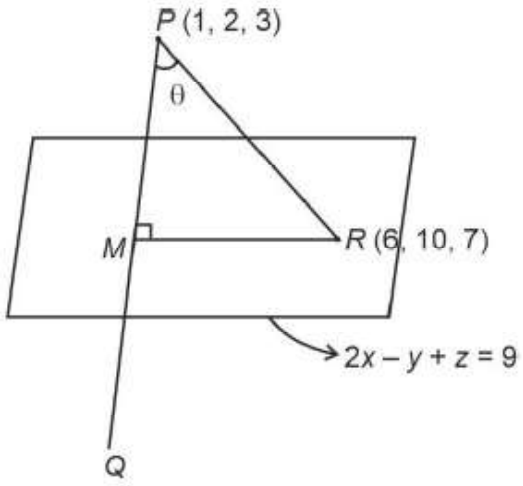
Now, $8\left|\sum_{\lambda\in S}\lambda\right|=|16-322|=306$

110. Let the image of the point P(1, 2, 3) in the plane 2x – y + z = 9 be Q. If the coordinates of the point R are (6, 10, 7), then the square of the area of the triangle PQR is _____. **[JEE(Main)April,2023]**

माना समतल 2x – y + z = 9 में बिंदु P(1, 2, 3) का प्रतिबिम्ब Q है। यदि बिंदु R के निर्देशांक (6, 10, 7) है, तब त्रिभुज PQR के {ksk Qy d kox Zgs _____ **[JEE(Main)April,2023]**

Ans. 594

Sol.



R lies on plane

$$PR = \sqrt{5^2 + 8^2 + 4^2} = \sqrt{105}$$

$$\cos\theta = \frac{(5\hat{i} + 8\hat{j} + 4\hat{k})(2\hat{i} - \hat{j} + \hat{k})}{\sqrt{105}\sqrt{6}}$$

$$= \frac{6}{\sqrt{630}}$$

$$\text{Area } (\Delta PQR) = 2\,\text{area}(\Delta PMR)$$

$$= 2 \cdot \frac{1}{2}(PR)^2 \sin\theta\cos\theta$$

$$= 105 \cdot \frac{6}{\sqrt{630}} \cdot \frac{\sqrt{594}}{\sqrt{630}}$$

$$= \sqrt{594}$$

111. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ and $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____. **[JEE(Main)April,2023]**

यदि रेखाएँ $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$ तथा $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ एक दूसरे को काटती है, तो $8\alpha\beta$ के न्यूनतम मान का परिमाण है _____ **[JEE(Main)April,2023]**

Ans. 18

Sol. If the lines $\frac{x-1}{2} = \frac{2-y}{-3} = \frac{z-3}{\alpha}$

And $\frac{x-4}{5} = \frac{y-1}{2} = \frac{z}{\beta}$ intersect

Point on first line (1, 2, 3) and point on second line (4, 1, 0).

Vector joining both points is $-3\hat{i} + \hat{j} + 3\hat{k}$

Now vector along first line is $2\hat{i} + 3\hat{j} + \alpha\hat{k}$

Also vector along second line is $5\hat{i} + 2\hat{j} + \beta\hat{k}$

Now these three vectors must be coplanar

$$\Rightarrow \begin{vmatrix} 2 & 3 & \alpha \\ 5 & 2 & \beta \\ -3 & 1 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 2(6-\beta) - 3(15+3\beta) + \alpha(11) = 0$$

$$\Rightarrow \alpha - \beta = 3$$

Now $\alpha = 3 + \beta$

Given expression $8(3+\beta)\cdot\beta = 8\left(\beta^2+3\beta\right)$

$=8\left(\beta^2+3\beta+\frac{9}{4}-\frac{9}{4}\right)=8\left(\beta+\frac{3}{2}\right)^2-18$

So magnitude of minimum value = 18

112. Let λ_1, λ_2 be the values of λ for which the points $\left(\frac{5}{2},1,\lambda\right)$ and (–2, 0, 1) are at equal distance from the plane $2x+3y-6z+7=0$. If $\lambda_1>\lambda_2$, then the distance of the point $(\lambda_1-\lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1}=\frac{y-1}{2}=\frac{z+7}{2}$ is _____. **[JEE(Main)April,2023]**

माना λ के मान, जिनके लिए बिंदु $\left(\frac{5}{2},1,\lambda\right)$ तथा (–2, 0, 1) समतल $2x+3y-6z+7=0$ से समान दूरी पर है, λ_1, λ_2 है। यदि $\lambda_1>\lambda_2$ है, तो बिंदु $(\lambda_1-\lambda_2, \lambda_2, \lambda_1)$ की रेखा $\frac{x-5}{1}=\frac{y-1}{2}=\frac{z+7}{2}$ से दूरी है _____

[JEE(Main)April,2023]

Ans. 9

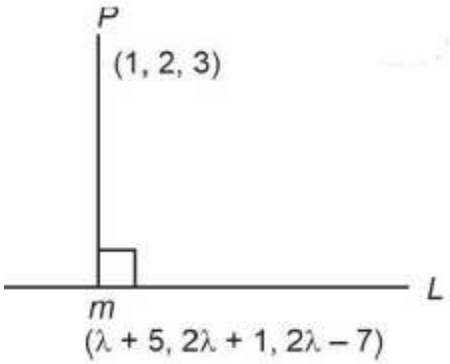
Sol. $\left|\frac{-4+0-6+7}{7}\right|=\left|\frac{15-6\lambda}{7}\right|$

$\frac{3}{7}=\left|\frac{15-6\lambda}{7}\right|$

$\lambda=2$ or 3

$\lambda_1=3, \lambda_2=2$

$\left(\lambda_1-\lambda_2,\lambda_2,\lambda_1\right)=(1,2,3)$



$\overrightarrow{PM}\cdot(\hat{i}+2\hat{j}+2k)=0$

$(\lambda+4)+2(2\lambda-1)+2(2\lambda-10)=0$

$\Rightarrow 9\lambda=18$ or $\lambda=2$

Distance $=\sqrt{6^2+3^2+6^2}=9$

113. Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0)$, $(2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the planes P_1 and P_2 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)April,2023]**

माना P_1 समतल $3x - y - 7z = 11$ है तथा बिंदुओं $(2, -1, 0)$, $(2, 0, -1)$ तथा $(5, 1, 1)$ से होकर जाने वाला समतल P_2 है। यदि बिंदु $(7, 4, -1)$ से समतलों P_1 तथा P_2 की प्रतिच्छेदन रेखा पर डाले गए लंब का पाद (α, β, γ) है, तो $\alpha + \beta + \gamma$ बराबर है : **[JEE(Main)April,2023]**

Ans. 11

Sol. $P_1 : 3x - y - 7z = 11$

$$P_2 : \begin{vmatrix} x-2 & y+1 & z \\ 0 & 1 & -1 \\ 3 & +2 & 1 \end{vmatrix} = 0$$

$3(x-2) - (y+1)(+3) - z(+3) = 0$

$P_2 : x - y - z = 3$

$\vec{P} = \vec{n_1} \times \vec{n_2}$

$\vec{n_1} = 3\hat{i} - \hat{j} - 7\hat{k}$

$\vec{n_2} = \hat{i} - \hat{j} - \hat{k}$

$\vec{n_1} \times \vec{n_2} = -6\hat{i} - 4\hat{j} - 2\hat{k}$

$\vec{P} = 3\hat{i} + 2\hat{j} + \hat{k}$ common point on the planes

Put $z = 0$

$3x - y = 11$

$x - y = 3$

$x = 4, y = 1, z = 0$

Line is $\frac{x-4}{3} = \frac{y-1}{2} = \frac{z-0}{1}$

Let point on this line $\equiv (3\lambda + y, 2\lambda + 1, \lambda) \equiv N$

given point is $A \equiv (7, 4, -1)$

$\overrightarrow{AN} = (3\lambda - 3)\hat{i} + (2\lambda - 3)\hat{j} + (\lambda + 1) = 0$

$\overrightarrow{AN} \cdot \vec{P} = 0$

$3(3\lambda - 3) + 2(2\lambda - 3) + 1(\lambda + 1) = 0$

$14\lambda - 14 = 0$ $\quad N \equiv (7, 3, 1)$

$\lambda = 1$ $\quad (\alpha, \beta, \gamma)$

$\alpha + \beta + \gamma = 11$

114. Let the foot of perpendicular from the point A(4, 3, 1) on the plane P : x – y + 2z + 3 = 0 be N. If B(5, α, β), α, β ∈ Z is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2+\beta^2+\alpha\beta$ is equal to _____. **[JEE(Main)April,2023]**

माना बिंदु A(4, 3, 1) से समतल P : x – y + 2z + 3 = 0 पर डाला गया लंबपाद N है। यदि समतल P पर एक बिंदु B(5, α, β), α, β ∈ Z इस प्रकार है कि त्रिभुज ABN का क्षेत्रफल $3\sqrt{2}$ है, तो $\alpha^2+\beta^2+\alpha\beta$ बराबर है :

[JEE(Main)April,2023]

Ans. 7

Sol. $5-\alpha+2\beta+3=0$

$\alpha-2\beta=8$

$\frac{x_1-4}{1}=\frac{y_1-3}{-1}=\frac{z_1-1}{2}=\frac{-(6)}{6}$

$N\equiv(x_1,y_1,z_1)=(3,4,-1)$

$B\equiv(5,\alpha,\beta)$

$\frac{1}{2}AN\times BN=3\sqrt{2}$

$AN\cdot BN=6\sqrt{2}$

$\frac{6}{\sqrt{6}}\cdot BN=6\sqrt{2}$

$BN=2\sqrt{3}$

$4+(\alpha-4)^2+(\beta+1)^2=12$

$(\alpha-4)^2+(\beta+1)^2=8$

$(2\beta+4)^2+(\beta+1)^2=8$

$5\beta^2+18\beta+9=0$

$\beta=-3,-\frac{3}{5}$

$\beta=-3,\alpha=2$

$\alpha^2+\beta^2+\alpha\beta=9+4-6=7$

115. Let the line $\ell: x=\frac{1-y}{-2}=\frac{z-3}{\lambda}, \lambda\in\mathbb{R}$ meet the plane P : x + 2y + 3z = 4 at the point (α, β, γ). If the angle between the line *l* and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha+2\beta+6\gamma$ is equal to _____.

[JEE(Main)April,2023]

माना रेखा $\ell : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ समतल $P : x + 2y + 3z = 4$ को बिंदु (α, β, γ) पर मिलती है। यदि रेखा *l* तथा समतल P के बीच का कोण $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$ है, तो $\alpha + 2\beta + 6\gamma$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 11

Sol. $\ell : x = \frac{y-1}{2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$

Dr's of line $\ell(1, 2, \lambda)$

Dr's of normal vector of plane $P : x + 2y + 3z = 4$ are (1, 2, 3)

Now, angle between line ℓ and plane P is given by

$$\sin\theta = \left|\frac{1+4+3\lambda}{\sqrt{5+\lambda^2}\cdot\sqrt{14}}\right| = \frac{3}{\sqrt{14}}\left(\text{ given } \cos\theta = \sqrt{\frac{5}{14}}\right)$$

$$\Rightarrow \lambda = \frac{2}{3}$$

Let variable point of line ℓ is $\left(t, 2t+1, \frac{2}{3}t+3\right)$

line of plane P.

$$\Rightarrow t = -1$$

$$\Rightarrow \left(-1, -1, \frac{7}{3}\right) \equiv (\alpha, \beta, \gamma)$$

$$\Rightarrow \alpha + 2\beta + 6\gamma = 11$$

116. Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocenter of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98(\alpha + \beta)^2$ is equal to _____. **[JEE(Main)April,2023]**

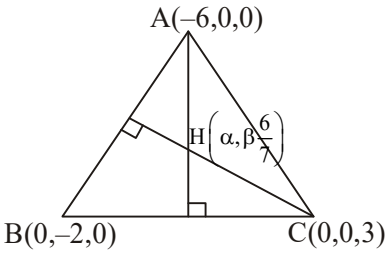
माना समतल $x + 3y - 2z + 6 = 0$ निर्देशांक अक्षों को बिंदुओं A, B, C पर मिलता है। यदि त्रिभुज ABC का लंबकेन्द्र $\left(\alpha, \beta, \frac{6}{7}\right)$ है, तो $98(\alpha + \beta)^2$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 288

Sol. A(–6, 0, 0) B(0, –2, 0) C = (0, 0, 3)

$\overrightarrow{AB} = 6\hat{i} - 2\hat{j}, \overrightarrow{BC} = 2\hat{j} + 3\hat{k}$,

$\overrightarrow{AC} = 6\hat{i} + 3\hat{k}$



$\overrightarrow{AH}\cdot\overrightarrow{BC}=0$

$\left(\alpha+6,\beta,\frac{6}{7}\right).(0,2,3)=0$

$\beta=\frac{-9}{7}$

$\overrightarrow{CH}\cdot\overrightarrow{AB}=0$

$\left(\alpha,\beta,\frac{-15}{7}\right).(6,-2,0)=0$

$6\alpha-2\beta=0$

$\alpha=\frac{-3}{7}$

$98(\alpha+\beta)^2=(98)\frac{(144)}{49}=288$

117. Let the image of the point $\left(\frac{5}{3},\frac{5}{3},\frac{8}{3}\right)$ in the plane x – 2y + z – 2 = 0 be P. If the distance of the point $Q(6,-2,\alpha),\alpha>0$, from P is 13, then α is equal to _____. **[JEE(Main)April,2023]**

माना समतल x – 2y + z – 2 = 0 में बिन्दु $\left(\frac{5}{3},\frac{5}{3},\frac{8}{3}\right)$ का प्रतिबिंब बिन्दु P है। यदि बिन्दु $Q(6,-2,\alpha),\alpha>0$ की बिन्दु P से 13 है, तो α बराबर है _____. **[JEE(Main)April,2023]**

Ans. 15

Sol. $\frac{x-\frac{5}{3}}{1}=\frac{y-\frac{5}{3}}{-2}=\frac{2-\frac{8}{3}}{1}=\frac{-2(-1)}{6}$

(x, y, z) = (2, 1, 3)

PQ = 13

$4^2+3^2+(\alpha-3)^2=169$

$|\alpha-3|=12$

$\alpha=15$

118. Let the plane P contain the line 2x + y – z – 3 = 0 = 5x –3y + 4z + 9 and be parallel to the line

$\frac{x+2}{2}=\frac{3-y}{-4}=\frac{z-7}{5}$. Then the distance of the point A(8, –1, –19) from the plane P measured parallel to the line $\frac{x}{-3}=\frac{y-5}{4}=\frac{2-z}{-12}$ is equal to _____. **[JEE(Main)April,2023]**

यदि समतल P रेखा $2x + y – z – 3 = 0 = 5x –3y + 4z + 9$ को रखता है तथा रेखा $\frac{x+2}{2}=\frac{3-y}{-4}=\frac{z-7}{5}$ के समान्तर है, तो fcUnq A(8, –1, –19) की रेखा $\frac{x}{-3}=\frac{y-5}{4}=\frac{2-z}{-12}$ के अनुदिश समतल P से दूरी होगी _____

[JEE(Main)April,2023]

Ans. 26

Sol. Plane $\equiv P_1 + \lambda P_2 = 0$

$(2x + y – z – 3) + \lambda (5x – 3y + 4z + 9) = 0$

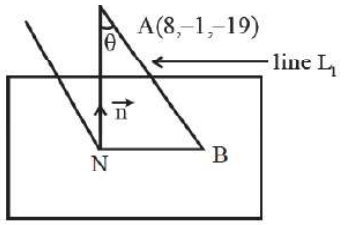
$(5\lambda + 2) x + (1 – 3\lambda) y + (4\lambda – 1) z + 9\lambda – 3 = 0$

$\vec{n}\cdot\vec{b} = 0$ where $\vec{b}(2,4,5)$

$2 (5\lambda + 2) + 4 (1 – 3\lambda) + 5 (4\lambda – 1) = 0$

$\lambda = -\frac{1}{6}$

Plane $7x + 9y – 10z – 27 = 0$



Equation of line AB is

$\frac{x-8}{-3}=\frac{y+1}{4}=\frac{z+19}{12}=\lambda$

Let B = $(8 – 3\lambda, –1 + 4\lambda, –19 + 12\lambda)$ lies on plane P

$\therefore 7 (8 – 3\lambda) + 9 (4\lambda – 1) – 10 (12\lambda – 19) = 27$

$\lambda = 2$

$\therefore$ Point B = (2,7,5)

$AB = \sqrt{6^2 + 8^2 + 24^2} = 26$

119. If the line x = y = z intersects the line $x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$, where A, B, C are the angles of a triangle ABC, then $80\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$ is equal to _____. **[JEE(Main)April,2023]**

यदि रेखा $x = y = z$ रेखाओं

$x\sin A + y\sin B + z\sin C - 18 = 0 = x\sin 2A + y\sin 2B + z\sin 2C - 9$ को प्रतिच्छेद करती है जहाँ A, B, C त्रिभुज ABC के कोण है, तो $80\left(\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$ का मान होगा _____ **[JEE(Main)April,2023]**

Ans. 5

Sol. $x = y = z = k$ (let

$\therefore k(\sin A + \sin B + \sin C) = 18$

$$\Rightarrow k\left(4\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2}\right) = 18 \quad ...(i)$$

$k(\sin 2A + \sin 2B + \sin 2C) = 9$

$$\Rightarrow k\left(4\sin A\cdot\sin B\cdot\sin C\right) = 9 \quad ...(ii)$$

(ii)/(i)

$$8\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2} = \frac{9}{18}$$

$$\Rightarrow 80\sin\frac{A}{2}\cdot\sin\frac{B}{2}\cdot\sin\frac{C}{2} = 5$$

120. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\left(\hat{i} - \hat{j} + \hat{k}\right)$. Let $\vec{c}$ be the vector such that $\vec{a}\times\vec{c} = \vec{b}$ and $\vec{a}\cdot\vec{c} = 3$. Then $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right) - \vec{b} - \vec{c}\right)$ is equal to : **[JEE(Main)Jan.2024]**

माना $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3\left(\hat{i} - \hat{j} + \hat{k}\right)$ है। माना एक सदिश $\vec{c}$ के लिए $\vec{a}\times\vec{c} = \vec{b}$ तथा $\vec{a}\cdot\vec{c} = 3$ हैं। तो $\vec{a}.\left(\left(\vec{c}\times\vec{b}\right) - \vec{b} - \vec{c}\right)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 36 (B) 24 (C) 32 (D) 20

Ans. B

Sol. $(\vec{a}\times\vec{c}).\vec{b} = \vec{b}\cdot\vec{b}$

$\left[\vec{a}\,\vec{c}\,\vec{b}\right] = 9(1+1+1) = 27$

$\vec{a}\cdot\vec{b} = 3(1-2+1) = 0$

$\vec{a}\cdot\vec{c} = 3$

$\vec{a}\cdot((\vec{c}\times\vec{b}) - \vec{b} - \vec{c}) = [\vec{a}\vec{c}\vec{b}] - \vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{c}$

$= 27 - 0 - 3 = 24$

121. Let the position vectors of the vertices A, B and C of a triangle be $2\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} + 2\hat{k}$ and $2\hat{i} + \hat{j} + 2\hat{k}$ respectively. Let ℓ_1, ℓ_2 and ℓ_3 be be the lengths of perpendiculars drawn from the ortho center of the triangle on the sides AB, BC and CA respectively, then $\ell_1^2 + \ell_2^2 + \ell_3^2$ equal : **[JEE(Main)Jan.2024]**

माना एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}+2\hat{k}$ तथा $2\hat{i}+\hat{j}+2\hat{k}$ हैं। माना इस त्रिभुज d sy Ec d sUnz sHkqtkvksa AB, BC तथा CA पर डाले गए लम्बों की लम्बाईयाँ क्रमशः ℓ_1, ℓ_2 तथा ℓ_3 हैं, तो $\ell_1^2+\ell_2^2+\ell_3^2$ बराबर है:

[JEE(Main)Jan.2024]

(A) $\frac{1}{5}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{3}$

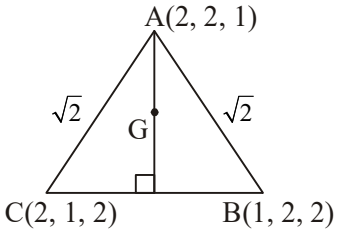
Ans. B

Sol. Equilateral Δ

$$\sqrt{2-\frac{1}{2}}=\sqrt{\frac{3}{2}}$$

$$\frac{1}{3}\times\sqrt{\frac{3}{2}}=\frac{1}{\sqrt{6}}$$

$$\ell_1^2+\ell_2^2+\ell_3^2=3\ell_1^2=3\times\frac{1}{6}=\frac{1}{2}$$



122. The position vectors of the vertices A, B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let *l* denotes the length of the angle bisector AD of ∠BAC where D is on the line segment BC, then $2l^2$ equals : **[JEE(Main)Jan.2024]**

एक त्रिभुज के शीर्षों A, B तथा C के स्थिति सदिश क्रमशः $2\hat{i}-3\hat{j}+3\hat{k}$, $2\hat{i}+2\hat{j}+3\hat{k}$ तथा $-\hat{i}+\hat{j}+3\hat{k}$ हैं। माना कोण ∠BAC के समद्विभाजक AD की लम्बाई *l* है, जहाँ बिंदु D, रेखाखण्ड BC पर है, तो $2l^2$ बराबर है : **[JEE(Main)Jan.2024]**

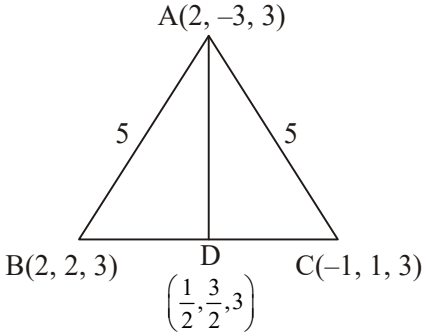
(A) 49 (B) 42 (C) 50 (D) 45

Ans. D

Sol. $\ell^2=\left(2-\frac{1}{2}\right)^2+\left(3+\frac{3}{2}\right)^2+0$

$$\ell^2=\frac{9}{4}+\frac{81}{4}$$

$$\ell^2=\frac{45}{2} \qquad 2\ell^2=45$$



123. Let O be the origin and the position vectors of A and B be $2\hat{i}+2\hat{j}+\hat{k}$ and $2\hat{i}+4\hat{j}+4\hat{k}$ respectively. If the internal bisector of ∠AOB meets the line AB at C, then the length of OC is :

[JEE(Main)Jan.2024]

माना O मूलबिन्दु है तथा A और B के स्थिति सदिश क्रमशः $2\hat{i}+2\hat{j}+\hat{k}$ और $2\hat{i}+4\hat{j}+4\hat{k}$ हैं। यदि ∠AOB की अंतः समद्विभाजक

रेखा AB को C पर मिलती है, तो OC की लंबाई है : **[JEE(Main)Jan.2024]**

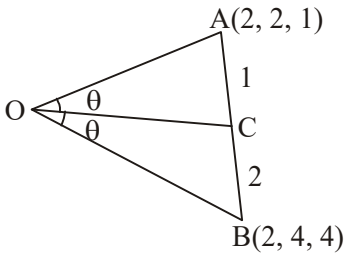
(A) $\frac{2}{3}\sqrt{34}$ (B) $\frac{3}{2}\sqrt{31}$ (C) $\frac{2}{3}\sqrt{31}$ (D) $\frac{3}{2}\sqrt{34}$

Ans. A

Sol. $\left|\overrightarrow{OA}\right| = \sqrt{4+4+1} = 3$

$\left|\overrightarrow{OB}\right| = \sqrt{4+16+16} = 6$

AC : CB = 3 : 6 = 1 : 2



$c\left(\frac{4+2}{3}, \frac{4+4}{3}, \frac{2+4}{3}\right)$

$c\left(2, \frac{8}{3}, 2\right)$

$\left|\overrightarrow{OC}\right| = \sqrt{4+\frac{64}{9}+4} = \frac{\sqrt{136}}{3} = \frac{2}{3}\sqrt{34}$

124. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors such that $\vec{b}$ and $\vec{c}$ are non-collinear. If $\vec{a}+5\vec{b}$ is collinear with $\vec{c}, \vec{b}+6\vec{c}$, is collinear with $\vec{a}$ and $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\vec{0}$ then $\alpha+\beta$ is equal to : **[JEE(Main)Jan.2024]**

माना $\vec{a}, \vec{b}$ तथा $\vec{c}$ तीन शून्येत्तर सदिश हैं तथा $\vec{b}$ और $\vec{c}$ संरेख नहीं हैं। यदि $\vec{a}+5\vec{b}$ और $\vec{c}$ संरेख हैं, $\vec{b}+6\vec{c}$ और $\vec{a}$ संरेख gS rFkk $\vec{a}+\alpha\vec{b}+\beta\vec{c}=\vec{0}$ है, तो $\alpha+\beta$ बराबर है : **[JEE(Main)Jan.2024]**

(A) – 25 (B) – 30 (C) 35 (D) 30

Ans. C

Sol. $\vec{a}+5\vec{b}=\lambda\vec{c}$

$\vec{b}+6\vec{c}=\mu\vec{a}$

Eliminating $\vec{a}$, $\lambda\vec{c}-5\vec{b}=\frac{6}{\mu}\vec{c}+\frac{1}{\mu}\vec{b}$

$\therefore \mu=\frac{-1}{5}, \lambda=-30$

$\alpha = 5, \beta = 30$

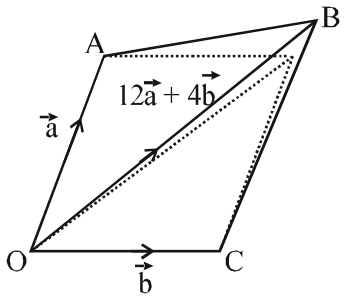
125. Let $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OB}=12\vec{a}+4\vec{b}$ and $\overrightarrow{OC}=\vec{b}$, where O is the origin. If S is the parallelogram with adjacent sides OA and OC, then $\frac{\text{area of the quadrilateral OABC}}{\text{area of S}}$ is equal to _____ . **[JEE(Main)Jan.2024]**

माना $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OB}=12\vec{a}+4\vec{b}$ तथा $\overrightarrow{OC}=\vec{b}$ हैं, जहाँ O मूल बिन्दु है। यदि संलग्न भुजाओं OA तथा OC का समांतर चतुर्भुज S है, तो $\frac{\text{चतुर्भूज OABC का क्षेत्र फल}}{\text{S का क्षेत्र फल}}$ बराबर है_____. **[JEE(Main)Jan.2024]**

(A) 7 (B) 6 (C) 10 (D) 8

Ans. D

Sol.



Area of parallelogram, $S=\left|\vec{a}\times\vec{b}\right|$

Area of quadrilateral $=\text{Area}(\Delta OAB)+\text{Area}(\Delta OBC)$

$=\frac{1}{2}\left\{\left|\vec{a}\times\left(12\vec{a}+4\vec{b}\right)\right|+\left|\vec{b}\times\left(12\vec{a}+4\vec{b}\right)\right|\right\}$

$=8\left|\left(\vec{a}\times\vec{b}\right)\right|$

$\text{Ratio}=\frac{8\left|\left(\vec{a}\times\vec{b}\right)\right|}{\left|\left(\vec{a}\times\vec{b}\right)\right|}=8$

126. Let a unit vector $\vec{u}=x\hat{i}+y\hat{j}+z\hat{k}$ make angles $\frac{\pi}{2}$, $\frac{\pi}{3}$ and $\frac{2\pi}{3}$ with the vectors $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ and $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ respectively. If $\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ then $\left|\hat{u}-\vec{v}\right|^2$ is equal to :

[JEE(Main)Jan.2024]

माना एक मात्रक सदिश $\vec{u}=x\hat{i}+y\hat{j}+z\hat{k}$, सदिशों $\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$, $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ तथा $\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$ से क्रमशः $\frac{\pi}{2}$, $\frac{\pi}{3}$ तथा $\frac{2\pi}{3}$ के कोण बनाता है। यदि $\vec{v}=\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}+\frac{1}{\sqrt{2}}\hat{k}$ है, तो $\left|\hat{u}-\vec{v}\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) $\frac{5}{2}$ (B) 7 (C) $\frac{11}{2}$ (D) 9

Ans. A

Sol. Unit vector $\hat{u}=x\hat{i}+y\hat{j}+z\hat{k}$

$\vec{p}_1 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k}$, $\vec{p}_2 = \frac{1}{\sqrt{2}}\hat{j} + \frac{1}{\sqrt{2}}\hat{k}$

$\vec{p}_3 = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

Now angle between $\hat{u}$ and $\vec{p}_1 = \frac{\pi}{2}$

$\hat{u} \cdot \vec{p}_1 = 0 \Rightarrow \frac{x}{\sqrt{2}} + \frac{z}{\sqrt{2}} = 0$

$\Rightarrow x + z = 0$

Now angle between $\hat{u}$ and $\vec{p}_2 = \frac{\pi}{3}$

$\hat{u} \cdot \vec{p}_2 = |\hat{u}| \cdot |\vec{p}_2| \cos\frac{\pi}{3}$

$\Rightarrow \frac{y}{\sqrt{2}} + \frac{z}{\sqrt{2}} = \frac{1}{2} \Rightarrow y + z = \frac{1}{\sqrt{2}}$

Now angle between $\hat{u}$ and $\vec{p}_3 = \frac{2\pi}{3}$

$\hat{u} \cdot \vec{p}_3 = |\hat{u}| \cdot |\vec{p}_3| \cos\frac{2\pi}{3}$

$\Rightarrow \frac{x}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{-1}{2}$

$\Rightarrow x + y = \frac{-1}{\sqrt{2}}$

from equation (i), (ii) and (iii) we get

$x = \frac{-1}{\sqrt{2}} \quad y = 0 \quad z = \frac{1}{\sqrt{2}}$

Thus $\hat{u} - \vec{v} = \frac{-1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{k} - \frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}n$

$\hat{u} - \vec{v} = \frac{-2}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}$

$\therefore |\hat{u} - \vec{v}|^2 = \left(\sqrt{\frac{4}{2} + \frac{1}{2}}\right)^2 = \frac{5}{2}$

127. Let P(3,2,3), Q(4,6,2) and R(7,3,2) be three vertices of ΔPQR. Then, the angle $\angle QPR$ is :

[JEE(Main)Jan.2024]

माना ΔPQR के शीर्ष P(3,2,3), Q(4,6,2) तथा R(7,3,2) हैं। तो $\angle QPR$ कराबर है : **[JEE(Main)Jan.2024]**

(A) $\cos^{-1}\left(\frac{1}{18}\right)$ (B) $\frac{\pi}{6}$ (C) $\cos^{-1}\left(\frac{7}{18}\right)$ (D) $\frac{\pi}{3}$

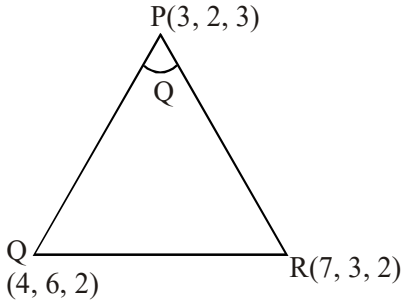
Ans. D

Sol. D.R of PQ = "1, 4, –1"

D.R of PR = "4, 1, –1"

$$\cos\theta = \left|\frac{1\times4+4\times1+-1\times-1}{\sqrt{1+16+1}\sqrt{16+1+1}}\right|$$

$$\Rightarrow \cos\theta = \frac{1}{2} \qquad \Rightarrow \theta = \frac{\pi}{3}$$



128. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ be two vectors such that $|\vec{a}| = 1$, $\vec{a}.\vec{b} = 2$ and $|\vec{b}| = 4$. If $\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$, then the angle between $\vec{b}$ and $\vec{c}$ is equal to : **[JEE(Main)Jan.2024]**

माना दो सदिशों $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ तथा $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ के लिए $|\vec{a}| = 1$, $\vec{a}.\vec{b} = 2$ तथा $|\vec{b}| = 4$ हैं। यदि $\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$ है, तो $\vec{b}$ तथा $\vec{c}$ के बीच कोण है : **[JEE(Main)Jan.2024]**

(A) $\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$ (C) $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Ans. C

Sol. Given $|\vec{a}| = 1, |\vec{b}| = 4, \vec{a}\cdot\vec{b} = 2$

$\vec{c} = 2\left(\vec{a}\times\vec{b}\right) - 3\vec{b}$

Dot product with $\vec{a}$ on both sides

$\vec{c}\cdot\vec{a} = -6$ ...(1)

Dot product with $\vec{b}$ on both sides

$\vec{b}\cdot\vec{c} = -48$ ...(2)

$\vec{c}\cdot\vec{c} = 4\left|\vec{a}\times\vec{b}\right| + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[|a|^2|b|^2 - \left(a\cdot\vec{b}\right)^2\right] + 9\left|\vec{b}\right|^2$

$\left|\vec{c}\right|^2 = 4\left[(1)(4)^2 - (4)\right] + 9(16)$

$\left|\vec{c}\right|^2 = 4[12] + 144$

$\left|\vec{c}\right|^2 = 192$

$$\therefore \cos\theta = \frac{\vec{b}\cdot\vec{c}}{|\vec{b}||\vec{c}|}$$

$$\therefore \cos\theta = \frac{-48}{\sqrt{192}\cdot 4}$$

$$\therefore \cos\theta = \frac{-48}{8\sqrt{3}\cdot 4}$$

$$\therefore \cos\theta = \frac{-3}{2\sqrt{3}}$$

$$\therefore \cos\theta = \frac{-\sqrt{3}}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{-\sqrt{3}}{2}\right)$$

129. Let A (2, 3, 5) and C (–3, 4, –2) be opposite vertices of a parallelogram ABCD. If the diagonal $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$, then the area of the parallelogram is equal to : **[JEE(Main)Jan.2024]**

माना एक समांतर चतुर्भुज ABCD के दो सम्मुख शीर्ष A (2, 3, 5) तथा C (–3, 4, –2) हैं। यदि इसका विकर्ण $\overrightarrow{BD} = \hat{i} + 2\hat{j} + 3\hat{k}$ है, तो समांतर चतुर्भुज का क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) $\frac{1}{2}\sqrt{586}$ (B) $\frac{1}{2}\sqrt{306}$ (C) $\frac{1}{2}\sqrt{410}$ (D) $\frac{1}{2}\sqrt{474}$

Ans. D

Sol. $$\frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{BD}\right| = \frac{1}{2}\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -5 & 1 & -7\\ 1 & 2 & 3\end{vmatrix}$$

$$= \frac{1}{2}\left|-17\hat{i} - 8\hat{j} + 11\hat{k}\right|$$

$$= \frac{1}{2}\sqrt{289 + 64 + 121} = \frac{1}{2}\sqrt{474}$$

130. Let $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$, $L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$ and $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$ be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2. Then, the point which lies on L_3 is : **[JEE(Main)Jan.2024]**

माना तीन रेखाऐं $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$, $L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$ तथा $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$ इस प्रकार हैं कि L_1 रेखा L_2 के लंबवत् है तथा L_3 दोनों L_1 तथा L_2 के लंबवत् हैं। तो निम्न में l sdlkS l kfcUnq L_3 पर है : **[JEE(Main)Jan.2024]**

(A) (1, 7, –4) (B) (–1, –7, 4) (C) (–1, 7, 4) (D) (1, –7, 4)

Ans. C

Sol. L_3 is

$L_3 \equiv \&\left(-\hat{i}+7\hat{j}+4\hat{k}\right)$

Since L_1 is perpendicular to L_2

$\Rightarrow \left(\hat{i}-\hat{j}+2\hat{k}\right)\cdot\left(3\hat{i}+\hat{j}+p\hat{k}\right)=0$

$3 - 1 + 2p = 0 \Rightarrow p = -1$

$L_3 \perp L_1 \& L_2$

$\Rightarrow L_3$ is parallel to $\left(\hat{i}-\hat{j}+2\hat{k}\right)\times\left(3\hat{i}+\hat{j}+p\hat{k}\right)$

$$\left(\hat{i}-\hat{j}+2\hat{k}\right)\times\left(3\hat{i}+\hat{j}+p\hat{k}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 2\\ 3 & 1 & -1\end{vmatrix}=-\hat{i}+7\hat{j}+4\hat{k}$$

$\ell = -1, m = 7, n = 4$

131. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{b}\right|=1$ and $\left|\vec{b}\times\vec{a}\right|=2$. Then $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2$ is equal to :

[JEE(Main)Jan.2024]

माना दो सदिश $\vec{a}$ तथा $\vec{b}$ इस प्रकार हैं कि $\left|\vec{b}\right|=1$ तथा $\left|\vec{b}\times\vec{a}\right|=2$ हैं। तो $\left|\left(\vec{b}\times\vec{a}\right)-\vec{b}\right|^2$ बराबर है :

[JEE(Main)Jan.2024]

(A) 3 (B) 1 (C) 5 (D) 4

Ans. C

Sol. $=\left|\vec{b}\times\vec{a}\right|^2+\left|\vec{b}\right|^2-2\left(\vec{b}\times\vec{a}\right)\cdot\vec{b}$

$= 4 + 1 - 0 = 5$

132. Let $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k},\alpha,\beta\in R$. Let a vector $\vec{b}$ be such that the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{4}$ and $\left|\vec{b}\right|^2=6$. If $\vec{a}\cdot\vec{b}=3\sqrt{2}$, then the value of $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ is : **[JEE(Main)Jan.2024]**

माना $\vec{a}=\hat{i}+\alpha\hat{j}+\beta\hat{k},\alpha,\beta\in R$ है। माना एक सदिश $\vec{b}$ इस प्रकार है कि $\vec{a}$ तथा $\vec{b}$ के बीच कोण $\frac{\pi}{4}$ है तथा $\left|\vec{b}\right|^2=6$ है। यदि $\vec{a}\cdot\vec{b}=3\sqrt{2}$ है, तो $\left(\alpha^2+\beta^2\right)\left|\vec{a}\times\vec{b}\right|^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 75 (B) 95 (C) 85 (D) 90

Ans. D

Sol. $\vec{a}\cdot\vec{b}=3\sqrt{2}$

$|\vec{a}||\vec{b}|\cos\frac{\pi}{4}=3\sqrt{2}$

$|\vec{a}|\sqrt{6}\cdot\frac{1}{\sqrt{2}}=3\sqrt{2}$

$|\vec{a}|=\sqrt{6}=\sqrt{1+\alpha^2+\beta^2}$

$\alpha^2+\beta^2=5$

$|\vec{a}\times\vec{b}|^2=|\vec{a}|^2|\vec{b}|^2\sin^2\frac{\pi}{4}$

$=6\times6\times\frac{1}{2}=18$

So, $\left(\alpha^2+\beta^2\right)|\vec{a}\times\vec{b}|^2=5\times18=90$

133. The distance of the point Q(0,2,–2) from the line passing through the point P(5,–4,3) and perpendicular to the lines $\vec{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\lambda\in R$ and $\vec{r}=\left(\hat{i}-2\hat{j}+\hat{k}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, $\mu\in R$ is :

[JEE(Main)Jan.2024]

बिन्दु P(5,–4,3) से होकर जाने वाली तथा रेखाओं $\vec{r}=\left(-3\hat{i}+2\hat{k}\right)+\lambda\left(2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\lambda\in R$ तथा $\vec{r}=\left(\hat{i}-2\hat{j}+\hat{k}\right)+\mu\left(-\hat{i}+3\hat{j}+2\hat{k}\right)$, $\mu\in R$ के लंबवत रेखा से बिन्दु Q(0,2,–2) की दूरी है :

[JEE(Main)Jan.2024]

(A) $\sqrt{20}$ (B) $\sqrt{54}$ (C) $\sqrt{74}$ (D) $\sqrt{86}$

Ans. C

Sol. A vector in the direction of the required line can be obtained by cross product of

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&3&5\\-1&3&2\end{vmatrix}$$

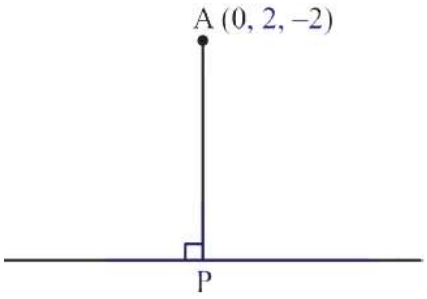
$=-9\hat{i}-9\hat{j}+9\hat{k}$

Required line,

$\vec{r}=\left(5\hat{i}-4\hat{j}+3\hat{k}\right)+\lambda'\left(-9\hat{i}-9\hat{j}+9\hat{k}\right)$

$\vec{r}=\left(5\hat{i}-4\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}+\hat{j}-\hat{k}\right)$

Now distance of (0,2,–2)



P.V. of $P \equiv (5+\lambda)\hat{i}+(\lambda-4)\hat{j}+(3-\lambda)\hat{k}$

$\overrightarrow{AP} = (5+\lambda)\hat{i}+(\lambda-6)\hat{j}+(5-\lambda)\hat{k}$

$\overrightarrow{AP}\cdot(\hat{i}+\hat{j}-\hat{k}) = 0$

$5 + \lambda + \lambda - 6 - 5 + \lambda = 0$

$\lambda = 2$

$\left|\overrightarrow{AP}\right| = \sqrt{49+16+9}$

$\left|\overrightarrow{AP}\right| = \sqrt{74}$

134. Let $\vec{a} = 3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b} = 4\hat{i}+\hat{j}+7\hat{k}$ and $\vec{c} = \hat{i}-3\hat{j}+4\hat{k}$ be three vectors. If a vector $\vec{p}$ satisfies $\vec{p}\times\vec{b} = \vec{c}\times\vec{b}$ and $\vec{p}.\vec{a} = 0$, then $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$ is equal to : **[JEE(Main)Jan.2024]**

माना तीन सदिश $\vec{a} = 3\hat{i}+\hat{j}-2\hat{k}$, $\vec{b} = 4\hat{i}+\hat{j}+7\hat{k}$ तथा $\vec{c} = \hat{i}-3\hat{j}+4\hat{k}$ हैं। यदि एक सदिश $\vec{p}$ के लिए $\vec{p}\times\vec{b} = \vec{c}\times\vec{b}$ तथा $\vec{p}.\vec{a} = 0$ हैं, तो $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 32 (B) 36 (C) 28 (D) 24

Ans. A

Sol. $(\vec{p}-\vec{c})\times\vec{b} = \vec{0}$ so, $(\vec{p}-\vec{c}) \| \vec{b}$

$\therefore \vec{p}-\vec{c} = \lambda\vec{b}$

$\vec{p} = \lambda\vec{b}+\vec{c}$

Now we have $\vec{p}\cdot\vec{a} = 0$

$\vec{p}\cdot\vec{a} = \lambda\vec{b}\cdot\vec{a}+\vec{c}\cdot\vec{a}$

$0 = \lambda(12+1-14)+(3-3-8)$

$\lambda = -8$

$\therefore \vec{p} = \vec{c}-8\vec{b} = \hat{i}-3\hat{j}+4\hat{k}-8(4\hat{i}+\hat{j}+7\hat{k}) = -31\hat{i}-11\hat{j}-52\hat{k}$

and $\vec{p}\cdot(\hat{i}-\hat{j}-\hat{k}) = -31 + 11 + 52 = 32$

135. Let $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$ and $\vec{c}=\left(\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right)\times\hat{i}\right)\times\hat{i}$. Then $\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)$ is equal to :

[JEE(Main)Jan.2024]

माना $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$ तथा $\vec{c}=\left(\left(\left(\vec{a}\times\vec{b}\right)\times\hat{i}\right)\times\hat{i}\right)\times\hat{i}$ हैं। तो $\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)$ बराबर है :

[JEE(Main)Jan.2024]

(A) –15 (B) –13 (C) –12 (D) –10

Ans. C

Sol. $\vec{a}=-5\hat{i}+\hat{j}-3\hat{k}$

$\vec{b}=\hat{i}+2\hat{j}-4\hat{k}$

$\left(\vec{a}\times\vec{b}\right)\times\hat{i}=\left(\vec{a}\cdot\hat{i}\right)\vec{b}-\left(\vec{b}\cdot\hat{i}\right)\vec{a}$

$=-5\vec{b}-\vec{a}$

$=\left(\left(\left(-5\vec{b}-\vec{a}\right)\times\hat{i}\right)\times\hat{i}\right)$

$=\left(\left(-11\hat{j}+23\hat{k}\right)\times\hat{i}\right)\times\hat{i}$

$\Rightarrow\left(11\hat{k}+23\hat{j}\right)\times\hat{i}$

$\Rightarrow\left(11\hat{j}-23\hat{k}\right)$

$\vec{c}\cdot\left(-\hat{i}+\hat{j}+\hat{k}\right)=11-23=-12$

136. Consider a ΔABC where A(1, 3, 2), B(–2, 8, 0) and C(3, 6, 7). If the angle bisector of $\angle BAC$ meets the line BC at D, then the length of the projection of the vector $\overrightarrow{AD}$ on the vector $\overrightarrow{AC}$ is :

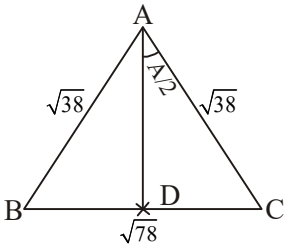
[JEE(Main)Jan.2024]

एक ΔABC, जहाँ A(1, 3, 2), B(–2, 8, 0) तथा C(3, 6, 7) हैं, का विचार कीजिए। यदि कोण BAC की समद्विभाजक रेखा, BC को D पर मिलती है, तो सदिश $\overrightarrow{AD}$ के सदिश $\overrightarrow{AC}$ पर प्रक्षेप की लम्बाई है : **[JEE(Main)Jan.2024]**

(A) $\frac{37}{2\sqrt{38}}$ (B) $\sqrt{19}$ (C) $\frac{\sqrt{38}}{2}$ (D) $\frac{39}{2\sqrt{38}}$

Ans. A

Sol.



Length of projection of $\overrightarrow{AD}$ a long $\overrightarrow{AC}$ $= AD\cos\dfrac{A}{2}$

$$= \left(\frac{2bc}{b+c}\right)\cos^2\frac{A}{2}$$

$$= \left(\frac{2bc}{b+c}\right)\left(\frac{1+\cos A}{2}\right)$$

$$= \frac{37}{2\sqrt{38}}$$

Method -II

$$D\left(\frac{1}{2}, 7, \frac{7}{2}\right)$$

$$\Rightarrow \overrightarrow{AD} = \frac{\hat{i} - 8\hat{j} - 3\hat{k}}{2}$$

$$\text{Length} = \left|\frac{\overrightarrow{AD}\,.\,\overrightarrow{AC}}{\left|\overrightarrow{AC}\right|}\right| = \frac{37}{2\sqrt{38}}$$

137. The least positive integral value of α, for which the angle between the vectors $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ and $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ is acute, is _____. **[JEE(Main)Jan.2024]**

α का न्यूनतम धनात्मक पूर्णांक मान, जिसके लिए सदिशों $\alpha\hat{i} - 2\hat{j} + 2\hat{k}$ तथा $\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k}$ के बीच का कोण, एक न्यून कोण है, बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $\cos\theta = \dfrac{(\alpha\hat{i} - 2\hat{j} + 2\hat{k})\cdot(\alpha\hat{i} + 2\alpha\hat{j} - 2\hat{k})}{\sqrt{\alpha^2 + 4 + 4}\sqrt{\alpha^2 + 4\alpha^2 + 4}}$

$$= \frac{\alpha^2 - 4\alpha - 4}{\sqrt{\alpha^2 + 8}\sqrt{5\alpha^2 + 4}} > 0$$

$\Rightarrow \alpha^2 - 4\alpha - 4 > 0$

$(\alpha - 2)^2 - 8 > 0$

$(\alpha - 2)^2 > 8$

$\alpha - 2 < -2\sqrt{2}$ or $\alpha - 2 > 2\sqrt{2}$

$\alpha < 2 - 2\sqrt{2}$ or $\alpha > 2 + 2\sqrt{2}$

$\alpha > 4.82$

$\alpha = 5$

138. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 4$, and $\vec{a}\cdot\vec{b} = 2$. If $\vec{c} = \left(2\vec{a}\times\vec{b}\right) - 3\vec{b}$ and the angle

between $\vec{b}$ and $\vec{c}$ is α, then $192\sin^2\alpha$ is equal to _____. **[JEE(Main)Jan.2024]**

माना दो सदिशों $\vec{a}$ तथा $\vec{b}$ के लिए $|\vec{a}|=1, |\vec{b}|=4$, तथा $\vec{a}\cdot\vec{b}=2$ हैं। यदि $\vec{c}=\left(2\vec{a}\times\vec{b}\right)-3\vec{b}$ है तथा $\vec{b}$ और $\vec{c}$ है के बीच कोण α है, तो $192\sin^2\alpha$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 48

Sol. $\vec{b}\cdot\vec{c}=\left(2\vec{a}\times\vec{b}\right)\cdot\vec{b}-3|b|^2$

$|b||c|\cos\alpha=-3|b|^2$

$|c|\cos\alpha=-12,\ \text{as}\ |b|=4$

$\vec{a}\cdot\vec{b}=2$

$\cos\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$

$|c|^2=\left|\left(2\vec{a}\times\vec{b}\right)-3\vec{b}\right|^2$

$=64\times\frac{3}{4}+144=192$

$|c|^2\cos^2\alpha=144$

$192\cos^2\alpha=144$

$192\sin^2\alpha=48$

139. Let $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c}$ be a vector such that $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ and $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$. Then $|\vec{c}|^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना $\vec{a}=3\hat{i}+2\hat{j}+\hat{k},\vec{b}=2\hat{i}-\hat{j}+3\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार हैं कि $\left(\vec{a}+\vec{b}\right)\times\vec{c}=2\left(\vec{a}\times\vec{b}\right)+24\hat{j}-6\hat{k}$ तथा $\left(\vec{a}-\vec{b}+\hat{i}\right).\vec{c}=-3$ हैं। तो $|\vec{c}|^2$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 38

Sol. $(\vec{a}+\vec{b})\times\vec{c}=2(\vec{a}\times\vec{b})+24\hat{j}-6\hat{k}$

$(5\hat{i}+\hat{j}+4\hat{k})\times\vec{c}=2(7\hat{i}-7\hat{j}-7\hat{k})+24\hat{j}-6\hat{k}$

$$\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\5&1&4\\x&y&z\end{vmatrix}=14\hat{i}+10\hat{j}-20\hat{k}$$

$\Rightarrow\hat{i}(z-4y)-\hat{j}(5z-4x)+\hat{k}(5y-x)=14\hat{i}+10\hat{j}-20\hat{k}$

$z-4y=14, 4x-5z=10, 5y-x=-20$

$(a-b+i)\cdot\vec{c}=-3$

$(2\hat{i}+3\hat{j}-2\hat{k})\cdot\vec{c}=-3$

$2x+3y-2z=-3$

$\therefore\ x=5, y=-3, z=2$

$|\vec{c}|^2=25+9+4=38$

140. Let the line of the shortest distance between the lines

$L_1:\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ and

$L_2:\vec{r}=\left(4\hat{i}+5\hat{j}+6\hat{k}\right)+\mu\left(\hat{i}+\hat{j}-\hat{k}\right)$

intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the mid point of the line segment PQ, then $2(\alpha+\beta+\gamma)$ is equal to _____. **[JEE(Main)Jan.2024]**

माना रेखाओं

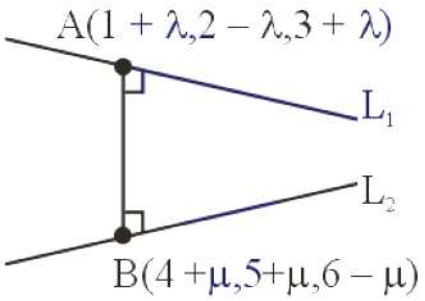
$L_1:\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right)$ तथा

$L_2:\vec{r}=\left(4\hat{i}+5\hat{j}+6\hat{k}\right)+\mu\left(\hat{i}+\hat{j}-\hat{k}\right)$

की न्यूनतम दूरी की रेखा, L_1 तथा L_2 को क्रमशः P तथा Q पर काटती है। यदि रेखाखण्ड PQ का मध्यबिंदु (α, β, γ) है, तो $2(\alpha+\beta+\gamma)$ बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 21

Sol.



$\vec{b}=\hat{i}-\hat{j}+\hat{k}$ (DR's of L_1)

$\vec{d}=\hat{i}+\hat{j}-\hat{k}$ (DR's of L_2)

$$\vec{b}\times\vec{d}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -1 & 1\\ 1 & 1 & -1\end{vmatrix}$$

$=0\hat{i}+2\hat{j}+2\hat{k}$ (DR's of Line perpendicular to L_1 and L_2)

DR of AB line

$= (0,2,2) = (3 + \mu - \lambda, 3 + \mu + \lambda, 3 - \mu - \lambda)$

$\frac{3+\mu-\lambda}{0} = \frac{3+\mu+\lambda}{2} = \frac{3-\mu-\lambda}{2}$

Solving above equation we get $\mu = -\frac{3}{2}$ and $\lambda = \frac{3}{2}$

point $A = \left(\frac{5}{2}, \frac{1}{2}, \frac{9}{2}\right)$

$B = \left(\frac{5}{2}, \frac{7}{2}, \frac{15}{2}\right)$

Point of $AB = \left(\frac{5}{2}, 2, 6\right) = (\alpha, \beta, \gamma)$

$2(\alpha, \beta, \gamma) = 5 + 4 + 12 = 21$

141. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ, then the greatest integer less than or equal to $\tan^2\theta$ is _____. **[JEE(Main)Jan.2024]**

माना तीन सदिशों $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k}$ तथा $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ के लिए $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$ हैं। यदि सदिश $\vec{c}$ तथा l fn' k $3\hat{i} + 4\hat{j} + \hat{k}$ के बीच का कोण θ हैं, तो महत्तम पूर्णांक $\leq \tan^2\theta$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 38

Sol. $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$

$\Rightarrow (\vec{c} - \vec{b}) \times \vec{a} = 0$

$\Rightarrow \vec{c} - \vec{b} = \lambda\vec{a}$

$\Rightarrow \vec{c} = \vec{b} + \lambda\vec{a}$

$\Rightarrow \vec{c} = (-1 + \lambda)\hat{i} + (-8 + \lambda)\hat{j} + (2 + \lambda)\hat{k}$

but $\Rightarrow \vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$

$\Rightarrow \lambda = 5, \quad C_2 = -3, C_3 = 7$

Now angle between $\Rightarrow \vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$ and $\Rightarrow 3\hat{i} + 4\hat{j} + \hat{k}$ is

$\Rightarrow \cos\theta = \frac{12 - 12 + 7}{\sqrt{16+9+49}\sqrt{9+16+1}} = \frac{7}{\sqrt{74}\sqrt{26}}$

$\Rightarrow \sec^2\theta = \frac{74 \times 26}{49} = 39.26$

$\Rightarrow \tan^2\theta = \sec^2\theta - 1 = 38.26$

$\Rightarrow \quad \left[\tan^2\theta\right] = 38$

142. The distance of the point (7, –2, 11) from the line $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$ is :

[JEE(Main)Jan.2024]

बिंदु (7, –2, 11) की रेखा $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ से रेखा $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$ के अनुदिश दूरी है :

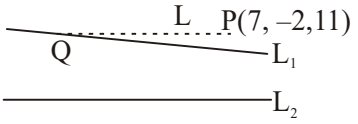
[JEE(Main)Jan.2024]

(A) 18 (B) 12 (C) 14 (D) 21

Ans. C

Sol. $L_1 : \frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$

$L_2 : \frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$



Equation of L is as $L_1 \parallel L_2$

$\frac{x-7}{2} = \frac{y+2}{-3} = \frac{z-11}{6}$

$\Rightarrow Q(2\lambda+7, -3\lambda-2, 6\lambda+11)$

Q lies on L_1

$\Rightarrow -3\lambda - 2 = 4$

$\Rightarrow -3\lambda = 6$

$\Rightarrow \lambda = -2$

$Q(3, 4, -1)$

$PQ = \sqrt{(7-3)^2 + (-2-4)^2 + (11+1}$

$= \sqrt{16+36+144}$

$= \sqrt{196} = 14$

143. If the shortest distance between the lines $\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

[JEE(Main)Jan.2024]

यदि रेखाओं $\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ तथा $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ के बीच न्यूनतम दूरी $\frac{6}{\sqrt{5}}$ है, तो λ के सभी संभव मानों का योग gS

[JEE(Main)Jan.2024]

(A) 5 (B) 7 (C) 10 (D) 8

Ans. D

Sol. $\frac{x-4}{1}=\frac{y+1}{2}=\frac{z}{-3}$

$\frac{x-\lambda}{2}=\frac{y+1}{4}=\frac{z-2}{-5}$

the shortest distance between the lines

$$=\left|\frac{(\vec{a}-\vec{b})\cdot\left(\overrightarrow{d_1}\times\overrightarrow{d_2}\right)}{\left|\overrightarrow{d_1}\times\overrightarrow{d_2}\right|}\right|$$

$$=\left|\frac{\begin{vmatrix}\lambda-4 & 0 & 2\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}}{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}}\right|$$

$$=\left|\frac{(\lambda-4)(-10+12)-0+2(4-4)}{|2\hat{i}-1\hat{j}+0\hat{k}|}\right|$$

$$\frac{6}{\sqrt{5}}=\left|\frac{2(\lambda-4)}{\sqrt{5}}\right|$$

$3 = |\lambda - 4|$

$\lambda - 4 = \pm 3$

$\lambda = 7, 1$

Sum of all possible values of λ is = 8

144. Let the image of the point (1,0,7) in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α, β, γ) Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ? **[JEE(Main)Jan.2024]**

माना रेखा $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ में बिन्दु (1,0,7) का प्रतिबिम्ब (α, β, γ) है। तो निम्न में से कौनसा बिंदु, बिंदु (α, β, γ) से होकर

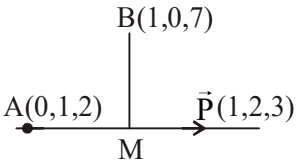
जाने वाली, y और z-अक्षों से क्रमशः $\frac{2\pi}{3}$ तथा $\frac{3\pi}{4}$ का कोण बनाने वाली तथा x-अक्ष से न्यून कोण बनाने वाली रेखा पर है?

[JEE(Main)Jan.2024]

(A) $\left(1,2,1-\sqrt{2}\right)$ (B) $\left(3,-4,3+2\sqrt{2}\right)$ (C) $\left(1,-2,1+\sqrt{2}\right)$ (D) $\left(3,4,3-2\sqrt{2}\right)$

Ans. D

Sol.



$M = (\lambda, 1 + 2\lambda, 2 + 3\lambda)$

$\overrightarrow{BM} = \left(\lambda - 1, 2\lambda + 1, 3\lambda - 5\right)$

$\vec{P} = (1, 2, 3)$

$\overrightarrow{BM} \cdot \vec{P} = 0$

$1(\lambda - 1) + 2(2\lambda + 1) + 3(3\lambda - 5) = 0$

$\lambda = 1$

$M = (1, 3, 5)$

Image (1, 6, 3)

$m = \cos 2\pi / 3 = -1/2$

$n = \cos 3\pi / 4 = -\frac{1}{\sqrt{2}}$

$m^2 + n^2 + \ell^2 = 1$

$\ell^2 = \frac{1}{4}$

$\ell = \pm\frac{1}{2}$

So, required line is

$$\frac{x-1}{\pm\frac{1}{2}} = \frac{y-6}{-\frac{1}{2}} = \frac{z-3}{-\frac{1}{\sqrt{2}}}$$

$$\frac{x-1}{\pm 1} = \frac{y-6}{1} = \frac{z-3}{\sqrt{2}}$$

145. Let PQR be a triangle with R (–1, 4, 2). Suppose M (2, 1, 2) is the mid point of PQ. The distance of the

centroid of ΔPQR from the point of intersection of the lines $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ and $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ is :

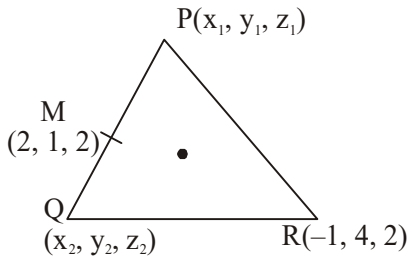
[JEE(Main)Jan.2024]

माना एक त्रिभुज PQR में R (–1, 4, 2) है। माना PQ का मध्य बिन्दु M (2, 1, 2) है। तो रेखाओं $\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}$ तथा $\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}$ के प्रतिच्छेदन बिन्दु से ΔPQR के केन्द्रक की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{99}$ (B) $\sqrt{69}$ (C) 69 (D) 9

Ans. B

Sol.



Given two lines are

$\frac{x-2}{0}=\frac{y}{2}=\frac{z+3}{-1}=\lambda$ (let)

$\frac{x-1}{1}=\frac{y+3}{-3}=\frac{z+1}{1}=\mu$ (let)

for point of intersection

$x = 2, y = 2\lambda, z = -\lambda - 3$ (General point of line 1)

$x = \mu + 1, y = -3\mu - 3, z = \mu - 1$ (General point of line 2)

$2 = \mu + 1 \Rightarrow \mu = 1$

$2\lambda = -3\mu - 3 \Rightarrow \lambda = -3$

Point of intersection of two lines is (2, –6, 0)

$\because$ M is the mid point of PQ,

$\therefore x_1 + x_2 = 4, y_1 + y_2 = 2, z_1 + z_2 = 4$

Centroid of ΔPQR

$=\left(\frac{x_1+x_2-1}{3}, \frac{y_1+y_2+4}{3}, \frac{z_1+z_2+2}{3}\right)$

= (1, 2, 2)

$\therefore$ Distance between (1, 2, 2) and (2, –6, 0)

$=\sqrt{1^2+8^2+2^2}=\sqrt{69}$

146. Λετ(α, β, γ) be the foot of perpendicular from the point (1, 2, 3) on the line $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. Then $19(\alpha+$ b + g) is equal to : **[JEE(Main)Jan.2024]**

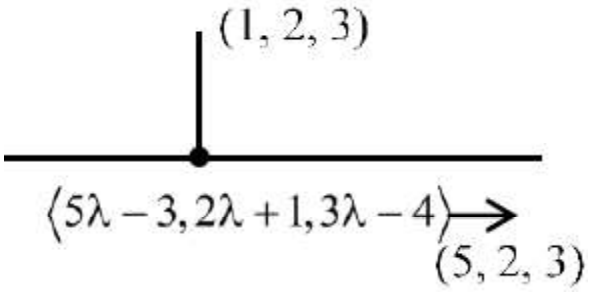
माना बिंदु (1,2,3) से रेखा $\frac{x+3}{5}=\frac{y-1}{2}=\frac{z+4}{3}$ पर लंब का पाद (α, β, γ) है। तो $19(\alpha+\beta+\gamma)$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 102 (B) 101 (C) 99 (D) 100

Ans. B

Sol. $\vec{r}=\langle -3,1-4\rangle+\lambda\langle 5,2,3\rangle$

Foot of $\perp^r \langle 5\lambda-3,2\lambda+1,3\lambda-4\rangle+\lambda\langle 5,2,3\rangle$



$\langle 5\lambda-4,2\lambda-1,3\lambda-7\rangle\cdot\langle 5,2,3\rangle=0$

$\Rightarrow 25\lambda-20+4\lambda-2+9\lambda-21=0$

$38\lambda=43$

$\lambda=\frac{43}{38}$

$\left\langle\frac{101}{38},\frac{124}{38},\frac{-23}{38}\right\rangle\Rightarrow 101$

147. The shortest distance, between lines L_1 and L_2, where $L_1:\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ and L_2 is the line, passing through the points A(–4, 4, 3), B(–1, 6, 3) and perpendicular to the line $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{1}$, is : **[JEE(Main)Jan.2024]**

यदि रेखा $L_1:\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+4}{2}$ है तथा बिन्दुओं A (–4, 4, 3), B (–1, 6, 3) से होकर जाने वाली और रेखा $\frac{x-3}{-2}=\frac{y}{3}=\frac{z-1}{1}$ के लंबवत रेखा L_2 है, तो L_1 तथा L_2 के बीच न्यूनतम दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{141}{\sqrt{221}}$ (B) $\frac{42}{\sqrt{117}}$ (C) $\frac{24}{\sqrt{117}}$ (D) $\frac{121}{\sqrt{221}}$

Ans. A

Sol. $L_2 : \frac{x+4}{3} = \frac{y-4}{2} = \frac{z-3}{0}$

$L_1 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+4}{2}$

$S \cdot D = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

$\vec{a}_1 = \hat{i} - \hat{j} - 4\hat{k} \quad \vec{b}_2 = 3\hat{i} + 2\hat{j} + 0\hat{k}$

$\vec{a}_2 = -4\hat{i} + 4\hat{j} + 3\hat{k} \quad \vec{b}_1 = 2\hat{i} - 3\hat{j} + 2\hat{k}$

$\vec{b}_1 \times \vec{b}_2 = -4\hat{i} + 6\hat{j} + 13\hat{k}$

$S \cdot D = \left| \frac{(-5\hat{i} + 5\hat{j} + 7\hat{k}) \cdot (-4\hat{i} + 6\hat{j} + 13\hat{k})}{\sqrt{221}} \right|$

$= \left| \frac{141}{\sqrt{221}} \right| = \frac{141}{\sqrt{221}}$

148. Λετ (α, β, γ) be the mirror image of the point (2, 3, 5) in the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Then $2\alpha + 3\beta + 4\gamma$ is equal to : **[JEE(Main)Jan.2024]**

माना बिन्दु (2, 3, 5) का रेखा $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ में दर्पण प्रतिबिंब (α, β, γ) है। तो $2\alpha + 3\beta + 4\gamma$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 34 (B) 32 (C) 31 (D) 33

Ans. D

Sol. Let $B(2\lambda+1, 3\lambda+2, 4\lambda+3)$

Direction rations of AB : $2\lambda - 1, 3\lambda - 1, 4\lambda - 2$

$\overline{AB}$ is perpendicular to given line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

A (2,3,5)

B

$\Rightarrow (2\lambda - 1) \times 2 + (3\lambda - 1)3 + (4\lambda - 2)4 = 0$

$29\lambda = 13$

$\lambda = 13/29$

$B\left(\frac{55}{29}, \frac{97}{29}, \frac{139}{29}\right)$

$\frac{\alpha+2}{\alpha} = \frac{55}{29}, \frac{\beta+3}{2} = \frac{97}{29}, \frac{\gamma+5}{2} = \frac{139}{29}$

$\alpha = \frac{52}{29}, \beta = \frac{167}{29}, \gamma = \frac{133}{29}$

Now $2\alpha + 3\beta + 4\lambda = \frac{104}{29} + \frac{321}{29} + \frac{532}{29} = \frac{957}{29} = 33$

149. If the shortest distance between the lines $\frac{x-\lambda}{-2} = \frac{y-2}{1} = \frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1} = \frac{y-1}{-2} = \frac{z-2}{1}$ is 1, then the sum of all possible values of λ is : **[JEE(Main)Jan.2024]**

यदि रेखाओं $\frac{x-\lambda}{-2} = \frac{y-2}{1} = \frac{z-1}{1}$ तथा $\frac{x-\sqrt{3}}{1} = \frac{y-1}{-2} = \frac{z-2}{1}$ के बीच न्यूनतम दूरी 1 है, तो λ के सभी संभव मानों का योग gS **[JEE(Main)Jan.2024]**

(A) $-2\sqrt{3}$ (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) 0

Ans. B

Sol. Shortest distance

$$= \left| \frac{\vec{V}_1 \times \vec{V}_2}{|\vec{V}_1 \times \vec{V}_2|} \cdot \overrightarrow{AB} \right| = 1$$

where $V_1 = -2\hat{i} + \hat{j} + \hat{k}$, $\overrightarrow{AB} = (\lambda - \sqrt{3}, 2-1, 1-2)$

$V_2 = \hat{i} - 2\hat{j} + \hat{k}$

$$\vec{V}_1 \times \vec{V}_2 = \begin{vmatrix} i & j & k \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix} = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

$\Rightarrow \vec{V}_1 \times \vec{V}_2 = 3(\hat{i} + \hat{j} + \hat{k})$

$\Rightarrow |\vec{V}_1 \times \vec{V}_2| = 3\sqrt{3}$

$$\Rightarrow \left| \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot ((\lambda - \sqrt{3})\hat{i} + \hat{j} - \hat{k})}{\sqrt{3}} \right| = 1$$

$$\left|\left(\lambda-\sqrt{3}\right)+1-1\right|=\sqrt{3}$$

$$\Rightarrow\left|\lambda-\sqrt{3}\right|=\sqrt{3}$$

$$\Rightarrow\lambda-\sqrt{3}=\sqrt{3}\text{ or }\lambda-\sqrt{3}=-\sqrt{3}$$

$$\Rightarrow\lambda=2\sqrt{3}\text{ or }0$$

$$\Rightarrow\text{sum of }\lambda=2\sqrt{3}$$

150. If the mirror image of the point P(3, 4, 9) in the line $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ is (α, β, γ), then $14(\alpha+\beta+\gamma)$ is:

[JEE(Main)Jan.2024]

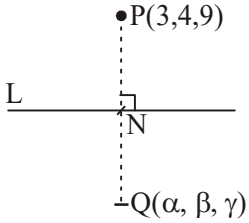
यदि बिन्दु P(3, 4, 9) का रेखा $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ में दर्पण प्रतिबिंब (α, β, γ) है, तो $14(\alpha+\beta+\gamma)$ बराबर है :

[JEE(Main)Jan.2024]

(A) 108 (B) 132 (C) 102 (D) 138

Ans. A

Sol.



$N \equiv$ f.o.p. from P on line

Let $N(1+3\lambda, 2\lambda-1, 2+\lambda)$

$$\overrightarrow{PN}.\left(3\hat{i}+2\hat{j}+\hat{k}\right)=0$$

$$\Rightarrow\lambda=\frac{23}{14}$$

$$\Rightarrow N\left(\frac{83}{14},\frac{32}{14},\frac{51}{14}\right)$$

$$\frac{\alpha+3}{2}=\frac{83}{14}\quad\Rightarrow\alpha=\frac{62}{7}$$

$$\frac{\beta+4}{2}=\frac{32}{14}\quad\Rightarrow\beta=\frac{4}{7}$$

$$\frac{\gamma+9}{2}=\frac{51}{14}\quad\Rightarrow\gamma=\frac{-12}{7}$$

$$14\left(\alpha+\beta+\gamma\right)=108$$

151. Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point R(1, 2, 3). If the centroid of the triangle PQR is (α, β, γ), then $\alpha^2+\beta^2+\gamma^2$ is : **[JEE(Main)Jan.2024]**

माना रेखा $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ पर बिंदु R(1, 2, 3) से 6 इकाई दूरी पर दो बिंदु P तथा Q हैं। यदि त्रिभुज PQR का केन्द्रक (α, β, γ) है, तो $\alpha^2+\beta^2+\gamma^2$ बराबर है : **[JEE(Main)Jan.2024]**

(A) 36 (B) 18 (C) 24 (D) 26

Ans. B

Sol. Line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$

$P(8\lambda-3, 2\lambda+4, 2\lambda-1)$

$PR = 6$

$\Rightarrow \lambda = 0, 1$

$\Rightarrow P(-3,4,-1)$ Q(5, 6, 1)

Controid of $\Delta PQR = (1, 4, 1)$

$\alpha = 1,\ \beta = 4,\ \gamma = 1$

$\Rightarrow \alpha^2+\beta^2+\gamma^2 = 18$

152. The line $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____. **[JEE(Main)Jan.2024]**

रेखाओं $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ तथा $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ का प्रतिच्छेदन बिंदु P है। यदि रेखा $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ से P की दूरी l है, तो $14l^2$ बराबर है _____। **[JEE(Main)Jan.2024]**

Ans. 108

Sol. Any point on L_1 (2a + 2, –2a, 16a + 7)

Any point on L_2 (4b – 3, 3b – 2, b – 2)

$$\begin{array}{l|l|l} 2a+2=4b-3 & -2a=3b-2 & 16a+7=b-2 \\ 2a-4b=-5 & -2a-3b=-2 & 16a-b=-9 \end{array}$$

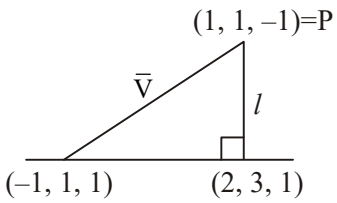
$-7b = -7$

$\boxed{b=1}$

$\boxed{a=-\frac{1}{2}}$

P(1, 1, –1)

$\bar{V} = (2, 0, -2)$



$\ell = \dfrac{|\bar{v} \times \bar{p}|}{|\bar{p}|} = \dfrac{\sqrt{108}}{\sqrt{14}} = \sqrt{\dfrac{54}{7}}$ $\quad\quad$ $\therefore 14\ell^2 = 14 \times \dfrac{54}{7} = \boxed{108}$ Ans.

153. A line with direction ratios 2, 1, 2 meets the lines $x = y + 2 = z$ and $x + 2 = 2y = 2z$ respectively at the points P and Q. If the length of the perpendicular from the point (1, 2, 12) to the line PQ is l, then l^2 is _____.

[JEE(Main)Jan.2024]

दिक् अनुपात 2, 1, 2 की एक रेखा, रेखाओं $x = y + 2 = z$ तथा $x + 2 = 2y = 2z$ को क्रमशः बिन्दुओं P तथा Q पर मिलती है। यदि fcUnq(1, 2, 12) से रेखा PQ पर लंब की लंबाई l है, तो l^2 बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 65

Sol. Let $P(t, t-2, t)$ and $Q(2s-2, s, s)$

D.R's of PQ are 2, 1, 2

$\dfrac{2s-2-t}{2} = \dfrac{s-t+2}{1} = \dfrac{s-t}{2}$

$\Rightarrow t = 6$ and $s = 2$

$\Rightarrow$ P(6, 4, 6) and Q(2, 2, 2)

$PQ : \dfrac{x-2}{2} = \dfrac{y-2}{1} = \dfrac{z-2}{2} = \lambda$

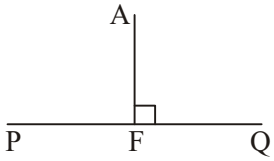
Let $F(2\lambda + 2, \lambda + 2, 2\lambda + 2)$

A(1, 2, 12)

$\overrightarrow{AF} \cdot \overrightarrow{PQ} = 0$

$\therefore \lambda = 2$

So F(6, 4, 6) and $AF = \sqrt{65}$



154. Let O be the origin and M and N be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{OM}\cdot\overrightarrow{ON}$ is equal to _____ . **[JEE(Main)Jan.2024]**

माना O मूल बिन्दु है तथा रेखाओं $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ तथा $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ पर क्रमशः बिन्दुओं M तथा N के लिए nhxbZ;s dk vYirke nwjh MN है। तो $\overrightarrow{OM}\cdot\overrightarrow{ON}$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 9

Sol. $L_1:\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}=\lambda$

Any point of $L_1:(4\lambda+5,\lambda+4,3\lambda+5):M$

$L_2:\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}=\mu$

Any point of $L_2:(12\mu-8,5\mu-2,9\mu-11):N$

DR of MN $MN<4\lambda-12\mu+13,\lambda-5\mu+6,3\lambda-9\mu+16>$

Now $4(4\lambda-12\mu+13)+(\lambda-5\mu+6)+3(3\lambda-9\mu+16)=0$

and $12(4\lambda-12\mu+13)+5(\lambda-5\mu+6)+9(3\lambda-9\mu+16)=0$

$\Rightarrow \lambda=-1$ and $\mu=1$

$\therefore$ M(1, 3, 2) adn N(4, 3, –2)

OM . ON = 1 . (4) + 3 . (3) + 2(–2)

=4 + 9 – 4 = 9

155. If d_1 is the shortest distance between the lines x + 1 = 2y = –12z, x = y + 2 = 6z – 6 and d_2 is the shortest distance between the lines $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$, $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}d_1}{d_2}$ is _____. **[JEE(Main)Jan.2024]**

यदि रेखाओं x + 1 = 2y = –12z, x = y + 2 = 6z – 6 के बीच की न्यूनतम दूरी d_1 है तथा रेखाओं $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$, $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$ के बीच न्यूनतम दूरी d_2 है, तो $\frac{32\sqrt{3}d_1}{d_2}$ का मान है _____।

[JEE(Main)Jan.2024]

Ans. 16

Sol. $L_1:\frac{x+1}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{-\frac{1}{12}}$, $L_2:\frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{\frac{1}{6}}$

d_1 = shortest distance between L_1 & L_2

$$= \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|(\vec{b}_1 \times \vec{b}_2)|} \right|$$

$d_1 = 2$

$L_3 : \frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}$, $L_4 : \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$

d_2 = shortest distance between L_3 & L_4

$d_2 = \frac{12}{\sqrt{3}}$

Hence $= \frac{32\sqrt{3}d_1}{d_2} = \frac{32\sqrt{3} \times 2}{\frac{12}{\sqrt{3}}} = 16$

156. Let a line passing through the point (–1, 2, 3) intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at M(α, β, γ) and $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ at N(a, b, c). Then the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equal _____.

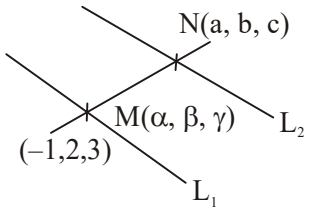
[JEE(Main)Jan.2024]

माना बिन्दु (–1, 2, 3) से होकर जाने वाली रेखा, रेखा $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ को M(α, β, γ) तथा रेखा $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ को N(a, b, c) पर काटती है। तो $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 196

Sol.



M(3λ + 1, 2λ + 2, –2λ – 1)

N(–3μ – 2, – 2μ + 2, 4μ + 1)

$\frac{3\lambda+2}{-3\mu-1} = \frac{2\lambda}{-2\mu} = \frac{-2\lambda-4}{4\mu-2} \Rightarrow \mu = 2, \lambda = 4$

So, $\begin{matrix} \alpha+\beta+\gamma = 14 \\ a+b+c = -1 \end{matrix} \Rightarrow \frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2} = 196$

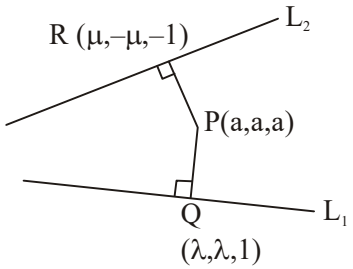
157. Let Q and R be the feet of perpendiculars from the point (a,a,a) on the lines x = y, z = 1 and x = –y, z = –1 respectively. If ∠QPR is a right angle, then $12a^2$ is equal to _____. **[JEE(Main)Jan.2024]**

माना बिन्दु P(a,a,a) से रेखाओं x = y, z = 1 तथा x = –y, z = –1 पर डाले गए लंबों के पाद क्रमशः Q तथा R हैं। यदि ∠QPR एक समकोण हैं, तो $12a^2$ बराबर है _____. **[JEE(Main)Jan.2024]**

Ans. 12

Sol. $L_1 : \frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = \lambda$ (let)

$L_2 : \frac{x}{1} = \frac{y}{-1} = \frac{z+1}{0} = \mu$ (let)



$PQ \perp L_1$

$(\lambda - a)1 + (\lambda - a)1 + (1 - a)0 = 0$

$\lambda = a$

$PR \perp L_2$

$1(\mu - a) + 1(\mu + a) + 0\,(-1 - a) = 0$

$\mu - a + \mu + a = 0$

$\mu = 0$

$\therefore$ Point Q(a,a,1) and R(0,0,–1)

According to question

$\angle QPR = 90^o$

Direction ratio of PR = (a, a, a + 1)

Direction ratiom of PQ = (0, 0, a – 1)

$a \times 0 + 0 \times a + (a + 1)(a - 1) = 0$

$a^2 = 1$

$\Rightarrow 12a^2 = 12 \times 1 = 12$

158. A line passes through A (4, –6, –2) and B (16, –2, 4). Then point P (a, b, c) , where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points P(a, b, c) and Q(4, –12, 3) is equal to _____. **[JEE(Main)Jan.2024]**

एक रेखा, बिन्दुओं A (4, –6, –2) तथा B (16, –2, 4) से होकर जाती है। रेखा AB पर बिन्दु P(a, b, c), जहाँ a, b, c ऋणोत्तर पूर्णांक gfcuq A से 21 इकाई की दूरी पर है। बिन्दुओं P(a, b, c) तथा Q(4, –12, 3) के बीच दूरी है _____

Ans. 22

Sol. $A(4-6-2)$ $B(16-2,4)$

Equation of line AB :

$$\frac{x-4}{6}=\frac{y+6}{2}=\frac{z+2}{3}$$

$$\frac{x-4}{\frac{6}{7}}=\frac{y+6}{\frac{2}{7}}=\frac{z+2}{\frac{3}{7}}=21$$

$x=22,\quad y=0,\quad z=7$

$P(22,0,7),\quad Q(4,-12,3)$

$PQ=\sqrt{(18)^2+(12)^2+(4)^2}$

$=\sqrt{484}=22$

159. Let a unit vector which makes an angle of 60º with $2\hat{i}+2\hat{j}-\hat{k}$ and an angle of 45º with $\hat{i}-\hat{k}$ be $\vec{C}$. Then $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ is : **[JEE(Main)April,2024]**

माना सदिश $2\hat{i}+2\hat{j}-\hat{k}$ से 60º का कोण तथा सदिश $\hat{i}-\hat{k}$ से 45° का कोण बनाने वाला एक मात्रक सदिश $\vec{C}$ है। तो $\vec{C}+\left(-\frac{1}{2}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{\sqrt{2}}{3}\hat{k}\right)$ है : **[JEE(Main)April,2024]**

(A) $\left(\frac{1}{\sqrt{3}}+\frac{1}{2}\right)\hat{i}+\left(\frac{1}{\sqrt{3}}-\frac{1}{3\sqrt{2}}\right)\hat{j}+\left(\frac{1}{\sqrt{3}}+\frac{\sqrt{2}}{3}\right)\hat{k}$

(B) $\frac{\sqrt{2}}{3}\hat{i}+\frac{1}{3\sqrt{2}}\hat{j}-\frac{1}{2}\hat{k}$

(C) $\frac{\sqrt{2}}{3}\hat{i}-\frac{1}{2}\hat{k}$

(D) $-\frac{\sqrt{2}}{3}\hat{i}+\frac{\sqrt{2}}{3}\hat{j}+\left(\frac{1}{2}+\frac{2\sqrt{2}}{3}\right)\hat{k}$

Ans. C

Sol. Let $\vec{C}=a\hat{i}+b\hat{j}+c\hat{k}$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(2\hat{i}+2\hat{j}-\hat{k}\right)=1\times3\times\frac{1}{2}$$

$$2a+2b-c=\frac{3}{2} \qquad \text{_____(1)}$$

$$\left(a\hat{i}+b\hat{j}+c\hat{k}\right)\cdot\left(\hat{i}-\hat{k}\right)=1\times\sqrt{2}\times\frac{1}{\sqrt{2}}$$

$a - c = 1$ _____(2)

$a^2 + b^2 + c^2 = 1$ _____(3)

Solving (1), (2) and (3)

$$a + 2b = \frac{1}{2}$$

$a^2 + b^2 + (a - 1)^2 = 1$

$2a^2 - 2a + b^2 = 0$

$$2a^2 - 2a + \left(\frac{2a-1}{4}\right)^2 = 0$$

$32a^2 - 32a + 4a^2 - 4a + 1 = 0$

$36a^2 - 36a + 1 = 0$

$$a = \frac{36 \pm \sqrt{(36)^2 - 4(36)}}{2 \times 36}$$

$$= \frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$b = \frac{1-2a}{4} \Rightarrow b = \frac{1 \pm \frac{2\sqrt{2}}{3} - 1}{4}$$

$$= - \pm \frac{1}{3\sqrt{2}}$$

$$C = -\frac{1}{2} \pm \frac{\sqrt{2}}{3}$$

$$C + \left(\frac{-1}{2}\hat{i} + \frac{1}{3\sqrt{2}}\hat{j} - \frac{\sqrt{2}}{3}\hat{k}\right)$$

$$= \frac{\sqrt{2}}{3}\hat{i} - \frac{1}{2}\hat{k}$$

160. For $\lambda > 0$, let θ be the angle between the vectors $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. If the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are mutually perpendicular, then the value of $(14 \cos \theta)^2$ is equal to : **[JEE(Main)April,2024]**

$\lambda > 0$ के लिए, माना सदिशों $\vec{a} = \hat{i} + \lambda\hat{j} - 3\hat{k}$ तथा $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ के बीच का कोण θ है। यदि सदिश $\vec{a} + \vec{b}$ तथा $\vec{a} - \vec{b}$ परस्पर y Ecor ~gSतो $(14 \cos \theta)^2$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) 20 (B) 25 (C) 40 (D) 50

Ans. B

Sol. $\left(\vec{a}-\vec{b}\right).\left(\vec{a}+\vec{b}\right)=0$

$\Rightarrow |a|^2=|b|^2 \quad \Rightarrow |a|^2=|b|^2 \quad \Rightarrow 1+\lambda^2+9$

$=9+1+4$

$\lambda=2$ and $\cos\theta=\dfrac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|} \quad \Rightarrow \cos\theta=\dfrac{-5}{14} \Rightarrow (14\cos\theta)^2=25$

161. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k}$, $x\in R$. If $\vec{d}$ is the unit vector in the direction of $\vec{b}+\vec{c}$ such that $\vec{a}\cdot\vec{d}=1$, then $\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ तथा $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k}$, $x\in R$ है। यदि $\vec{b}+\vec{c}$ की दिशा में एक इकाई सदिश $\vec{d}$ इस i dk gS d $\vec{a}\cdot\vec{d}=1$, तो $\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}$ बराबर है : **[JEE(Main)April,2024]**

(A) 11 (B) 3 (C) 6 (D) 9

Ans. A

Sol. $\vec{d}=\lambda\left((2+x)\hat{i}+6\hat{j}-2\hat{k}\right)$

$\vec{a}\cdot\vec{d}=1\Rightarrow\lambda(x+6)=1$ _____(1)

and $|\vec{d}|=1\Rightarrow\lambda^2\left((x+2)^2+36+4\right)=1$ _____(2)

(A) and (2) : x = 1

$\vec{c}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{a}\times\vec{b}=-9\hat{i}+7\hat{j}+2\hat{k}$

$\left(\vec{a}\times\vec{b}\right)\cdot\vec{c}=-9+14+6=11$

162. If A (1, –1, 2), B(5, 7, –6), C(3, 4 –10) and D(–1, –4, –2) are the vertices of quadrilateral ABCD, then its are is : **[JEE(Main)April,2024]**

यदि एक चतुर्भुज ABCD के शीर्ष A (1, –1, 2), B(5, 7, –6), C(3, 4 –10) तथा D(–1, –4, –2) हैं, तो इसका क्षेत्रफल है : **[JEE(Main)April,2024]**

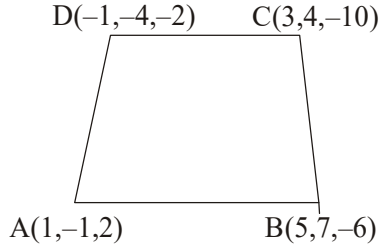
(A) $24\sqrt{29}$ (B) $12\sqrt{29}$ (C) $48\sqrt{7}$ (D) $24\sqrt{7}$

Ans. B

Sol. Diagonal $\overrightarrow{AC}=2\hat{i}+5\hat{j}-12\hat{k}$ and $\overrightarrow{BD}=-6\hat{i}-11\hat{j}+4\hat{k}$

$$\overrightarrow{AC}\times\overrightarrow{BD}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&5&-12\\-6&-11&4\end{vmatrix}=8\left(-14\hat{i}+8\hat{j}+\hat{k}\right)$$

$$\text{Area}=\frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{BD}\right|=4\sqrt{196+64+1}=12\sqrt{29}$$



163. Consider three vectors $\vec{a},\vec{b},\vec{c}$. Let $|\vec{a}|=2,|\vec{b}|=3$ and $\vec{a}=\vec{b}\times\vec{c}$. If $\alpha\in\left[0,\frac{\pi}{3}\right]$ is the angle between the vectors $\vec{b}$ and $\vec{c}$, then the minimum value of $27|\vec{c}-\vec{a}|^2$ is equal to : **[JEE(Main)April,2024]**

तीन सदिशों $\vec{a},\vec{b},\vec{c}$ का विचार कीजिए। माना $|\vec{a}|=2,|\vec{b}|=3$ तथा $\vec{a}=\vec{b}\times\vec{c}$ हैं। यदि सदिशों $\vec{b}$ तथा $\vec{c}$ के बीच का कोण $\alpha\in\left[0,\frac{\pi}{3}\right]$ है, तो $27|\vec{c}-\vec{a}|^2$ का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) 105 (B) 124 (C) 121 (D) 110

Ans. B

Sol. $\vec{a}=\vec{b}\times\vec{c}$

$|\vec{a}|=2,\ |\vec{b}|=3$

$\vec{a}.\vec{b}=0$ and $\vec{a}.\vec{c}=0$

$|\vec{c}-\vec{a}|^2=|\vec{c}|^2+|\vec{a}|^2-2\vec{c}\cdot\vec{a}$

$=4+|\vec{c}|^2$

$|\vec{a}|=|\vec{b}\times\vec{c}|=|\vec{b}|\sin\alpha|\vec{c}|$

$\Rightarrow \sin\alpha|\vec{c}|=\frac{2}{3}$

$\Rightarrow \sin^2\alpha=\frac{4}{9|\vec{c}|^2}$

$\Rightarrow |\vec{c}|^2=\frac{4}{9\sin^2\alpha}$

$\Rightarrow |\vec{c}-\vec{a}|^2 = 4 + \dfrac{4}{9\sin^2\alpha}$

For $|\vec{c}-\vec{a}|^2$ to be minimum for $\alpha \in \left[0, \dfrac{\pi}{3}\right]$

$\sin\alpha = \dfrac{\sqrt{3}}{2}$

$27\,|\vec{c}-\vec{a}|^2 = 27\left[4+\dfrac{4.4}{9.3}\right]$

$= 27\left[\dfrac{124}{27}\right] = 124$

164. Let $\vec{a} = 2\hat{i}+5\hat{j}-\hat{k}, \vec{b} = 2\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{c}$ be three vectors such that $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}\right) = \vec{a}\times\left(\vec{c}+\hat{i}\right)$. If $\vec{a}\ .\ \vec{c} = -29$, then $\vec{c}\cdot\left(-2\hat{i}+\hat{j}+\hat{k}\right)$ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिशों $\vec{a} = 2\hat{i}+5\hat{j}-\hat{k}, \vec{b} = 2\hat{i}-2\hat{j}+2\hat{k}$ तथा $\vec{c}$ के लिए $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}\right) = \vec{a}\times\left(\vec{c}+\hat{i}\right)$ है। यदि $\vec{a}\ .\ \vec{c} = -29$ है, तो $\vec{c}\cdot\left(-2\hat{i}+\hat{j}+\hat{k}\right)$ बराबर है : **[JEE(Main)April,2024]**

(A) 10 (B) 5 (C) 12 (D) 15

Ans. B

Sol. $\left(\vec{c}+\hat{i}\right)\times\left(\vec{a}+\vec{b}+\hat{i}+\vec{a}\right) = 0$

$\Rightarrow \vec{c} + \hat{i} = \lambda(\vec{a} + \vec{b} + \hat{i} + \vec{a})$

$= \lambda(2\vec{a} + \vec{b} + \hat{i})$

$= \lambda(7\hat{i} + 8\hat{j})$

$\Rightarrow \vec{c} = (7\lambda - 1)\hat{i} + 8\lambda\hat{j}$

$\vec{c}\cdot\vec{a} = -29$

$\Rightarrow 14\lambda - 2 + 40\lambda = -29$

$\Rightarrow 54\lambda = -27$

$\Rightarrow \lambda = -\dfrac{1}{2}$

$\therefore\ \vec{c} = \left(\dfrac{-7}{2}-1\right)\hat{i} - 4\hat{j} = \dfrac{-9}{2}\hat{i} - 4\hat{j}$

$\vec{c}.\left(-2\hat{i}+\hat{j}+\hat{k}\right)=9-4=5$

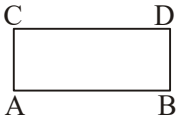
165. If $A(3,1,-1), B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right), C(2,2,1)$ and $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ are the vertices of a quadrilateral ABCD, then its area is: **[JEE(Main)April,2024]**

यदि $A(3,1,-1), B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right), C(2,2,1)$ और $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ चतुर्भुज ABCD के शीर्ष हैं, तो इसका क्षेत्रफल : **[JEE(Main)April,2024]**

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{5\sqrt{2}}{3}$ (C) $2\sqrt{2}$ (D) $\frac{4\sqrt{2}}{3}$

Ans. D

Sol.



Area $=\frac{1}{2}\left|\overline{BD}\times\overline{AC}\right|$

$\overline{BD}=\frac{5}{3}\hat{i}-\frac{5}{3}\hat{j}-\frac{2}{3}\hat{k}$

$\overline{AC}=\hat{i}-\hat{j}-2\hat{k}$

166. Let $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If $\vec{c}$ is a is vector such that $|\vec{c}|\geq 6, \vec{a}\cdot\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 60°, then $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=6\hat{i}+\hat{j}-\hat{k}$ तथा $\vec{b}=\hat{i}+\hat{j}$ है। यदि एक सदिश $\vec{c}$ इस प्रकार है कि $|\vec{c}|\geq 6, \vec{a}\cdot\vec{c}=6|\vec{c}|, |\vec{c}-\vec{a}|=2\sqrt{2}$ तथा $\vec{a}\times\vec{b}$ और $\vec{c}$ के मध्य का कोण 60° है, तो $\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|$ बराबर है : **[JEE(Main)April,2024]**

(A) $\frac{9}{2}\left(6-\sqrt{6}\right)$ (B) $\frac{9}{2}\left(6+\sqrt{6}\right)$ (C) $\frac{3}{2}\sqrt{3}$ (D) $\frac{3}{2}\sqrt{6}$

Ans. B

Sol. $|\left(\vec{a}\times\vec{b}\right)\times\vec{c}|=|\vec{a}\times\vec{b}||\vec{c}|\frac{\sqrt{3}}{2}$

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 6 & 1 & -1\\ 1 & 1 & 0\end{vmatrix}=\hat{i}-\hat{j}+5\hat{k}$$

$|\vec{a}\times\vec{b}|=\sqrt{27}$

$\therefore\ |\vec{c}-\vec{a}|=2\sqrt{2}$

$|\vec{c}|^2+|\vec{a}|-2\vec{c}.\vec{a}=8$

$|\vec{c}|^2+|\vec{a}|-12|\vec{c}|=8$

$|\vec{c}|^2-12|\vec{c}|+30=0$

$|\vec{c}|=6+\sqrt{6}$

$\therefore\ |\vec{c}|>6$

$$\left|\left(\vec{a}\times\vec{b}\right)\times\vec{c}\right|=\sqrt{27}\times\left(6+\sqrt{6}\right)\times\frac{\sqrt{3}}{2}=\frac{9}{2}\left(6+\sqrt{6}\right)$$

167. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k},\vec{b}=\left(\left(\vec{a}\times\left(\hat{i}+\hat{j}\right)\right)\times\hat{i}\right)\times\hat{i}$. Then the square of the projection of $\vec{a}$ on $\vec{b}$ is :

[JEE(Main)April,2024]

माना $\vec{a}=2\hat{i}+\hat{j}-\hat{k},\vec{b}=\left(\left(\vec{a}\times\left(\hat{i}+\hat{j}\right)\right)\times\hat{i}\right)\times\hat{i}$ हैं। तो $\vec{b}$ पर $\vec{a}$ के प्रक्षेप का वर्ग है : **[JEE(Main)April,2024]**

(A) $\frac{2}{3}$ (B) $\frac{1}{5}$ (C) $\frac{1}{3}$ (D) 2

Ans. D

Sol. $$\vec{a}\times\left(\hat{i}+\hat{j}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 1 & -1\\ 1 & 1 & 0\end{vmatrix}=\hat{i}-\hat{j}+\hat{k}$$

$\left(\vec{a}\left(\hat{i}+\hat{j}\right)\right)\times i=k+\hat{j}\quad\Rightarrow|z_2|=\left(\left(\vec{a}\times\left(1+\hat{i}\right)\right)\times\hat{i}\right)\times\hat{i}=\hat{j}-\hat{k}=\hat{b}$

Projection of $\vec{a}$ on $\vec{b}=\frac{\vec{a}.\vec{b}}{|\vec{b}|}=\frac{2}{\sqrt{2}}=\sqrt{2}$

168. Let P(x, y, z) be a point in the first octant, whose projection in the xy-plane is the point Q. Let OP = γ; the angle between OQ and the positive x-axis is θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is : **[JEE(Main)April,2024]**

माना P(x, y, z) प्रथम अष्टांशक में एक बिन्दु है जिसका xy-तल में प्रक्षेप बिन्दु Q है। माना OP = γ है तथा OQ और धनात्मक x-axis कोण θ है एवं OP तथा धनात्मक z-अक्ष के बीच कोण ϕ है, जहाँ O मूल बिन्दु है, तब P की x-अक्ष से दूरी होगी :

[JEE(Main)April,2024]

(A) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$ (B) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

(C) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$ (D) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

Ans. C

Sol. $\overrightarrow{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

$\overrightarrow{OQ} = x\hat{i} + y\hat{j}$

$|OP| = \gamma = \sqrt{x^2 + y^2 + z^2}$

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}} \Rightarrow \cos^2\theta = \frac{x^2}{\gamma^2 - z^2} = \frac{x^2}{\gamma^2 - \gamma^2\cos^2\phi}$$

$$\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}} = \frac{z}{\gamma}$$

Distance of P from x-axis = $\sqrt{y^2+z^2}$

$d = \sqrt{\gamma^2 - x^2}$

$\Rightarrow x^2 = \gamma^2 \sin^2\phi \cos^2\theta$

$\Rightarrow d = \sqrt{\gamma^2 - \gamma^2\sin^2\phi\cos^2\theta}$

$= \gamma\sqrt{1-\sin^2\phi\cos^2\theta}$

169. The set of all α, for which the vectors $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t \in R$, is : **[JEE(Main)April,2024]**

α के सभी मानों का समुच्चय, जिसके लिए सदिशों $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$ तथा $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$ के मध्य अधिक कोण है जहाँ $t \in R$ है : **[JEE(Main)April,2024]**

(A) $\left(-\frac{4}{3}, 1\right)$ (B) $\left(-\frac{4}{3}, 0\right]$ (C) (–2,0] (D) [0,1)

Ans. B

Sol. $\vec{a} = \alpha t\hat{i} + 6\hat{j} - 3\hat{k}$

$\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t\hat{k}$

So, $\vec{a}.\vec{b} < 0$, $\forall\ t \in R$

$\alpha t^2 - 12 + 6\alpha t < 0$

$\alpha t^2 + 6\alpha t - 12 < 0$, $\forall\ t \in R$

$\alpha < 0$, and $D < 0$

$36\alpha^2 + 48\alpha < 0$

$12\alpha(3\alpha + 4) < 0$

$$\frac{-4}{3} < \alpha < 0$$

also for a = 0, $\vec{a}.\vec{b} < 0$

hence a $\alpha \in \left(-\frac{4}{3}, 0\right]$

170. If the shortest distance between the lines

$$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \quad \lambda \in R$$

$$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \quad \mu \in R$$

is $\frac{m}{\sqrt{n}}$, where gcd(m, n) = 1, then the value of m + n equals : **[JEE(Main)April,2024]**

माना रेखाओं के बीच की न्यूनतम दूरी

$$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \quad \lambda \in R$$

$$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \quad \mu \in R$$

$\frac{m}{\sqrt{n}}$ है, जहाँ gcd(m, n) = 1, तब m + n बराबर है : **[JEE(Main)April,2024]**

(A) 387 (B) 390 (C) 377 (D) 384

Ans. A

Sol. $L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}$

$$L_1 = 2\hat{i} + \hat{j} + 3\hat{k} + \lambda(\hat{i} - 3\hat{j} + 4\hat{k})$$

$$L_2 : \vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + \hat{k})$$

$$\vec{a}_1 = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{a}_2 = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = 2\hat{j} + 2\hat{k}$$

$\vec{b}_1 = \hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{b}_2 = 2\hat{i} + 3\hat{j} + \hat{k}$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 4 \\ 2 & 3 & 1 \end{vmatrix}$$

$\hat{i}(-3-12) - \hat{j}(1-8) + \hat{k}(3+6)$

$= -15\hat{i} + 7\hat{j} + 9\hat{k}$

$\left|\vec{b}_1 \times \vec{b}_2\right| = \sqrt{225+49+81}$

$$\left|\frac{(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1 \times \vec{b}_2)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| = \frac{14+18}{\sqrt{355}} = \frac{32}{\sqrt{355}}$$

$m + n = 387$

171. Let $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$. If $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$ then $|\vec{c}|^2$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 11\hat{i} - \hat{j} + \hat{k}$ है तथा एक सदिश $\vec{c}$ के लिए $(\vec{a} + \vec{b}) \times \vec{c} = \vec{c} \times (-2\vec{a} + 3\vec{b})$ है। यदि $(2\vec{a} + 3\vec{b}) \cdot \vec{c} = 1670$ है, तो $|\vec{c}|^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 1618 (B) 1609 (C) 1600 (D) 1627

Ans. A

Sol. $\vec{c} \times (-\vec{a} - \vec{b}) + \vec{c} \times (2\vec{a} - 3\vec{b}) = 0$

$\vec{c} \times (\vec{a} - 4\vec{b}) = 0$

$\vec{c} = \lambda(\vec{a} - 4\vec{b})$

& $(2\vec{a} + 3\vec{b}).\vec{c} = 1670$

$(2\vec{a} + 3\vec{b}).(\lambda)(\vec{a} - 4\vec{b}) = 1670$

$\lambda\left(2|\vec{a}|^2 - 5\vec{a}\cdot\vec{b} - 12|\vec{b}|^2\right) = 1670$

$\lambda(2 \times 18 - 5(46) - 12\ (123)) = 1670$

$\lambda = -1$

$\vec{c} = 4\vec{b} - \vec{a} = 40\hat{i} - 3\hat{j} + 3\hat{k}$

$|\vec{c}|^2 = 1600 + 9 + 9 = 1618$

172. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ be three vectors. Let $\vec{r}$ be a unit vector along $\vec{b} + \vec{c}$. If $\vec{r}\cdot\vec{a} = 3$, then 3λ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिश $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ तथा $\vec{c} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ हैं। माना $\vec{b} + \vec{c}$ के अनुदिश एक मात्रक सदिश $\vec{r}$ है। ; fn $\vec{r}\cdot\vec{a} = 3$ है, तो 3λ बराबर है : **[JEE(Main)April,2024]**

(A) 30 (B) 21 (C) 25 (D) 27

Ans. C

Sol. $\vec{r} = k\left(\vec{b} + \vec{c}\right)$

$\vec{r}\cdot\vec{a} = 3$

$\vec{r}\cdot\vec{a} = k\left(\vec{b}\cdot\vec{a} + \vec{c}\cdot\vec{a}\right)$

$3 = k(2 + 6 - 15 + 3 - 2 + 3\lambda)$

$3 = k(-3 + 3\lambda)$ ...(1)

$\vec{r} = k\left(5\hat{i} - 2\hat{j} - (5 - \lambda)\hat{k}\right)$

$|\vec{r}| = k\sqrt{25 + 4 + 25 + \lambda^2 - 10\lambda}$

$k = \dfrac{3}{-6 + 3\lambda} = \dfrac{1}{-2 + \lambda}$ put in (1)

$4 + \lambda^2 - 4\lambda = (54 + \lambda^2 - 10\lambda)$

$6\lambda = 50$

$3\lambda = 25$

173. Let three vectors $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}, \vec{c} = x\hat{i} + y\hat{i} + z\hat{k}$ form a triangle such that $\vec{c} = \vec{a} - \vec{b}$ and the area of the triangle is $5\sqrt{6}$. If α is a positive real number, then $|\vec{c}|^2$ is equal to : **[JEE(Main)April,2024]**

माना तीन सदिश $\vec{a} = \alpha\hat{i} + 4\hat{j} + 2\hat{k}, \vec{b} = 5\hat{i} + 3\hat{j} + 4\hat{k}, \vec{c} = x\hat{i} + y\hat{i} + z\hat{k}$ एक त्रिभुज बनाते है, जो इस प्रकार है कि $\vec{c} = \vec{a} - \vec{b}$ है r Hkkbl f=Hkqt dk {ks=Qy $5\sqrt{6}$ है। यदि α एक धनात्मक वास्तविक संख्या है, तो $|\vec{c}|^2$ बराबर है :

[JEE(Main)April,2024]

(A) 14 (B) 10 (C) 12 (D) 16

Ans. A

Sol. $c = (\alpha - 5)\hat{i} + \hat{j} - 2\hat{k}$

Area of Δ ABC $= \left|\frac{1}{2}\left(\vec{a}\times\vec{b}\right)\right| = 5\sqrt{6} \Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 4 & 2 \\ 5 & 3 & 4 \end{vmatrix} = 10\sqrt{6}$

$\left|10\hat{i}-(4\alpha-10)\hat{j}+(3\alpha-20)\hat{k}\right| = 10\sqrt{6}$

$\Rightarrow \sqrt{100+(10-4\alpha)^2+(3\alpha-20)^2} = 10\sqrt{6}$

$\Rightarrow 25\alpha^2 - 200\alpha + 600 = 600$

$\Rightarrow \alpha = 8 \qquad (\because a \in R^+)$

$\vec{c} = 3\hat{i}+\hat{j}-2\hat{k}$

$\Rightarrow |\vec{c}|^2 = 9+1+4 = 14$

174. Let $\overrightarrow{OA} = 2\vec{a}, \overrightarrow{OB} = 6\vec{a}+5\vec{b}$ and $\overrightarrow{OC} = 3\vec{b}$, where O is the origin. If the area of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is 15 sq. units, then the area (in sq. units) of the quadrilateral OABC is equal to:

[JEE(Main)April,2024]

माना $\overrightarrow{OA} = 2\vec{a}, \overrightarrow{OB} = 6\vec{a}+5\vec{b}$ तथा $\overrightarrow{OC} = 3\vec{b}$ है, जहाँ O मूल बिन्दु है। यदि $\overrightarrow{OA}$ तथा $\overrightarrow{OC}$ संलग्न भुजाओं वाले समांतर pr d k{ksk Qy 15 वर्ग इकाई है, तो चतुर्भुज OABC का क्षेत्रफल (वर्ग इकाई में) बराबर है :

[JEE(Main)April,2024]

(A) 40 (B) 35 (C) 38 (D) 32

Ans. B

Sol. Area of parallelogram having sides

$\overrightarrow{OA}$ & $\overrightarrow{OC} = \left|\overrightarrow{OA}\times\overrightarrow{OC}\right| = \left|2\vec{a}\times3\vec{b}\right| = 15$

$6\left|\vec{a}\times\vec{b}\right| = 15$

$\Rightarrow |\vec{a}\times\vec{b}| = \frac{5}{2}$ .....(1)

Area of quadrilateral

$OABC = \frac{1}{2}\left|\vec{d}_1\times\vec{d}_2\right|$

$= \frac{1}{2}\left|\overrightarrow{AC}\times\overrightarrow{OB}\right| = \frac{1}{2}\left|\left(3\vec{b}-2\vec{a}\right)\times\left(6\vec{a}+5\vec{b}\right)\right|$

$$= \frac{1}{2}\left|18\vec{b}\times\vec{a} - 10\vec{a}\times\vec{b}\right| = 14\,|\,\vec{a}\times\vec{b}\,|$$

$$= 14\times\frac{5}{2} = 35$$

175. Between the following two statements :

Statement I : Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$. Then the vector $\vec{r}$ satisfying $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ and $\vec{a}\cdot\vec{r} = 0$ is of magnitude $\sqrt{10}$.

Statement II : In a triangle ABC, $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$. **[JEE(Main)April,2024]**

(A) Statement I is correct but Statement II is incorrect.

(B) Statement I is incorrect but Statement II is correct.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

निम्नलिखित दो कथनों :

कथन I : माना $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ और $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ हैं। तो $\vec{a}\times\vec{r} = \vec{a}\times\vec{b}$ तथा $\vec{a}\cdot\vec{r} = 0$ को संतुष्ट करने वाले सदिश $\vec{r}$ का परिमाण $\sqrt{10}$ है।

कथन II : एक त्रिभुज ABC में $\cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$ है। **[JEE(Main)April,2024]**

(A) कथन I सही है परन्तु कथन II गलत है।

(B) कथन I गलत है पररन्तु II सही है।

(C) कथन I तथा कथन II दोनों सही है।

(D) कथन I तथा कथन II गलत है।

Ans. B

Sol. $\forall$ two vectors $\vec{c}$ and $\vec{d}$

$$\left|\vec{c}\times\vec{d}\right|^2 = |\vec{c}|^2\left|\vec{d}\right|^2 - \left(\vec{c}\cdot\vec{d}\right)^2$$

replacing $\vec{c} = \vec{a}$ and $\vec{d} = \vec{r}$

$$\Rightarrow \left|\vec{a}\times\vec{r}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2 - \left(\vec{a}\cdot\vec{r}\right)^2$$

$$\Rightarrow \quad \left|\vec{a}\times\vec{b}\right| = \left|\vec{a}\right|^2\left|\vec{r}\right|^2$$

$\left(\because \vec{a}\times\vec{r}=\vec{a}\times\vec{b} \text{ and } \vec{a}\cdot\vec{r}=0\right)$

$\Rightarrow 35=14|\vec{r}|^2$

$\Rightarrow |\vec{r}|=\sqrt{\frac{35}{14}}=\sqrt{\frac{5}{2}}\neq\sqrt{10}$

$\therefore$ Statement {I} is incorrect Statement II is correct

$\left(\text{i.e., } \cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}\right)$

Proof: $\because \left(\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}\right)\geq 0$ _____(1)

and $\left|\overrightarrow{OA}\right|^2=\left|\overrightarrow{OB}\right|^2=\left|\overrightarrow{OC}\right|^2=R^2$ _____(2)

Now, using (1), we get $\left|\overrightarrow{OA}\right|^2+\left|\overrightarrow{OB}\right|^2+\left|\overrightarrow{OC}\right|^2+2\left(\overrightarrow{OA}\cdot\overrightarrow{OB}+\overrightarrow{OB}\cdot\overrightarrow{OC}+\overrightarrow{OC}\cdot\overrightarrow{OA}\right)\geq 0$

$\Rightarrow 3R^2+2R^2(\cos 2A+\cos 2B+\cos 2C)\geq 0$

$\Rightarrow \cos 2A+\cos 2B+\cos 2C\geq -\frac{3}{2}$

176. Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b}=-\hat{i}+\hat{k}$, $\vec{c}=\beta\hat{j}-\hat{k}$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α, β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1, β_1) and (α_2, β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to : **[JEE(Main)April,2024]**

माना $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k}$, $\vec{b}=-\hat{i}+\hat{k}$, $\vec{c}=\beta\hat{j}-\hat{k}$ हैं, जहाँ α और β पूर्णांक हैं और $\alpha\beta=-6$ है। माना क्रमित युग्म (α, β) के मान ft ud sfy, fod.kkZ $\vec{a}+\vec{b}$ और $\vec{b}+\vec{c}$ के समांतर चतुर्भुज का क्षेत्रफल $\frac{\sqrt{21}}{2}$ है, (α_1, β_1) और (α_2, β_2) हैं। तो $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ बराबर है : **[JEE(Main)April,2024]**

(A) 21 (B) 17 (C) 19 (D) 24

Ans. C

Sol. Area of parallelogram $=\frac{1}{2}\left|\vec{d}_1\times\vec{d}_2\right|$

$A=\frac{1}{2}\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|$

$=\frac{\sqrt{21}}{2}$

so, $\vec{a}+\vec{b}=\hat{i}+\alpha\hat{j}+2\hat{k}$

$\vec{b}+\vec{c}=-\hat{i}+\beta\hat{j}$

$$\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & \alpha & 2\\ -1 & \beta & 0\end{vmatrix}$$

$=\hat{i}(-2\beta)-\hat{j}(2)+\hat{k}(\beta+\alpha)$

$\left|\left(\vec{a}+\vec{b}\right)\times\left(\vec{b}+\vec{c}\right)\right|=\sqrt{4\beta^2+4+(\alpha+\beta)^2}=\sqrt{21}$

$4\beta^2 + 4 + \alpha^2 + \beta^2 + 2\alpha\beta = 21$

$\alpha^2 + 5\beta^2 - 12 = 17$

$\alpha^2+5\beta^2=29$ and $\alpha\beta=-6$

and $\alpha\beta = -6$

and given $\alpha_i\beta$ are integers

so,

$\alpha = -3,\ \beta = 2$

or

$\alpha = 3,\ \beta = -2$

$(\alpha_1, \beta_1) = (-3, 2)$

$(\alpha_2, \beta_2) = (3, -2)$

$\alpha_1^2+\beta_1^2-\alpha_2\beta_2 = 9 + 4 + 6 = 19$

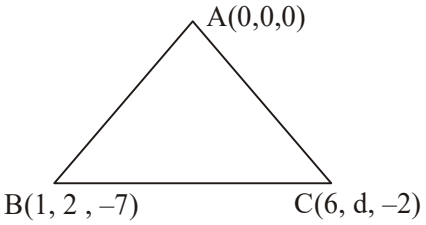
177. Let ABC be a triangle of area $15\sqrt{2}$ and the vectors $\overrightarrow{AB}=\hat{i}+2\hat{j}-7\hat{k}$, $\overrightarrow{BC}=a\hat{i}+b\hat{j}+c\hat{k}$ and $\overrightarrow{AC}=6\hat{i}+d\hat{j}-2\hat{k}$, $d>0$. Then the square of the length of the largest side of the triangle ABC is _____.

[JEE(Main)April,2024]

माना एक त्रिभुज ABC का क्षेत्रफल $15\sqrt{2}$ है तथा सदिश $\overrightarrow{AB}=\hat{i}+2\hat{j}-7\hat{k}$, $\overrightarrow{BC}=a\hat{i}+b\hat{j}+c\hat{k}$ और $\overrightarrow{AC}=6\hat{i}+d\hat{j}-2\hat{k}$, $d>0$ हैं। तो त्रिभुज ABC की सबसे बड़ी भुजा की लम्बाई का वर्ग है _____ **[JEE(Main)April,2024]**

Ans. 54

Sol.



$$\frac{1}{2}\begin{vmatrix} i & j & k \\ 1 & 2 & -7 \\ 6 & d & -2 \end{vmatrix} = \pm 30\sqrt{2}$$

$$\hat{i}(-4+7d) - \hat{j}(-2+42) + \hat{k}(d-12) = \pm 30\sqrt{2}$$

$\Rightarrow (7d-4)^2 + 40^2 + (d-12)^2 = 1800$

$49d^2 + 16 - 56d + 1600 + d^2 + 144 - 24d = 1800$

$50d^2 - 80d - 40 = 0$

$5d^2 - 8d - 4 = 0$

$5d^2 - 10d + 2d - 4 = 0$

$(5d+2)(d-2) = 0$

$d = 2$

$AB = \sqrt{54}$

$AC = \sqrt{36+4+4} = \sqrt{44}$

$BC = \sqrt{25+0+25}$

178. Let $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{i} + \hat{k}$ and $\vec{c}$ be a vector such that $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$. If $\vec{a} \cdot \vec{c} = 130$, then $\vec{b} \cdot \vec{c}$ is equal to_____. **[JEE(Main)April,2024]**

माना $\vec{a} = \hat{i} - 3\hat{j} + 7\hat{k}, \vec{b} = 2\hat{i} - \hat{i} + \hat{k}$ तथा एक सदिश $\vec{c}$ के लिए $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$ हैं। यदि $\vec{a} \cdot \vec{c} = 130$ है, तो $\vec{b} \cdot \vec{c}$ बराबर है_____ **[JEE(Main)April,2024]**

Ans. 30

Sol. $(\vec{a} + 2\vec{b}) \times \vec{c} = 3(\vec{c} \times \vec{a})$

$\Rightarrow \vec{b} \times \vec{c} + 2(\vec{a} \times \vec{c}) = 0$

$(\vec{b} + 2\vec{a}) \times \vec{c} = 0$

$\vec{c} = \lambda(\vec{b} + 2\vec{a})$

$\vec{c}.\vec{a} = 130 \Rightarrow \lambda = 1$

$\vec{c} = 4\hat{i} - 7\hat{j} + 15\hat{k}$

$\vec{b}.\vec{c} = 30$

179. Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}, \vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times \vec{c} = \hat{i} + 8\hat{j} + 13\hat{k}$. If $\vec{a} \cdot \vec{c} = 13$, then $(24 - \vec{b} \cdot \vec{c})$ is equal to _____. **[JEE(Main)April,2024]**

माना $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k},\vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ तथा सदिश $\vec{c}$ इस प्रकार है कि $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$ है। यदि $\vec{a}\cdot\vec{c}=13$, तो $\left(24-\vec{b}\cdot\vec{c}\right)$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 46

Sol. $\vec{a}\times\vec{b}+\vec{a}\times\vec{c}+\vec{b}\times\vec{c}=(1,8,13)$

$\vec{a}\times\left(\vec{a}\times\vec{b}\right)+\vec{a}\times\left(\vec{a}\times\vec{c}\right)+\vec{a}\times\left(\vec{b}\times\vec{c}\right)$

$=\vec{a}\times\left(\hat{i}+8\hat{j}+13k\right)$

$\left(\vec{a}\cdot\vec{b}\right)\vec{a}-a^2\vec{b}+\left(\vec{a}\cdot\vec{c}\right)\vec{a}-a^2\vec{c}+\left(\vec{a}\cdot\vec{c}\right)\vec{b}-\left(\vec{a}\cdot\vec{b}\right)\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow -26\vec{a}-29\vec{b}+13\vec{a}-29\vec{c}+13\vec{b}+26\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow \quad -13\vec{a}-16\vec{b}-3\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)$

$\Rightarrow \quad -13\vec{a}\cdot\vec{b}-16b^2-3\vec{b}\cdot\vec{c}=\vec{a}\times\left(\hat{i}+8\hat{j}+13\hat{k}\right)\cdot\vec{b}$

$$\Rightarrow \quad (-13)(-26)-16(50)-3\vec{b}\cdot\vec{c}=\begin{vmatrix}2 & -3 & 4\\ 1 & 8 & 13\\ 3 & 4 & -5\end{vmatrix}$$

$\Rightarrow -462-3\vec{b}\cdot\vec{c}=-396$

$\Rightarrow \vec{b}\cdot\vec{c}=-22$

Hence $24-\vec{b}\cdot\vec{c}=46$

180. Let $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$, $\vec{b}=3\hat{i}+7\hat{j}-13\hat{k}$ and $\vec{c}=17\hat{i}-2\hat{j}+\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\left(\vec{b}+\vec{c}\right)\times\vec{a}$ and $\vec{r}\cdot\left(\vec{b}-\vec{c}\right)=0$, then $\dfrac{\left|593\vec{r}+67\vec{a}\right|^2}{(593)^2}$ is equal to _____.

[JEE(Main)April,2024]

माना $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$, $\vec{b}=3\hat{i}+7\hat{j}-13\hat{k}$ तथा $\vec{c}=17\hat{i}-2\hat{j}+\hat{k}$ तीन सदिश दिये गये है यदि $\vec{r}$ एक सदिश इस प्रकार है fd $\vec{r}\times\vec{a}=\left(\vec{b}+\vec{c}\right)\times\vec{a}$ तथा $\vec{r}\cdot\left(\vec{b}-\vec{c}\right)=0$ है, तो $\dfrac{\left|593\vec{r}+67\vec{a}\right|^2}{(593)^2}$ बराबर होगा _____

[JEE(Main)April,2024]

Ans. 569

Sol. $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}$

$\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$

$\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$

$\vec{r} \times \vec{a} = \left(\vec{b} + \vec{c}\right) \times \vec{a}$

$\left(\vec{r} - \left(\vec{b} + \vec{c}\right)\right) \times \vec{a} = 0$

$\Rightarrow \vec{r} = \left(\vec{b} + \vec{c}\right) + \lambda\vec{a}$

$\vec{r} = \left(20\hat{i} + 5\hat{j} - 12\hat{k}\right) + \lambda\left(9\hat{i} - 13\hat{j} + 25\hat{k}\right)$

$= \left(20 + 9\lambda\right)\hat{i} + \left(5 - 13\lambda\right)\hat{j} + \left(25\lambda - 12\right)\hat{k}$

Now, $\vec{r} \cdot \left(\vec{b} - \vec{c}\right) = 0$

$\vec{r} \cdot \left(-14\hat{i} + 9\hat{j} - 14\hat{k}\right) = 0$

Now $-14(20 + 9\lambda) + 9(5 - 13\lambda) - 14(25\lambda - 12) = 0$

$-593\lambda - 67 = 0$

$\lambda = -\dfrac{67}{593}$

$\therefore \vec{r} = \left(\vec{b} + \vec{c}\right) - \dfrac{67}{593}\vec{a}$

$$\frac{\left|593\vec{r} + 67\vec{a}\right|^2}{\left|593\right|^2} = \left|\vec{b} + \vec{c}\right|^2 = \left|20\hat{i} + 5\hat{j} - 12\hat{k}\right|^2$$

$= 569$

181. Let the point, on the line passing through the points P(1, –2,3) and Q(5, –4, 7), farther from the origin and at a distance of 9 units from the point P, be (α, β, γ). Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

[JEE(Main)April,2024]

माना बिन्दुओं P(1, –2,3) तथा Q(5, –4, 7) से होकर जाने वाली रेखा पर, मूल बिन्दु से ज्यादा दूर तथा बिन्दु P से 9 इकाई दूरी पर, स्थित बिन्दु (α, β, γ) है, तो $\alpha^2 + \beta^2 + \gamma^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 155 (B) 150 (C) 165 (D) 160

Ans. A

Sol. PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

pt(4t + 1, –2t – 2, 4t + 3)

distance^2 = $16t^2 + 4t^2 + 16t^2 = 81$

$t = \pm\frac{3}{2}$

pt(7, –5, 9)

$\alpha^2 + \beta^2 + \gamma^2 = 155$

182. Let P be the point of intersection of the lines $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$ and $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$. Then, the shortest distance of P from the line $4x = 2y = z$ is : **[JEE(Main)April,2024]**

माना रेखाओं $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$ तथा $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$ का प्रतिच्छेदी बिन्दु P है, तो रेखा $4x = 2y = z$ से बिन्दु P की

U;wure nyjhgS **[JEE(Main)April,2024]**

(A) $\frac{6\sqrt{14}}{7}$ (B) $\frac{3\sqrt{14}}{7}$ (C) $\frac{5\sqrt{14}}{7}$ (D) $\frac{\sqrt{14}}{7}$

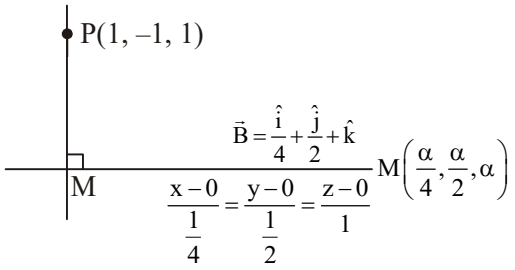
Ans. B

Sol. For point of intersection

$P(\lambda + 2, 5\lambda + 4, \lambda + 2) = (2\mu + 3, 3\mu + 2, 2\mu + 3)$

$\lambda + 2 = 2\mu + 3$, $5\lambda + 4 = 3\mu + 2$, $\lambda + 2 = 2\mu + 3$

$\lambda = -1$, $\mu = -1$ point P = Q (1, – 1, 1)



$\overrightarrow{PM}.\vec{B} = 0$

$\frac{1}{4}\left(\frac{\alpha}{4} - 1\right) + \frac{1}{2}\left(\frac{\alpha}{2} + 1\right) + 1(\alpha - 1) = 0 \qquad \Rightarrow \alpha = \frac{4}{7}$

$M = \left(\frac{1}{7}, \frac{2}{7}, \frac{4}{7}\right)$

Shortest distance PM = $\sqrt{\left(\frac{1}{7}-1\right)^2+\left(\frac{2}{7}+1\right)^2+\left(\frac{4}{7}-1\right)^2}=\frac{3\sqrt{14}}{7}$

183. If the line $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$ makes a right angle with the line $\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7}$ then $4\lambda+9\mu$ is equal to :

[JEE(Main)April,2024]

यदि रेखा $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$, रेखा $\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7}$ से एक समकोण बनाती है, तो $4\lambda + 9\mu$ बराबर है :

[JEE(Main)April,2024]

(A) 6 (B) 13 (C) 4 (D) 5

Ans. A

Sol. $\frac{2-x}{3}=\frac{3y-2}{4\lambda+1}=4-z$ .....(1)

$$\frac{2-x}{(-3)}=\frac{y-\frac{2}{3}}{\left(\frac{4\lambda+1}{3}\right)}=\frac{z-4}{(-1)}$$

$$\frac{x+3}{3\mu}=\frac{1-2y}{6}=\frac{5-z}{7} \quad .....(2)$$

$$\frac{x+3}{3\mu}=\frac{y-\frac{1}{2}}{(-3)}=\frac{z-5}{(-7)}$$

Right angle $\Rightarrow (-3)(3\mu)+\left(\frac{4\lambda+1}{3}\right)(-3)+(-1)(-7)=0$

$-9\mu-4\lambda-1+7=0$

$4\lambda+9\mu=6$

184. Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point (7, 8, 9). Then $d^2 + 6$ is equal to :

[JEE(Main)April,2024]

माना रेखाओं $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ तथा $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ के प्रतिच्छेदन बिंदु की बिंदु (7, 8, 9) से दूरी d है। तो $d^2 + 6$ बराबर है :

[JEE(Main)April,2024]

(A) 69 (B) 72 (C) 75 (D) 78

Ans. C

Sol. $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}=\lambda$ _____(1)

$x=3\lambda-6, y=2\lambda, z=\lambda-1$

$\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}=\mu$ _____(2)

$x=4\mu+7, y=3\mu+9, z=2\mu+4$

$3\lambda-6=4\mu+7 \Rightarrow 3\lambda-4\mu=13$ _____(3) × 2

$2\lambda=3\mu+9 \Rightarrow 2\lambda-3\mu=9$ _____(4) × 3

$6\lambda-8\mu=26$

$6\lambda-9\mu=27$

$- \quad + \quad -$

$\mu=-1$

$\Rightarrow 3\lambda-4(-1)=13$

$3\lambda=9$

$\lambda=3$

int. point (3, 6, 2) ; (7, 8, 9)

$d^2=16+4+49=69$

Ans. $d^2+6=69+6=75$

185. Let (α, β, γ) be the image of the point (8, 5, 7) in the line $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-2}{5}$. Then $\alpha+\beta+\gamma$ is equal to:

[JEE(Main)April,2024]

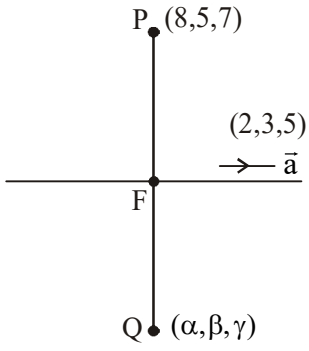
माना बिन्दु (8, 5, 7) का रेखा $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-2}{5}$ में प्रतिबिंब (α, β, γ) है। तो $\alpha+\beta+\gamma$ बराबर है :

[JEE(Main)April,2024]

(A) 14 (B) 20 (C) 18 (D) 16

Ans. A

Sol.



Coordinate of F <$2\lambda + 1, 3\lambda - 1, 5\lambda + 2$>

$P\vec{F} = < 2\lambda - 7, 3\lambda - 6, 5\lambda - 5 >$

$\Pi\vec{F} . \vec{a} = 0 \Rightarrow \lambda = \frac{3}{2}$

$\frac{\alpha+8}{2} = 2\lambda+1, \frac{\beta+5}{2} = 3\lambda-1, \frac{\gamma+7}{2} = 5\lambda+2$

$\Rightarrow \alpha = 4\lambda - 6, \beta = 6\lambda - 7, \gamma = 10\lambda - 3$

$\Rightarrow \alpha + \beta + \gamma = 20\lambda - 16 = 14$

186. The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is :

[JEE(Main)April,2024]

रेखाओं $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ और $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ के बीच की न्यूनतम दूरी है :

[JEE(Main)April,2024]

(A) $8\sqrt{3}$ (B) $6\sqrt{3}$ (C) $4\sqrt{3}$ (D) $5\sqrt{3}$

Ans. C

Sol. $\vec{a}_1 = 3\hat{i} - 15\hat{j} + 9\hat{k}$

$\vec{a}_2 = -\hat{i} + \hat{j} + 9\hat{k}$

$\vec{b} = 2\hat{i} - 7\hat{j} + 5\hat{k}$

$\vec{d} = 2\hat{i} + \hat{j} - 3\hat{k}$

Shortest distance $= \left| \frac{(\vec{a}_1 - \vec{a}_2) \cdot (\vec{b} \times \vec{d})}{(\vec{b} \times \vec{d})} \right|$

$\left| \frac{4(16) + (-16) \times 16 + 0 \times 16}{\sqrt{16^2 + 16^2 + 16^2}} \right|$ (where $\vec{b} \times \vec{d} = 16\hat{i} + 16\hat{j} + 16\hat{k}$)

$= \frac{16 \times 12}{16\sqrt{3}} = 4\sqrt{3}$

187. Let P(α, β, γ) be the image of the point Q(3, –3, 1) in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point(2, 5, –1). If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to :

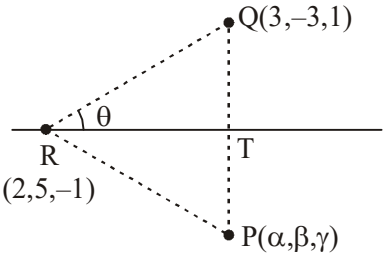
[JEE(Main)April,2024]

माना रेखा $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ में बिन्दु Q(3, –3, 1) का प्रतिबिंब बिन्दु P(α, β, γ) है और R(2, 5, –1) एक अन्य बिन्दु है। यदि PQR का क्षेत्रफल λ है तथा $\lambda^2 = 14K$, तो K बराबर है : **[JEE(Main)April,2024]**

(A) 18 (B) 36 (C) 72 (D) 81

Ans. D

Sol.



$$QR=\sqrt{(3-2)^2+(-3-5)^2+(1+1)^2}$$

$$=\sqrt{1+64+4}=\sqrt{69}$$

$$\overrightarrow{RQ}=\hat{i}-8\hat{j}+2\hat{k}$$

$$\overrightarrow{RT}=\hat{i}+\hat{j}-\hat{k}$$

$$\cos\theta=\frac{1-8-2}{\sqrt{1+64+4}\sqrt{1+1+1}}=\frac{-9}{\sqrt{69}}=\frac{-9}{3\sqrt{23}}=\frac{-3}{\sqrt{23}}$$

$$\sin\theta=\frac{\sqrt{14}}{\sqrt{23}} \text{ and } \cos\theta=\frac{RS}{RQ}=\frac{3}{\sqrt{23}}$$

$$QS=\frac{\sqrt{14}}{23}\times\sqrt{69}=\sqrt{42} \qquad RS=\frac{3}{\sqrt{23}}\times\sqrt{69}=3\sqrt{3}$$

$$\text{Area of } \Delta PQR=\frac{1}{2}(PQ)(RS)$$

$$=\frac{1}{2}\times 2\sqrt{42}\times 3\sqrt{3}$$

$$\lambda=9\sqrt{14}$$

$\lambda^2 = 81 \times 14 = 14K$

$\therefore K = 81$

188. If the shortest distance between the lines $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ and $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is : **[JEE(Main)April,2024]**

यदि रेखाओं $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ तथा $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ के बीच की न्यूनतम दूरी $\frac{13}{\sqrt{29}}$ है, तो λ का एक मान है :

[JEE(Main)April,2024]

(A) $\frac{13}{25}$ (B) 1 (C) $-\frac{13}{25}$ (D) –1

Ans. B

Sol. Given lines are parallel

$$d=\frac{\left|\left(\vec{b}-\vec{a}\right)\times\vec{p}\right|}{|\vec{p}|}$$

$$\vec{b}-\vec{a}=(2-\gamma)\hat{i}+\hat{j}+4\hat{k}$$

$$\vec{p}=2\hat{i}+3\hat{j}+4\hat{k}$$

$$\left(\vec{b}-\vec{a}\right)\times\vec{P}=-12\hat{i}-(8-4\lambda-8)\hat{j}+(6-3\lambda-0)\hat{k}\ =-12\hat{i}+4\lambda\hat{j}+3(2-\lambda)\hat{k}$$

$$d=\frac{\sqrt{144+16\lambda^2+9(\lambda-2)^2}}{\sqrt{4+9+16}}=\frac{13}{\sqrt{29}}$$

$144 + 16\lambda^2 + 9\lambda^2 – 36\lambda + 36 = 169$

$25\lambda^2 – 36\lambda + 11 = 0$

$(25\lambda – 11)\ (\lambda – 1) = 0$

$\lambda=\frac{11}{25}$ or $\lambda = 1$

189. Let the line L intersect the lines $x – 2 = – y = z –1$, $2(x + 1) = 2\ (y – 1) = z\ + 1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. Then which of the following points lies on L? **[JEE(Main)April,2024]**

माना रेखा L, रेखाओं $x – 2 = – y = z –1$, $2(x + 1) = 2\ (y – 1) = z\ + 1$ को काटती है, तथा रेखा $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$ के समान्तर है। तो निम्न में से कौनसा बिन्दु L पर है? **[JEE(Main)April,2024]**

(A) $\left(-\frac{1}{3},-1,-1\right)$ (B) $\left(-\frac{1}{3},-1,1\right)$ (C) $\left(-\frac{1}{3},1,-1\right)$ (D) $\left(-\frac{1}{3},1,1\right)$

Ans. C

Sol. $\frac{x-2}{1}=\frac{y}{-1}=\frac{z-1}{1},\frac{x+1}{1/2}=\frac{y-1}{1/2}=\frac{z+1}{1}$

$(k+2, -k, k+1), \left(\frac{\alpha}{2}-1, \frac{\alpha}{2}+1, \alpha-1\right)$

$$\frac{\frac{\alpha}{2}-1-k-2}{3}=\frac{\frac{\alpha}{2}+1+k}{1}\frac{\alpha-1-k-1}{2}$$

$$k=-\frac{4}{3}$$

$\therefore$ L passes through point $\left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$.

$$L: \frac{x-\frac{2}{3}}{3}=\frac{y-\frac{4}{3}}{1}=\frac{z+\frac{1}{3}}{2}$$

$\left(-\frac{1}{3}, 1, -1\right)$ lies on L

190. The shortest distance between the lines $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is :

[JEE(Main)April,2024]

रेखाओं $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ तथा $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ के बीच की न्यूनतम दूरी है :

[JEE(Main)April,2024]

(A) $\frac{185}{\sqrt{563}}$ (B) $\frac{187}{\sqrt{563}}$ (C) $\frac{178}{\sqrt{563}}$ (D) $\frac{179}{\sqrt{563}}$

Ans. B

Sol. Given lines are

$\vec{r}=\left(3\hat{i}-7\hat{j}+\hat{k}\right)+\lambda\left(4\hat{i}-11\hat{j}+5\hat{k}\right)$ and $\vec{r}=\left(5\hat{i}+9\hat{j}-2\hat{k}\right)+\mu\left(3\hat{i}-6\hat{j}+\hat{k}\right)=\vec{a}_2+\mu\vec{b}_2$

form shortest distance between two lines

$$\left|\frac{\left(\vec{a}_2-\vec{a}_1\right).\left(\vec{b}_1\times\vec{b}_2\right)}{|\vec{b}_1\times\vec{b}_2|}\right|$$

$$\left|\frac{\left(2\hat{i}+16\hat{j}-3\hat{k}\right).\left(19\hat{i}+11\hat{j}+9\hat{k}\right)}{\sqrt{(19)^2+(11)^2+(9)^2}}\right| \text{ here } \vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 4 & -11 & 5\\ 3 & -6 & 1\end{vmatrix}=19\hat{i}+11\hat{j}+9\hat{k}$$

$$=\frac{38+176-27}{\sqrt{563}}$$

$$=\frac{187}{\sqrt{563}}$$

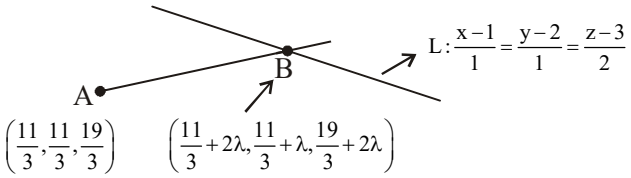
191. Consider the line L passing through the points (1,2,3) and (2,3,5). The distance of the point $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ is equal to : **[JEE(Main)April,2024]**

बिन्दुओं (1,2,3) तथा (2,3,5) से होकर जाने वाली रेखा L का विचार कीजिए। रेखा $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ के अनुदिश fcUnq $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ की रेखा L से दूरी है : **[JEE(Main)April,2024]**

(A) 6 (B) 5 (C) 3 (D) 4

Ans. C

Sol. $A\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ lies on line $\frac{x-\frac{11}{3}}{2}=\frac{y-\frac{11}{3}}{1}=\frac{z-\frac{19}{3}}{2}=\lambda$



B lies on L so $\frac{11}{3}+2\lambda-1=\frac{11}{3}+\lambda-2=\frac{\frac{19}{3}+2\lambda-3}{2}$

$\Rightarrow \lambda=-1$

So, $B\left(\frac{11}{3}-2,\frac{11}{3}-1,\frac{19}{3}-2\right)$ So, AB $=\sqrt{4+1+4}=3$

192. Consider a line L passing through the points P(1, 2, 1) and Q(2, 1, –1). If the mirror image of the point A(2, 2, 2) in the line L is (α, β, γ), then $\alpha+\beta+6\gamma$ is equal to _____. **[JEE(Main)April,2024]**

बिन्दुओं P(1, 2, 1) तथा Q(2, 1, –1) से गुजरने वाली रेखा L पर विचार कीजिए। यदि बिन्दु A(2, 2, 2) का दर्पण प्रतिचित्रण रेखा L में (α, β, γ) है, तो $\alpha + \beta + 6\gamma$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 6

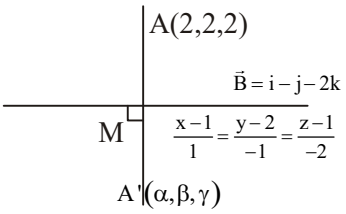
Sol. $M(\lambda + 1, -\lambda + 2, -2\lambda + 1)$

$\overrightarrow{PM} \cdot \vec{B} = 0$

$1(\lambda - 1) - 1(-\lambda) - 2(-2\lambda - 1) = 0$

$\lambda = -\frac{1}{6}$

$M\left(\frac{5}{6}, \frac{13}{6}, \frac{4}{3}\right), \left(\frac{\alpha+2}{2}, \frac{\beta+2}{2}, \frac{\gamma+2}{2}\right)$



$\alpha = -\frac{1}{3}, \beta = \frac{7}{3}, \gamma = \frac{2}{3}$

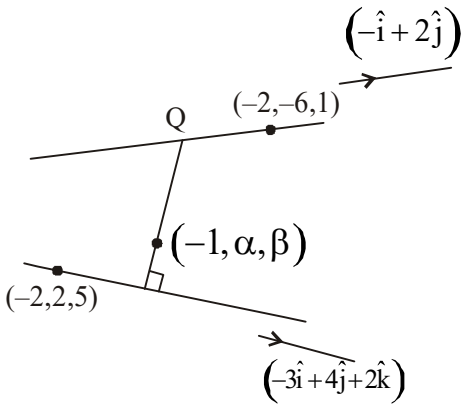
$\Rightarrow \alpha + \beta + 6\gamma = 6$

193. Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____. **[JEE(Main)April,2024]**

माना बिन्दु $(-1, \alpha, \beta)$ रेखाओं $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ तथा $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$ के बीच न्यूनतम दूरी की रेखा पर है। तो $(\alpha - \beta)^2$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 25

Sol.



$P(-3\lambda - 2, 4\lambda + 2, 2\lambda + 5)$

$Q(-\mu-2, 2\mu-6, 1)$

DRS of PQ = $(3\lambda-\mu, 2\mu-4\lambda-8, -2\lambda-4)$

$$\text{DRS of PQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 0 \\ -3 & 4 & 2 \end{vmatrix} = \left(4\hat{i}+2\hat{j}+2\hat{k}\right)$$

OR

(2, 1, 1)

$$\frac{3\lambda-\mu}{2} = \frac{2\mu-4\lambda-8}{1} = \frac{-2\lambda-4}{1}$$

$\Rightarrow \mu = \lambda + 2$ & $7\lambda = \mu - 8$

$\boxed{\lambda = -1}$ $\boxed{\mu = 1}$

Q : (–3, –4, 1)

$$L_{PQ} = \frac{x+3}{2} = \frac{y+4}{1} = \frac{z-1}{1}$$

$$(-1, \alpha, \beta) \Rightarrow 1 = \frac{\alpha+4}{1} = \frac{\beta-1}{1}$$

$\Rightarrow \alpha = -3, \beta = 2$

$(\alpha-\beta)^2 = 25$

194. Let P be the point (10, –2, –1) and Q be the foot of the perpendicular drawn from the point R(1, 7, 6) on the line passing through the points (2, –5, 11) and (–6, 7, –5). Then the length of the line segment PQ is equal to _____.

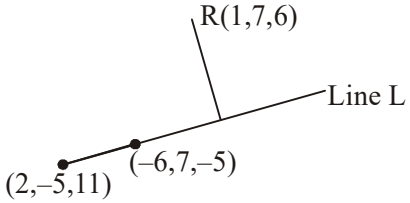
[JEE(Main)April,2024]

माना बिन्दु P के निर्देशांक (10, –2, –1) है तथा बिन्दुओं (2, –5, 11) और (–6, 7, –5) से गुजरने वाली रेखा पर बिन्दु R(1, 7, 6) से डाले गए लंब का पाद Q है, तो रेखाखण्ड PQ की लम्बाई बराबर है _____

[JEE(Main)April,2024]

Ans. 13

Sol.



equation of L

$$\frac{x-2}{-6-2}=\frac{y+5}{7+5}=\frac{z-11}{-5-11}$$

$$\frac{x-2}{-8}=\frac{y+5}{12}=\frac{z-11}{-16}$$

$$\frac{x-2}{-2}=\frac{y+5}{3}=\frac{z-11}{-4}=\lambda \text{ (Let)}$$

Let $Q(-2\lambda+2, 3\lambda-5, -4\lambda+11)$

DR's of QR

$(-2\lambda+2-1, 3\lambda-5-7, -4\lambda+11-6)$

$(-2\lambda+1, 3\lambda-12, -4\lambda+5)$

$\because$ RQ $\perp$ Line L

$\therefore -2(-2\lambda+1)+3(3\lambda+12)-4(-4\lambda+5)=0$

$\Rightarrow 4\lambda-2+9\lambda-36+16\lambda-20=0$

$\therefore 29\lambda=58 \Rightarrow \lambda=2$

$\therefore (-2\times2+2, 3\times2-5, -4\times2+11)$

Q(–2, 1, 3)

$$\therefore PQ=\sqrt{(10+2)^2+(-2-1)^2+(-1-3)^2}$$

$$=\sqrt{144+9+16}$$

$$=\sqrt{169}$$

PQ = 13

195. If the shortest distance between the lines $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to _____. **[JEE(Main)April,2024]**

यदि रेखाओं $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ तथा $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ के बीच की न्यूनतम दूरी $\frac{44}{\sqrt{30}}$ है, तो $|\lambda|$ का अधिकतम संभव eku cjkcj gS_____ **[JEE(Main)April,2024]**

Ans. 43

Sol. $L_1: \frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$

$L_2: \frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$

$$n_1 \times n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix}$$

$$= -6\hat{i} - 15\hat{j} + 3\hat{k}$$

$$d = \left| \frac{\left[(\lambda+2)\hat{i} + 7\hat{j} - 3\hat{k}\right]\left[-6\hat{i} - 15\hat{j} + 3\hat{k}\right]}{|-6\hat{i} - 15\hat{j} + 3\hat{k}|} \right| = \frac{44}{\sqrt{30}}$$

$$\left| \frac{-6\lambda - 12 - 105 - 9}{\sqrt{270}} \right| = \frac{44}{\sqrt{30}}$$

$|6\lambda + 126| = 132$

$|\lambda + 21| = 22$

$\lambda + 21 = +22$

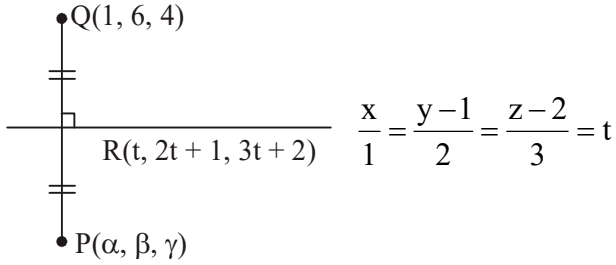
$|\lambda|_{max} = 43$

196. Let $P(\alpha, \beta, \gamma)$ be the image of the point Q(1,6,4) in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Then $2\alpha + \beta + \gamma$ is equal to _____. **[JEE(Main)April,2024]**

माना रेखा $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ में बिन्दु Q(1,6,4) का प्रतिबिंब $P(\alpha, \beta, \gamma)$ है। तो $2\alpha + \beta + \gamma$ बराबर है _____. **[JEE(Main)April,2024]**

Ans. 11

Sol.



$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$

$$\overrightarrow{QR} \cdot \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) = 0$$

$(t - 1) + (2t - 5) \times 2 + (3t - 2) \times 3 = 0 \qquad \Rightarrow t = \frac{17}{14}$

$$\Rightarrow R \equiv \left(\frac{17}{14}, \frac{48}{14}, \frac{79}{14}\right)$$

$$\Rightarrow \frac{\alpha+1}{2} = \frac{17}{14}, \frac{\beta+6}{2} = \frac{48}{14}, \frac{\gamma+4}{2} = \frac{79}{14}$$

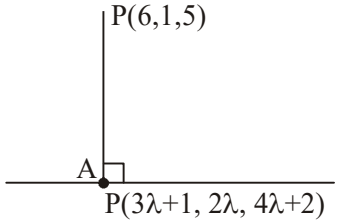
$$2\alpha + \beta + \gamma = \frac{68}{14} - 2 + \frac{96}{14} - 6 + \frac{158}{14} - 4 = 11$$

197. The square of the distance of the image of the point (6,1,5) in the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$, from the origin is _____. **[JEE(Main)April,2024]**

बिन्दु (6,1,5) के रेखा $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$ में प्रतिबिंब की मूल बिन्दु से दूरी का वर्ग है _____. **[JEE(Main)April,2024]**

Ans. 62

Sol.



$$\overrightarrow{PA} = (3\lambda - 5)\hat{i} + (2\lambda - 1)\hat{j} + (4\lambda - 3)\hat{k}$$

$(3\lambda - 5)3 + (2\lambda - 1)2 + (4\lambda - 3)4 = 0$

$\Rightarrow 9\lambda - 15 + 4\lambda - 2 + 16\lambda - 12 = 0$

$\Rightarrow 29\lambda = 29$

$\therefore \lambda = 1$

$\therefore$ A(4, 2, 6)

$\therefore$ P' : mirror image of P

$\Rightarrow$ P'(2, 3, 7)

$(OP')^2 = 4 + 9 + 49$

$= 62$